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Artificial Intelligence (AI) and Machine Learning (ML) on Big Data Seismology

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Abstract

The exponential expansion of continuous seismic recording archives and the deployment of dense regional networks have created severe operational bottlenecks for traditional earthquake monitoring workflows. Classical automated phase pickers, such as Short-Time Average/Long-Time Average (STA/LTA) and autoregressive Akaike Information Criterion (AIC) formulations, degrade substantially when confronted with low Signal-to-Noise Ratio (SNR), emergent phase onsets, overlapping seismic codas, and anthropogenic noise. Consequently, regional seismic observatories remain heavily dependent on labor-intensive manual analyst review, introducing latency and limiting catalog completeness at microseismic scales.

This dissertation presents the design, implementation, and empirical validation of an end-to-end, procedural machine learning pipeline for automated earthquake pick detection, phase association, hypocentral relocation, and magnitude characterization, specifically engineered for scalable execution on modern High Performance Computing (HPC) architectures. The computational framework leverages the open-source SeisBench ecosystem to deploy deep-learning phase picking architectures, including the 1D U-Net convolutional network PhaseNet (PhaseNet) Zhu et al., 2018 and the hierarchical convolutional-transformer network Earthquake Transformer (EQTransformer) Mousavi et al., 2020. To systematically investigate domain adaptation and model transferability, these pickers are evaluated across diverse regional and global benchmark training datasets: the Italian seismic dataset Italian Seismic Dataset for Machine Learning (INSTANCE) Michelini et al., 2021; the global seismic dataset STanford EArthquake Dataset (STEAD) Mousavi et al., 2019; the Southern California seismic dataset Southern California Earthquake Data Center (SCEDC) Southern California Earthquake Center, 2013 archive; and the original author training sets.

The target case study comprises continuous three-component waveform streams recorded by the Sistema di Monitoraggio terrestre dell'Italia Nord Orientale (SMINO), managed by the Istituto Nazionale di Oceanografia e di Geofisica Sperimentale (OGS), augmented by seventeen regional and transboundary networks in Veneto, Trentino-Alto Adige, Austria, and Slovenia. The network configuration encompasses 266 stations and over 790 channels sampled at 100 Hz. The observation horizon spans from DATE RANGE, centered

on the March 27, 2024, M_L 4.6 (M_w 4.5) Socchieve earthquake in Carnia, Friuli—one of the most energetic seismic episodes in Northeastern Italy since the destructive 1976 sequence.

Detected phase arrivals are clustered and associated into discrete seismic events using two complementary state-of-the-art algorithms: the Bayesian Gaussian Mixture Model Associator (GaMMA) Zhu et al., 2022 and the high-throughput octree-based associator High-Throughput Octree-Based Seismic Phase Associator (PyOcto) Münchmeyer, 2024. Associated events are relocated using a 1D probabilistic non-linear search via Non-Linear Earthquake Location Algorithm (NonLinLoc) Lomax et al., 2000, followed by a parametric local magnitude (M_L) inversion. To establish an unbiased comparative baseline, the OGS reference catalog for the observation window was independently relocated under the identical 1D NonLinLoc velocity model, yielding `NUM MATCHED EVENTS` and `NUM MATCHED PICKS` manually curated phase picks (`NUM MATCHED P PICKS` P and `NUM MATCHED S PICKS` S). Performance is evaluated using a physically bounded Bipartite Graph Matching Algorithm (BPGMA), resolving matched, missed, and proposed events and picks with strict spatiotemporal and phase-type conservation.

Distributed execution on the CINECA Leonardo supercomputer (EuroHPC Booster partition, utilizing NVIDIA A100 GPUs and multi-threaded CPU pipelines) demonstrates near-linear scalability, reducing continuous waveform processing times by orders of magnitude compared to legacy serial workflows. Systematic evaluation reveals that PhaseNet trained on the regional INSTANCE dataset achieves the highest pick recovery, optimal phase residual distributions, and the most reliable association completeness, demonstrating that domain-specific regional noise representation outweighs sheer dataset volume for regional seismotectonic surveillance. The proposed infrastructure provides operational observatories with a scalable, reproducible foundation for high-resolution historical catalog reprocessing.

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Acronyms

ADSAI Applied Data Science and Artificial Intelligence

AI Artificial Intelligence

AIC Akaike Information Criterion

ARSO Agencija Republike Slovenije za okolje

BPGMA Bipartite Graph Matching Algorithm

CINECA Consorzio Interuniversitario del Nord-Est per il Calcolo Automatico

CNN Convolutional Neural Network

CRS Centro di Ricerche Sismologiche

DBSCAN Density-Based Spatial Clustering of Applications with Noise

DL Deep Learning

EDT Equal Differential Time

EIDA European Integrated Data Archive

EM Expectation-Maximization

EQTransformer Earthquake Transformer

FDSN International Federation of Digital Seismograph Networks

GaMMA Gaussian Mixture Model Associator

HPC High Performance Computing

HypoDD Hypocenter Double-Difference

INGV Istituto Nazionale di Geofisica e Vulcanologia

INSTANCE Italian Seismic Dataset for Machine Learning

KPI Key Performance Indicator

LSTM Long Short-Term Memory

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MHPC Master in High Performance Computing

MIGe Dipartimento di Matematica, Informatica e Geoscienze

ML Machine Learning

MPI Message Passing Interface

NonLinLoc Non-Linear Earthquake Location Algorithm

NUMA Non-Uniform Memory Access

OGS Istituto Nazionale di Oceanografia e di Geofisica Sperimentale

ORFEUS Observatories and Research Facilities for European Seismology

PhaseNet PhaseNet

PNRR Piano Nazionale di Ripresa e Resilienza

PyOcto High-Throughput Octree-Based Seismic Phase Associator

QoS Quality of Service

RAN Rete Accelerometrica Nazionale

SCEDC Southern California Earthquake Data Center

SED Swiss Seismological Service

SMINO Sistema di Monitoraggio terrestre dell'Italia Nord Orientale

SNR Signal-to-Noise Ratio

STA/LTA Short-Time Average/Long-Time Average

STEAD STanford EArthquake Dataset

TeRABIT Terabit Network for Research and Academic Big Data in Italy

UniTS Università degli Studi di Trieste

Introduction

0.1 Seismological Foundations and Wave Propagation Physics

Earthquakes represent transient mechanical disturbances generated by the sudden release of accumulated elastic strain energy within the Earth's lithosphere. The lithosphere behaves as a brittle-elastic medium under tectonic differential stress; when shear stresses acting along a pre-existing fault plane exceed the frictional resistance governed by Byerlee's law or rate-and-state friction, dynamic shear failure nucleates and ruptures along the fault surface Byerlee, 1978; Dieterich, 1979. This catastrophic dislocation radiates transient elastodynamic stress waves that propagate through the heterogeneous interior and along the free surface of the Earth Kanamori et al., 2004.

In a continuous, isotropic, linearly elastic medium characterized by mass density $\rho(\mathbf{x})$ and Lamé elastic moduli $\lambda(\mathbf{x})$ and $\mu(\mathbf{x})$, the infinitesimal elastodynamic equation of motion expressing conservation of linear momentum is formulated as:

$$\rho(\mathbf{x}) \frac{\partial^2 u_i}{\partial t^2}(\mathbf{x}, t) = \frac{\partial \sigma_{ij}}{\partial x_j}(\mathbf{x}, t) + f_i(\mathbf{x}, t) \quad (1)$$

where $\mathbf{u}(\mathbf{x}, t) = [u_1, u_2, u_3]^T$ denotes the displacement vector field, $\boldsymbol{\sigma}(\mathbf{x}, t)$ is the symmetric Cauchy stress tensor, and $\mathbf{f}(\mathbf{x}, t)$ represents external body force distributions per unit volume (such as equivalent double-couple source representations of fault rupture). Under the generalized Hooke's law for isotropic elasticity, the constitutive stress-strain relation is given by:

$$\sigma_{ij} = \lambda \delta_{ij} \epsilon_{kk} + 2\mu \epsilon_{ij}, \quad (2)$$

where δ_{ij} is the Kronecker delta, $\epsilon_{ij} = (\partial_{x_j} u_i + \partial_{x_i} u_j) / 2$ is the infinitesimal strain tensor, and Einstein summation convention over repeated indices is implied. For the homogeneous-medium specialization used below, let $\rho > 0$ and $\lambda, \mu > 0$ be spatially constant. Substituting the constitutive law Eq. (2) into the equation of motion Eq. (1) then yields the Navier–Cauchy

elastodynamic wave equation:

$$\rho \frac{\partial^2 \mathbf{u}}{\partial t^2} = (\lambda + \mu) \nabla (\nabla \cdot \mathbf{u}) + \mu \nabla^2 \mathbf{u} + \mathbf{f}. \quad (3)$$

For source-free body-wave modes, set $\mathbf{f} = 0$. On an unbounded domain, assume that the displacement and potentials decay sufficiently rapidly at infinity. With the Coulomb gauge $\nabla \cdot \boldsymbol{\psi} = 0$, Helmholtz's decomposition represents the displacement as an irrotational (curl-free) scalar potential $\phi(\mathbf{x}, t)$ and a solenoidal (divergence-free) vector potential $\boldsymbol{\psi}(\mathbf{x}, t)$:

$$\mathbf{u} = \nabla \phi + \nabla \times \boldsymbol{\psi}, \quad \text{with} \quad \nabla \cdot \boldsymbol{\psi} = 0. \quad (4)$$

The gauge and decay conditions remove harmonic residuals in the potential equations. Substituting decomposition Eq. (4) into the source-free homogeneous equation of motion Eq. (3) therefore decouples the wavefield into two independent wave equations propagating through the medium at distinct characteristic phase velocities:

$$\frac{\partial^2 \phi}{\partial t^2} = \alpha^2 \nabla^2 \phi, \quad \frac{\partial^2 \boldsymbol{\psi}}{\partial t^2} = \beta^2 \nabla^2 \boldsymbol{\psi}, \quad (5)$$

where α and β represent the longitudinal compressional (Primary, P) and transverse shear (Secondary, S) wave propagation velocities, respectively:

$$\alpha = \sqrt{\frac{\lambda + 2\mu}{\rho}}, \quad \beta = \sqrt{\frac{\mu}{\rho}}. \quad (6)$$

The physical characteristics and particle motion trajectories of these seismic phases govern the observational records captured by modern seismological sensor systems:

- (a) **Primary (P) Waves:** Compressional or dilatational longitudinal waves that propagate via volumetric deformation ($\nabla \cdot \mathbf{u} \neq 0, \nabla \times \mathbf{u} = 0$). Particle motion is strictly parallel to the direction of wave propagation. Because $\lambda > 0$ and $\mu > 0$, $\alpha > \beta$ holds universally, ensuring that P waves represent the fastest seismic phase and form the first onset arrival observed on seismograms. P waves propagate through solids, fluids, and gases.
- (b) **Secondary (S) Waves:** Equivoluminal transverse shear waves that propagate via shear deformation without volume change ($\nabla \cdot \mathbf{u} = 0, \nabla \times \mathbf{u} \neq 0$). Particle displacement occurs perpendicular to the ray propagation vector. In a Cartesian coordinate system aligned with the incidence plane, S waves are decomposed into vertically polarized shear motion (SV) and horizontally polarized shear motion (SH). Because fluids have zero shear modulus ($\mu = 0$), S waves propagate exclusively through solid elastic

media.

In an ideal Poissonian solid ($\lambda = \mu$, equivalent to Poisson's ratio $\nu = 0.25$), the kinematic velocity ratio satisfies:

$$\frac{\alpha}{\beta} = \sqrt{\frac{\lambda + 2\mu}{\mu}} = \sqrt{3} \approx 1.732. \quad (7)$$

In complex geological crustal formations, lithological variations, microcrack porosity, and fluid saturation cause the observed V_P/V_S ratio to vary typically between 1.65 and 1.90.

At discontinuous geological interfaces and the Earth's stress-free surface ($\sigma_{i3}|_{z=0} = 0$), mode conversion and interference between body waves generate **surface waves**. These propagate horizontally along the crustal waveguide with amplitudes decaying exponentially with depth:

- **Rayleigh Waves:** Resulting from constructive P-SV interference at the free boundary, exhibiting retrograde elliptical particle motion in the vertical-radial propagation plane, with phase velocities $c_R \approx 0.92\beta$.
- **Love Waves:** Resulting from constructive SH wave trapping within low-velocity surface layers overlying a higher-velocity half-space, exhibiting purely horizontal transverse motion perpendicular to the propagation path.

Modern digital seismological observatories record these vector ground motions along three mutually orthogonal spatial components: Vertical (Z), North-South (N), and East-West (E). Observational stations deploy high-gain broadband seismometers (*e.g.*, measuring velocity $\dot{u}(t)$ over flat bandwidths from 0.01 to 50 Hz) and strong-motion accelerometers (*e.g.*, measuring ground acceleration $\ddot{u}(t)$ without clipping during high-amplitude near-source shaking). Data are continuously digitized at high sampling rates (100 Hz) and packaged into standard miniSEED (MSEED) containers, with instrumental transfer functions, calibration poles, zeros, and geodetic station coordinates formally cataloged in StationXML schemas.

0.2 Seismotectonic Context of Northeastern Italy and the 1976 Legacy

Northeastern Italy, embracing the Friuli Venezia Giulia (FVG) and Veneto regions, represents one of the most tectonically active and seismically hazardous sectors of the entire European Alpine arc. The geodynamic evolution of the region is governed by the active collision and northward indentation of the Adriatic microplate (Adria) into the stable Eurasian plate. Geodetic measurements from continuous GNSS networks indicate that

the Adria microplate rotates counterclockwise around an Eulerian pole in the Western Po Plain, producing active crustal shortening across the Eastern Southern Alps at rates of 1.5–3.0 mm/yr in a NNW-SSE orientation Anderson et al., 1987; Handy et al., 2014.

Structurally, Northeastern Italy constitutes a complex tectonic junction where two major collisional orogenic belts converge:

1. The **Eastern Southern Alps**: A south-verging thrust-and-fold retro-wedge characterized by E-W to WNW-ESE striking thrust faults and ramp-anticlines (including the Periadriatic Lineament, the Fella-Sava fault, the Maniago thrust, and the Buia-Tricesimo thrust system) that accommodate crustal shortening through north-dipping blind and exposed thrusting.
2. The **External Dinarides**: A dextral strike-slip structural system extending from Slovenia into Eastern Friuli, dominated by NW-SE trending transcurrent fault structures such as the Idrija and Ravne fault networks. Bressan et al., 1998

This tectonic configuration results in concentrated intraplate seismicity characterized by shallow hypocenters (typically 5–15 km depth), predominantly compressive thrust-faulting mechanisms with subordinate strike-slip kinematics, and recurrent destructive earthquake sequences.

The defining modern seismic episode in this region is the catastrophic **1976 Friuli earthquake sequence**. On May 6, 1976, at 20:00:12 UTC, a major earthquake of moment magnitude M_w 6.4 (M_L 6.5) nucleated near Gemona del Friuli, producing epicentral intensities of $I_0 = IX-X$ on the Mercalli-Cancani-Sieberg (MCS) scale. The mainshock ruptured blind thrust faults along the Southalpine front Aoudia et al., 2000. The sequence was compounded in September 1976 by two devastating aftershocks on September 11 (M_w 5.9) and September 15 (M_w 6.0). In total, the 1976 Friuli seismic crisis claimed 989 lives, injured over 2,800 people, left over 100,000 homeless, and destroyed entire medieval villages across the Tagliamento and Fella river basins.

The tragedy of the 1976 disaster catalyzed a profound transformation in European seismology, earthquake engineering, civil protection organization, and disaster response legislation. Crucially, it led the Regional Government of Friuli Venezia Giulia to mandate the establishment of a dedicated permanent seismological monitoring infrastructure. In 1977, the Istituto Nazionale di Oceanografia e di Geofisica Sperimentale (OGS) established the Centro di Ricerche Sismologiche (CRS) in Udine and Trieste. Since its inception, OGS CRS has maintained continuous, high-resolution seismic monitoring across Northeastern Italy, serving as the official technical-scientific body supporting the regional Civil Protection agencies of Friuli Venezia Giulia and Veneto.

0.3 The OGS-SMINO Monitoring Network Infrastructure

Over four decades of technological development, the initial sparse analog network deployed after the 1976 earthquake evolved into the modern Sistema di Monitoraggio terrestre dell'Italia Nord Orientale (SMINO) (Bragato et al., 2021). The SMINO monitoring infrastructure represents a state-of-the-art, multi-parametric, real-time observation system covering Friuli Venezia Giulia, Veneto, and surrounding border areas.

The regional monitoring infrastructure integrates:

- A dense permanent backbone of digital three-component broadband and very-broadband seismometers (principally Streckeisen STS-2, Nanometrics Trillium, and Guralp sensors) installed in vaults, tunnels, and boreholes to minimize cultural and atmospheric noise.
- High-dynamic-range strong-motion accelerometers (Kinematics Episensor) collocated at broadband sites and deployed across urban centers and critical engineering infrastructures to ensure on-scale recordings of strong ground motions.
- Cross-border data exchange with neighboring Italian, Austrian, and Slovenian institutions, creating a unified transboundary seismic monitoring domain.

In operational practice, seismic monitoring in Northeastern Italy is not conducted in isolation; wavefield propagation respects no administrative boundaries. As described in Chapter 1, the complete regional monitoring aperture analyzed in this study encompasses **266 stations** originating from 18 distinct regional, national, and international networks (including OGS, Istituto Nazionale di Geofisica e Vulcanologia (INGV), GeoSphere Austria, the Agencija Republike Slovenije za okolje (ARSO), the Swiss Seismological Service (SED), and the Rete Accelerometrica Nazionale (RAN)). The continuous recording from this dense collective array provides the empirical foundation for our machine learning and computational investigations.

0.4 Classical Automated Picking and Association Limitations

Transforming continuous, multi-channel ground motion velocity streams into discrete, scientifically verified earthquake catalogs requires identifying the exact onset times of P- and S-wave phase arrivals (phase picking) and grouping arrivals originating from the same physical hypocentral rupture (phase association).

For over four decades, automated seismological processing has relied on classical heuristic algorithms:

1. **Short-Time Average over Long-Time Average (STA/LTA):** Introduced in foundational form by Allen (1978), the method calculates the ratio of energy or amplitude within a short sliding time window (T_{STA}) to that within a preceding long time window (T_{LTA}) over a characteristic function $CF(t)$:

$$R(t) = \frac{STA(t)}{LTA(t)} = \frac{\frac{1}{T_{\text{STA}}} \int_{t-T_{\text{STA}}}^t CF(\tau) d\tau}{\frac{1}{T_{\text{LTA}}} \int_{t-T_{\text{LTA}}}^t CF(\tau) d\tau}, \quad (8)$$

where $CF(t)$ is commonly defined as $y(t)^2$ or $y(t)^2 + C\dot{y}(t)^2$. A pick trigger is declared when $R(t) \geq \eta_{\text{on}}$, and deactivated when $R(t) \leq \eta_{\text{off}}$.

2. **Autoregressive Akaike Information Criterion (AIC):** Applied to seismic phase timing by Leonard et al. (1999), the AIC models the seismogram before and after an arbitrary split point k as two distinct stationary autoregressive processes. The minimum of the AIC function identifies the optimal transition between background noise and seismic onset:

$$AIC(k) = k \log(\hat{\sigma}_1^2) + (N - k) \log(\hat{\sigma}_2^2), \quad (9)$$

where $\hat{\sigma}_1^2$ and $\hat{\sigma}_2^2$ are the variance estimators of the prediction errors for intervals $[1, k]$ and $[k + 1, N]$, respectively.

Despite their enduring historical utility, classical algorithms suffer from fundamental physical and algorithmic limitations:

- **Vulnerability to Noise and False Triggers:** STA/LTA is highly sensitive to non-stationary transient noise, cultural vibrations, electrical glitches, and meteorological disturbances, causing excessive false trigger rates in densely populated or industrial valleys.
- **Emergent and Low-SNR Degradation:** Onsets of small-magnitude earthquakes ($M_L < 2.0$) or emergent arrivals traveling through attenuative, heterogeneous crustal paths exhibit low signal-to-noise ratios (SNR). STA/LTA triggers systematically late, while AIC minimization frequently settles into local minima created by noise fluctuations.
- **Phase Misidentification:** Neither STA/LTA nor AIC natively distinguishes between compressional (P) and shear (S) phases without ad-hoc, brittle horizontal-to-vertical

amplitude ratios or polarization filters. S-wave arrivals arriving within the high-amplitude, reverberating P-wave coda are regularly missed or misclassified.

- **Combinatorial Association Bottlenecks:** Grid-search associators that cluster picks by evaluating travel-time residuals across spatial velocity grids exhibit exponential computational scaling $\mathcal{O}(N_{\text{picks}}^k)$ with pick volume. When spurious noise picks enter the pipeline, association algorithms suffer combinatoric explosion or form hallucinated, unphysical hypocenters.

Because of these limitations, operational seismological observatories have historically required dedicated human analysts to inspect, refine, or manually pick phase arrivals for every candidate event. This manual intervention introduces severe operational latency (often hours to days), creates subjective timing inconsistencies between analysts, and establishes an insurmountable operational ceiling that forces observatories to discard microseismic events below completeness thresholds ($M_c \approx 1.5$).

0.5 The Big Data Era and Deep Learning in Seismology

Over the past decade, the rapid expansion of permanent networks, dense nodal geophone arrays, and real-time transboundary telemetry has propelled observational seismology into the era of “Big Data.” Continuous seismic archives at regional and national data centers ingest tens of gigabytes per day and routinely accumulate terabytes to petabytes of continuous waveform data annually. Analyzing these continuous, multi-channel data streams exhaustively is beyond human manual processing capacity.

This data deluge has coincided with revolutionary advances in Deep Learning (DL). Deep neural networks possess the unique ability to learn rich, non-linear, hierarchical feature representations directly from continuous, uncurated sensor time series without requiring manual feature engineering. Applied to seismic waveforms, deep convolutional and attention-based architectures learn the multi-scale spatio-temporal signatures of wave propagation, dispersion, impedance contrasts, and noise textures Mousavi et al., 2022.

Key architectural breakthroughs include:

- **PhaseNet** (Zhu et al., 2018): A deep 1D Convolutional Neural Network (Convolutional Neural Network (CNN)) based on the U-Net architecture. PhaseNet employs an encoder-decoder topology with skip connections, processing three-component waveforms through successive downsampling and upsampling convolutional stages to produce continuous probability distributions for P-wave arrivals, S-wave arrivals, and noise.

- **Earthquake Transformer (EQTransformer)** (Mousavi et al., 2020): A sophisticated multi-task deep network that integrates 1D convolutional layers, bidirectional Long Short-Term Memory (LSTM) recurrent units, and multiple multi-head self-attention mechanisms. EQTransformer models both local waveform geometries and long-range temporal dependencies across 60-second windows, simultaneously outputting detection probabilities for earthquake presence, P-wave arrival, and S-wave arrival.

To overcome software fragmentation and provide standardized interfaces for benchmarking, Woollam et al. (2022) introduced **SeisBench** (Woollam et al., 2022), an open-source platform offering unified Python APIs for machine learning in seismology. SeisBench provides standardized data loaders, benchmark training datasets, and pre-trained weights for state-of-the-art architectures, dramatically enhancing the reproducibility of deep-learning seismic workflows. In this thesis, SeisBench serves as the core foundation for evaluating pre-trained phase pickers across diverse training corpora.

0.6 The March 27, 2024 Socchieve Earthquake Sequence

To test the operational efficacy, robustness, and scientific fidelity of deep-learning pipelines under demanding real-world conditions, this research focuses on an intense seismic sequence that struck Friuli in early 2024.

On **March 27, 2024, at 21:19:37 UTC** (22:19:37 local time), a magnitude M_L 4.6 ± 0.3 (M_w 4.5) earthquake occurred in the Carnic Prealps, approximately 5 km south-southwest of Socchieve, Udine province (hypocentral coordinates: Latitude 46.3583° N, Longitude 12.8080° E, focal depth 10.22 km). The earthquake was strongly felt across the entire Friuli Venezia Giulia and Veneto regions, as well as in neighboring Austria and Slovenia, causing localized structural damage, masonry cracking, and rockfalls in the epicentral area.

This earthquake constitutes the **most energetic tectonic event to occur in Friuli since the 1976–1980 crisis**. It initiated a vigorous aftershock sequence comprising hundreds of minor earthquakes and microseismic events that decayed over subsequent weeks and months. The spatial proximity of dense seismic instrumentation, combined with the presence of complex Alpine topography, variable focal mechanisms, and realistic operational cultural noise, renders the Socchieve sequence a pristine natural testbed for stress-testing machine learning detection, phase association, and hypocentral relocation against an expert analyst ground truth.

0.7 Research Objectives and Scope of the Dissertation

The overarching scientific objective of this doctoral dissertation is to construct, optimize, deploy, and rigorously validate a complete, automated, procedural machine learning pipeline for earthquake pick detection, phase association, and event characterization, specifically engineered for scalable execution on HPC infrastructures.

The key research questions addressed in this thesis are:

1. **Domain Adaptation and Generalization:** How effectively do deep-learning phase pickers (PhaseNet and EQTransformer) trained on large external datasets (INSTANCE in Italy, STEAD globally, SCEDC in California) generalize to the complex geological and noise environment of Northeastern Italy? Does regional training data (INSTANCE) outperform massive but geographically foreign corpora (SCEDC, STEAD)?
2. **Associator Scaling and Performance:** How do advanced phase association algorithms—specifically the Bayesian Gaussian Mixture Model Associator (GaMMA) and the high-throughput octree-based associator (PyOcto)—compare in association recall, computational efficiency, and robustness against false-positive pick contamination?
3. **Physically Bounded Performance Validation:** How can model-generated earthquake catalogs be compared against manual analyst reference catalogs without introducing spatiotemporal bias or duplicate assignment artifacts?
4. **HPC Scalability and Operational Viability:** Can the entire processing workflow—from continuous multi-client FDSN waveform ingestion to deep inference, association, and relocation—be parallelized across modern GPU-accelerated supercomputing architectures (such as the CINECA Leonardo Tier-0 system) to support near-real-time operational monitoring and rapid historical archive reprocessing?

0.8 Dissertation Structure and Chapter Roadmap

The remainder of this dissertation is organized into eight interconnected chapters, followed by a dedicated Data and Resources statement and technical appendices:

- **Chapter 1 (Datasets):** Details the operational case study data acquired across Northeastern Italy (DATE RANGE, STATIONS stations, NUM EVENTS relocated OGS reference events with 7,129 picks) and reviews the four benchmark training corpora (INSTANCE, STEAD, SCEDC, and Original sets).

- **Chapter 2 (Data Engineering):** Presents the software architecture of the pipeline, detailing the central `ogsconstants` configuration, weight-to-uncertainty mappings, automated FDSN multi-client waveform downloading, multi-format legacy catalog parsing (HPL, DAT, TXT, PUN), catalog aggregation, geodetic math, and visualization frameworks.
- **Chapter 3 (Computational Pipeline):** Formulates the complete procedural workflow, detailing continuous waveform preprocessing, deep-learning phase picking inference, Bayesian (GaMMA) and octree (PyOcto) phase association, 1D NonLinLoc probabilistic relocation, and local magnitude (M_L) inversion.
- **Chapter 4 (Pipeline Evaluation Framework):** Develops the formal evaluation methodology, defining bipartite graph maximum-weight matching (BPGMA) for both picks and events, the construction of domain-specific confusion matrices, and the mathematical derivation of recall, precision, and F_1 scores.
- **Chapter 5 (HPC Implementation):** Analyzes the high-performance computing implementation deployed on the CINECA Leonardo supercomputer (EuroHPC Booster partition), detailing SLURM job orchestration, MPI and GPU process allocation, parallel I/O throughput, and execution scaling profiles.
- **Chapter 6 (Results):** Delivers a comprehensive empirical evaluation of the benchmarking campaign on the Northeastern Italy dataset, presenting pick detection recall curves, timing residual distributions, hypocentral relocation offsets, magnitude fidelity, and comparative associator analysis.
- **Chapter 7 (Conclusion):** Synthesizes the core scientific and engineering findings, discusses the implications for modern operational seismological observatories, outlines current limitations, and charts future directions for AI-driven seismological monitoring.
- **Data and Resources:** Declares the persistent digital object identifiers (DOIs), software repositories, and access endpoints for all code, models, and catalogs developed in this work.
- **Appendix:** Documents supplementary mathematical derivations, configuration schemas, and detailed per-station performance tables.

Supplementary Material: Introduction

Derivation of the Isotropic Elastodynamic Wave Equation

Let $\Omega \subseteq \mathbb{R}^3$ be the spatial domain while time $t \in \mathbb{R}$, with indices $i, j, k \in \{1, 2, 3\}$ following the Einstein summation convention (*i.e.*, repeated indices are summed). The displacement is $\mathbf{u}(\mathbf{x}, t)$; the density $\rho(\mathbf{x}) > 0$, and the Lamé parameters $\lambda(\mathbf{x}), \mu(\mathbf{x}) > 0$ are spatially constant material parameters. Thus, the derivation concerns a homogeneous, isotropic, linear-elastic medium; it follows the standard continuum-elastodynamics treatment in Aki et al., 1980.

$$\nabla \cdot \boldsymbol{\sigma} + \mathbf{f} = \rho \frac{\partial^2 \mathbf{u}}{\partial t^2} \quad (10)$$

Kinematics and constitutive relation

The infinitesimal strain, its trace, and isotropic Hooke law are

$$\begin{aligned} \epsilon_{ij} &= \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) = \epsilon_{ji}, & \epsilon_{kk} &= \frac{\partial u_k}{\partial x_k} = \nabla \cdot \mathbf{u}, \\ \sigma_{ij} &= \lambda \delta_{ij} \epsilon_{kk} + 2\mu \epsilon_{ij} \\ &= \lambda \delta_{ij} \frac{\partial u_k}{\partial x_k} + \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right), \end{aligned}$$

where δ_{ij} is the Kronecker delta.

Equation of motion

Since $\partial_{x_j}\lambda = \partial_{x_j}\mu = 0$, differentiation with respect to x_j gives

$$\begin{aligned}\frac{\partial\sigma_{ij}}{\partial x_j} &= \frac{\partial}{\partial x_j} \left[\lambda\delta_{ij}\frac{\partial u_k}{\partial x_k} + \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \right] \\ &= \lambda \frac{\partial}{\partial x_i} \left(\frac{\partial u_k}{\partial x_k} \right) + \mu \frac{\partial^2 u_i}{\partial x_j \partial x_j} + \mu \frac{\partial}{\partial x_i} \left(\frac{\partial u_j}{\partial x_j} \right) \\ &= (\lambda + \mu) \frac{\partial}{\partial x_i} \left(\frac{\partial u_k}{\partial x_k} \right) + \mu \nabla^2 u_i.\end{aligned}\tag{11}$$

The interchange $\partial_{x_j}\partial_{x_i}u_j = \partial_{x_i}\partial_{x_j}u_j$ used in the second line is valid because the displacement components are assumed to be twice continuously differentiable, $u_j \in C^2(\Omega \times \mathbb{R})$. Spatially constant λ and μ separately justify taking those coefficients outside the spatial derivatives.

Newton's second law applied to a differential volume gives a force balance per unit volume. Its inertial term is mass density times acceleration, $\rho\partial_t^2 u_i$. The stress-divergence term $\partial_{x_j}\sigma_{ij}$ is the i th component of the net internal force transmitted across the volume boundary. The remaining term f_i is an externally prescribed body force per unit volume, for example an equivalent force representation of a seismic source. Thus, for each $i \in \{1, 2, 3\}$, conservation of linear momentum gives

$$\begin{aligned}\rho \frac{\partial^2 u_i}{\partial t^2} &= \frac{\partial\sigma_{ij}}{\partial x_j} + f_i, \\ \rho \frac{\partial^2 \mathbf{u}}{\partial t^2} &= (\lambda + \mu) \nabla(\nabla \cdot \mathbf{u}) + \mu \nabla^2 \mathbf{u} + \mathbf{f}.\end{aligned}\tag{12}$$

The first line is the component-wise balance. Substituting Eq. (11) into it and collecting the three components produces the second line, the vector Navier–Cauchy equation in Eq. (3) of the main text.

P- and S-wave projections

For source-free body-wave modes, set $\mathbf{f} = 0$. In an unbounded homogeneous medium (*i.e.*, one with no physical boundaries and with a smooth displacement field that decays sufficiently rapidly at infinity), the Helmholtz decomposition expresses $\mathbf{u}(\mathbf{x}, t)$ through a scalar potential ϕ and a vector potential $\boldsymbol{\psi}$ Morse et al., 1953. The Coulomb gauge $\nabla \cdot \boldsymbol{\psi} = 0$, together with the stated decay condition, excludes nonzero harmonic residuals:

$$\begin{aligned}\mathbf{u} &= \mathbf{u}_P + \mathbf{u}_S = \nabla\phi + \nabla \times \boldsymbol{\psi}, & \nabla \cdot \boldsymbol{\psi} &= 0, \\ \nabla \cdot \mathbf{u} &= \nabla^2 \phi, & \nabla \times \mathbf{u} &= -\nabla^2 \boldsymbol{\psi}.\end{aligned}$$

Taking the divergence of Eq. (12) gives

$$\begin{aligned}\rho \frac{\partial^2}{\partial t^2}(\nabla \cdot \mathbf{u}) &= (\lambda + \mu) \nabla^2(\nabla \cdot \mathbf{u}) + \mu \nabla^2(\nabla \cdot \mathbf{u}) \\ &= (\lambda + 2\mu) \nabla^2(\nabla \cdot \mathbf{u}), \\ \frac{\partial^2}{\partial t^2}(\nabla^2 \phi) &= \frac{\lambda + 2\mu}{\rho} \nabla^2(\nabla^2 \phi).\end{aligned}$$

Equivalently,

$$\nabla^2 \left[\frac{\partial^2 \phi}{\partial t^2} - \frac{\lambda + 2\mu}{\rho} \nabla^2 \phi \right] = 0.$$

The bracketed residual is harmonic. The gauge and decay conditions exclude a nonzero harmonic residual, so the scalar potential satisfies

$$\frac{\partial^2 \phi}{\partial t^2} = \alpha^2 \nabla^2 \phi, \quad \alpha^2 = \frac{\lambda + 2\mu}{\rho}. \quad (13)$$

Taking the curl of Eq. (12) gives

$$\begin{aligned}\rho \nabla \times \frac{\partial^2 \mathbf{u}}{\partial t^2} &= (\lambda + \mu) \nabla \times \nabla(\nabla \cdot \mathbf{u}) + \mu \nabla \times \nabla^2 \mathbf{u} \\ &= \mathbf{0} + \mu \nabla^2(\nabla \times \mathbf{u}) \\ &= \mu \nabla^2(\nabla \times \mathbf{u}).\end{aligned}$$

The first term vanishes because the curl of a gradient is zero (*i.e.*, $\nabla \times \nabla f = 0$ for any smooth scalar field f). The temporal derivatives commute with curl, and curl commutes with the Laplacian because the field is sufficiently smooth. Therefore,

$$\begin{aligned}\rho \frac{\partial^2}{\partial t^2}(\nabla \times \mathbf{u}) &= \mu \nabla^2(\nabla \times \mathbf{u}), \\ \rho \frac{\partial^2}{\partial t^2}(-\nabla^2 \psi) &= \mu \nabla^2(-\nabla^2 \psi), \\ -\rho \nabla^2 \frac{\partial^2 \psi}{\partial t^2} &= -\mu \nabla^4 \psi.\end{aligned}$$

Equivalently,

$$\nabla^2 \left[\frac{\partial^2 \psi}{\partial t^2} - \frac{\mu}{\rho} \nabla^2 \psi \right] = \mathbf{0}.$$

The bracketed vector residual is harmonic. Under the same gauge and decay conditions, it vanishes, so the shear potential satisfies

$$\frac{\partial^2 \psi}{\partial t^2} = \beta^2 \nabla^2 \psi, \quad \beta^2 = \frac{\mu}{\rho}. \quad (14)$$

Thus the P mode depends on the bulk-plus-shear stiffness $\lambda + 2\mu$ whereas the S mode only depends on the shear modulus μ . Under the main text's conventional conditions $\rho > 0$, $\lambda > 0$, and $\mu > 0$,

$$\alpha^2 - \beta^2 = \frac{\lambda + \mu}{\rho} > 0 \implies \alpha > \beta.$$

Alternative curl–curl formulation

The preceding divergence and curl projections establish the scalar and vector potential equations directly. The equivalent curl–curl formulation below reconstructs the same separation at the displacement level, making explicit how volumetric and rotational motions enter the source-free Navier–Cauchy equation under the same gauge and decay conditions.

Using the vector identity for the Laplacian of a vector field,

$$\nabla^2 \mathbf{u} = \nabla(\nabla \cdot \mathbf{u}) - \nabla \times (\nabla \times \mathbf{u})$$

the source-free Navier–Cauchy equation Eq. (12) becomes

$$\begin{aligned} \rho \frac{\partial^2 \mathbf{u}}{\partial t^2} &= (\lambda + \mu) \nabla(\nabla \cdot \mathbf{u}) + \mu \nabla^2 \mathbf{u} \\ &= (\lambda + \mu) \nabla(\nabla \cdot \mathbf{u}) + \mu [\nabla(\nabla \cdot \mathbf{u}) - \nabla \times (\nabla \times \mathbf{u})] \\ &= (\lambda + 2\mu) \nabla(\nabla \cdot \mathbf{u}) - \mu \nabla \times (\nabla \times \mathbf{u}). \end{aligned} \tag{15}$$

The first term is the irrotational force associated with volumetric dilation or compression. The second is the rotational force associated with equivoluminal shear deformation.

Substitution into Eq. (15) gives

$$\begin{aligned} \rho \frac{\partial^2}{\partial t^2} (\nabla \phi + \nabla \times \boldsymbol{\psi}) &= (\lambda + 2\mu) \nabla(\nabla^2 \phi) - \mu \nabla \times \nabla \times (\nabla \times \boldsymbol{\psi}), \\ \nabla \times \nabla \times \boldsymbol{\psi} &= \nabla(\nabla \cdot \boldsymbol{\psi}) - \nabla^2 \boldsymbol{\psi} = -\nabla^2 \boldsymbol{\psi}, \\ \nabla \times \nabla \times (\nabla \times \boldsymbol{\psi}) &= -\nabla^2 (\nabla \times \boldsymbol{\psi}). \end{aligned}$$

Consequently, the irrotational and solenoidal contributions satisfy

$$\nabla \left[\rho \frac{\partial^2 \phi}{\partial t^2} - (\lambda + 2\mu) \nabla^2 \phi \right] + \nabla \times \left[\rho \frac{\partial^2 \boldsymbol{\psi}}{\partial t^2} - \mu \nabla^2 \boldsymbol{\psi} \right] = \mathbf{0}. \tag{16}$$

Taking the divergence of Eq. (16) gives the harmonic scalar residual derived above. Taking its curl gives a vanishing double curl of the vector residual; the Coulomb gauge makes that residual divergence-free, so the vector identity implies that it is harmonic as well. Under the stated decay condition, both residuals vanish. Therefore, Eq. (16) recovers the previously

obtained independent potential equations:

$$\rho \frac{\partial^2 \phi}{\partial t^2} = (\lambda + 2\mu) \nabla^2 \phi, \quad \rho \frac{\partial^2 \psi}{\partial t^2} = \mu \nabla^2 \psi. \quad (17)$$

Plane-wave check

Let $\mathbf{k}_\phi, \mathbf{k}_\psi \in \mathbb{R}^3$ be the wave vectors, the angular frequencies be $\omega_\phi > 0$ and $\omega_\psi > 0$, the scalar amplitude be $A \in \mathbb{C}$, and $\mathbf{B} \in \mathbb{C}^3$ the vector amplitude with $\mathbf{k}_\psi \cdot \mathbf{B} = 0$. For

$$\begin{aligned} \phi &= A \exp(i(\mathbf{k}_\phi \cdot \mathbf{x} - \omega_\phi t)), & \psi &= \mathbf{B} \exp(i(\mathbf{k}_\psi \cdot \mathbf{x} - \omega_\psi t)), \\ \frac{\partial^2 \phi}{\partial t^2} &= -\omega_\phi^2 \phi, & \nabla^2 \phi &= -\|\mathbf{k}_\phi\|^2 \phi, \\ \frac{\partial^2 \psi}{\partial t^2} &= -\omega_\psi^2 \psi, & \nabla^2 \psi &= -\|\mathbf{k}_\psi\|^2 \psi. \end{aligned}$$

Substitution into the two potential equations (Eq. (13), Eq. (14)) gives

$$\omega_\phi^2 = \alpha^2 \|\mathbf{k}_\phi\|^2, \quad v_P = \frac{\omega_\phi}{\|\mathbf{k}_\phi\|} = \alpha. \quad (18)$$

The same substitution for ψ gives

$$\omega_\psi^2 = \beta^2 \|\mathbf{k}_\psi\|^2, \quad v_S = \frac{\omega_\psi}{\|\mathbf{k}_\psi\|} = \beta. \quad (19)$$

These relations reproduce the P- and S-wave velocities stated in Eq. (6). The result is limited to the homogeneous, isotropic, linear-elastic, source-free body-wave derivation; interfaces, attenuation, anisotropy, and free-surface effects require additional boundary conditions and constitutive terms.

Chapter 1

Datasets

This chapter details the empirical seismological datasets underpinning this dissertation. Section 1.1 presents the target operational domain: the continuous waveform recordings and analyst-curated reference catalog acquired across Northeastern Italy during the three-month observation window of the spring 2024 Socchieve earthquake crisis. Section 1.2 presents an in-depth comparative review of the four benchmark training corpora utilized to train and evaluate the deep-learning phase pickers: INSTANCE (Italy), STEAD (global), SCEDC (Southern California), and the original author training sets of PhaseNet and EQ-Transformer. These collections span localized regional to global tectonic environments and exhibit diverse signal-to-noise ratios, noise textures, and labeling standards, establishing a testbed for evaluating the domain-adaptation hypothesis.

1.1 Target Regional Case Study: Northeastern Italy

The primary observational dataset serves as the target domain for evaluating the end-to-end performance, accuracy, and operational throughput of the procedural machine learning pipeline. The dataset comprises continuous multi-channel seismic recordings acquired across Northeastern Italy and adjacent border sectors in Austria, Slovenia, Veneto, and Trentino-Alto Adige.

The spatial, instrumental, and seismological parameters characterizing the case study include:

- **Geographical Bounds and Spatial Polygon:** The study region is formally bounded by the OGS operational polygon (`OGS_POLY_REGION`) defined in the central configuration module `ogsconstants` (see Section 2.1.1). The polygon encloses an eight-point convex domain spanning longitudes 10.0° E to 14.5° E and latitudes 44.5° N to 47.0° N, capturing the active thrust fronts of the Eastern Southern Alps and the

transcurrent structures of the Western Dinaric junction. The extended bounding box for waveform queries covers longitudes 9.5°–15.0° E and latitudes 44.3°–47.5° N (OGS_STUDY_REGION).

- **Instrumental Network Aperture:** Waveform data were retrieved from **266 distinct seismic stations** belonging to 18 regional, national, and international networks (Figure 1.1). The sensor network comprises high-sensitivity three-component broadband velocimeters (*e.g.*, Streckeisen STS-2, Nanometrics Trillium 120/240, Guralp CMG-3T/40T) and high-dynamic-range strong-motion accelerometers (Kinematics Episensor). Over 790 continuous channels were monitored, providing dense spatial sampling across the epicentral zone and surrounding Alpine valleys.
- **Temporal Window and Observation Horizon:** The target observation window spans **93 continuous days**, from DATE RANGE (00:00:00 UTC to 23:59:59 UTC). This NUM YEARS horizon captures pre-seismic background microseismicity, the energetic mainshock and immediate aftershock sequence, and the subsequent rate decay of aftershocks over seasonal environmental transitions.
- **The March 27, 2024 Socchieve Earthquake:** The focal seismic event anchoring this study occurred on **March 27, 2024, at 21:19:37 UTC** (22:19:37 local Italian time). The earthquake nucleated in the Carnic Prealps approximately 5 km south-southwest of Socchieve, Udine province (focal parameters: Latitude 46.3583° N, Longitude 12.8080° E, hypocentral depth 10.22 km, local magnitude M_L 4.6 ± 0.3 , moment magnitude M_w 4.5). This event represents the most energetic tectonic earthquake in Friuli since the 1976–1980 seismic crisis and generated extensive macroseismic shaking (EMS-98 intensity V–VI) throughout Friuli and Veneto, triggering rockfalls, masonry cracking, and emergency civil protection deployments.
- **Continuous Ingestion and Temporal Availability:** All continuous waveform data were retrieved in standardized miniSEED (MSEED) format directly from distributed International Federation of Digital Seismograph Networks (FDSN) data centers using the automated `ogsdownloader` utility (see Section 2.3). Data centers queried include OGS internal servers, INGV, GFZ, IRIS, ETH, and Observatories and Research Facilities for European Seismology (ORFEUS). Station availability over the observation horizon is illustrated in Figure 1.3, demonstrating sustained telemetry uptime across the core regional network. Instrument transfer functions, sensor orientations, gains, and geographic coordinates were parsed and validated from StationXML metadata files.

- **Sampling Rate Standardization:** All ingested continuous waveforms were resampled from their native acquisition frequencies (50, 100, or 200 Hz) to a standardized frequency of **100 Hz** ($\Delta t = 0.01$ s) via polyphase anti-aliasing decimation, ensuring exact alignment with the input tensors expected by SeisBench deep-learning models.

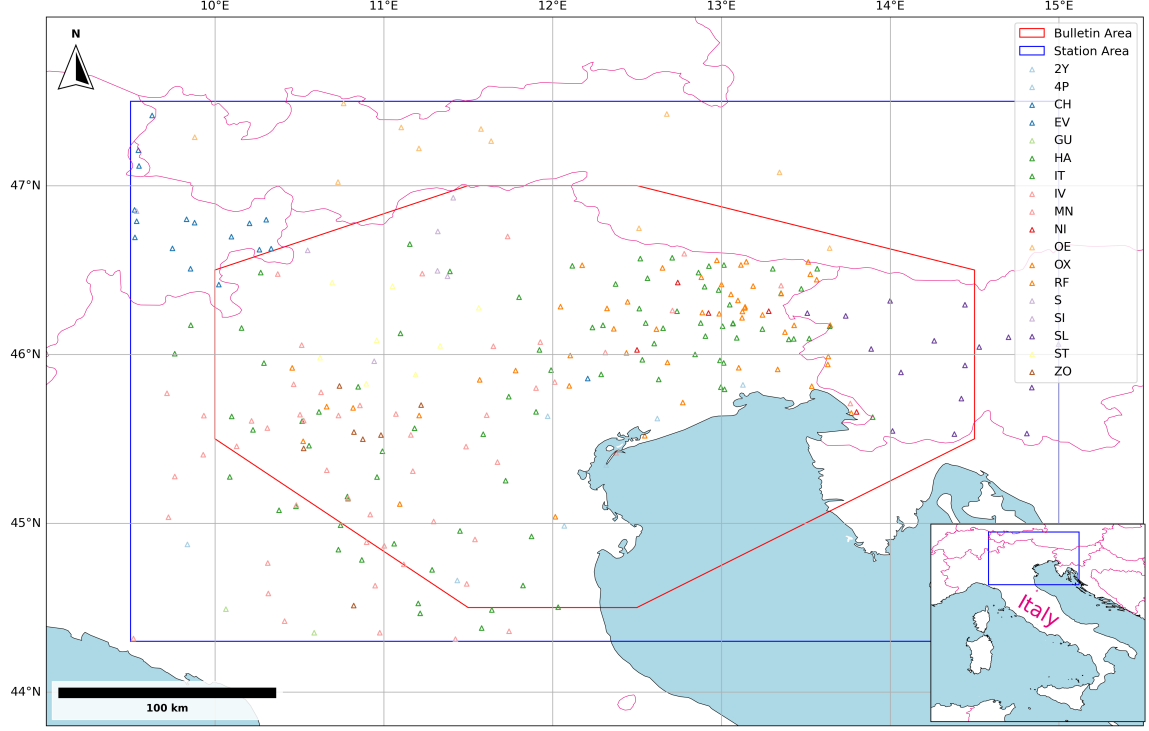


Figure 1.1 Spatial distribution of the 266 seismological and accelerometric stations across Northeastern Italy and adjacent border regions analyzed in this study. The reader should notice the high station density along the Friuli Prealpine belt and the transboundary coverage extending into Austria, Slovenia, and the Veneto plain. Station coordinates and metadata verified against official FDSN StationXML inventories from 18 participating networks.

1.1.1 Analyst-Curated OGS Reference Catalog

To establish a baseline reference for evaluating the machine-learning pipeline, we used an analyst-curated subset of the revised OGS Centro di Ricerche Sismologiche (CRS) seismic bulletin catalog Magrin et al. (2026). The archive is a reviewed operational reference, not an observation of every physical earthquake. In standard operational practice, OGS seismic analysts manually pick phase arrivals, assign onset quality weights (see Table 2.1), verify event associations, and compute event hypocenters using Hypo71 or HypoEllipse algorithms. The source files used here in the supported .hpl, .dat, .txt, and .pun formats were parsed and merged by the DataCatalog implementation in `OGS/src/ogsparser.py`; its output is a consolidated catalog of standardized EVENTS and PICKS tables.

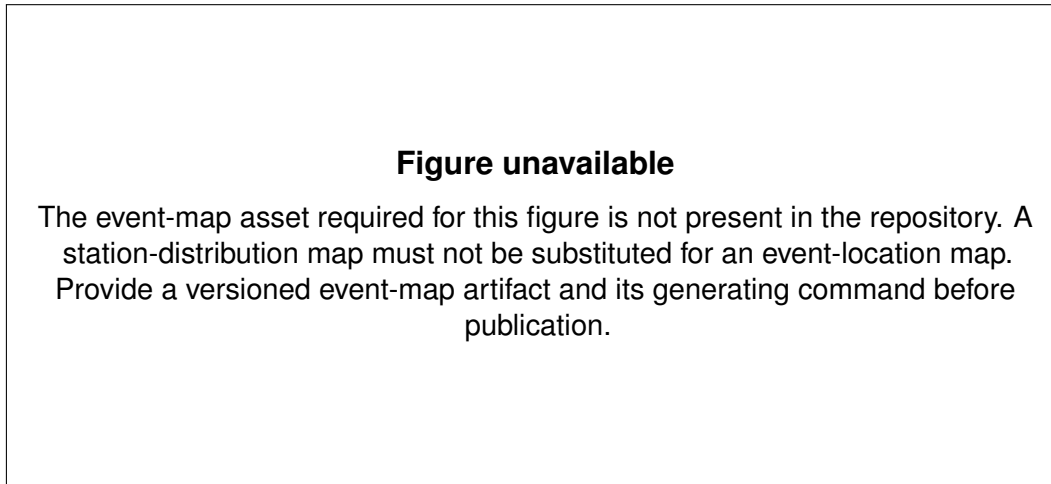


Figure 1.2 Placeholder for the event-location map within the OGS operational polygon over the stated observation interval. The intended analysis is the spatial relation between analyst-curated reference events and the Socchieve sequence; no such relation is asserted while the map artifact is unavailable. Evidence limitation: the required versioned event-map file and its run provenance are absent from this checkout.

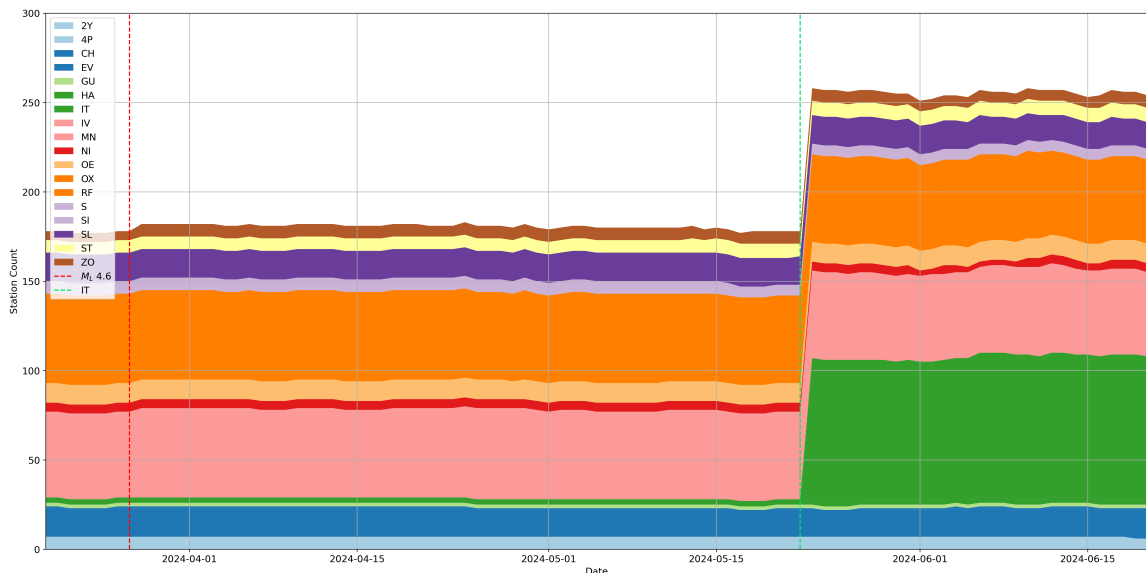


Figure 1.3 Temporal station availability and daily trace acquisition metrics across the 93-day observation period (March 20 to June 20, 2024). The reader should notice the sustained operational continuity across all networks, with minor transient drops resolved via automatic multi-client FDSN fallback. Data generated from continuous miniSEED audit logs parsed by `ogsutils.waveforms`.

However, comparing machine-learning hypocenter locations derived from modern probabilistic non-linear search algorithms against legacy Hypo71 locations would introduce systematic biases arising from differing inversion formulations, crustal velocity discretizations, and station elevation corrections. Therefore, to isolate the performance of the machine-learning picking and association components, the OGS reference events were independently relocated using the identical **1D NonLinLoc** (Lomax et al., 2000) velocity model and inversion parameterization employed in our computational pipeline (Magrin, personal communication).

The resulting relocated OGS reference catalog contains:

- **NUM EVENTS** **confirmed seismic events** strictly contained within the regional polygon boundary OGS_POLY_REGION during the **NUM YEARS** observation window.
- Local magnitudes ranging continuously from microseismic events ($M_L = -0.2$) up to the M_L 4.6 Socchieve mainshock, with completeness threshold $M_c \approx 1.2$.
- Focal depths predominantly confined between 5.0 km and 14.0 km, reflecting the shallow seismogenic brittle layer of the Southern Alps.
- **NUM PICKS** **manually verified phase picks**, comprising **NUM P PICKS** **Primary (P)** wave arrivals and **NUM S PICKS** **Secondary (S)** wave arrivals. Each pick includes analyst-assigned onset polarity, weight, and timing uncertainty.

This high-quality reference catalog provides the baseline for the bipartite graph matching assessment (BPGMA) developed in Chapter 5.

1.2 Benchmark Machine-Learning Training Corpora

The deep-learning phase pickers evaluated in this thesis—PhaseNet and EQTransformer—rely on supervised training over massive corpora of three-component seismic waveforms labeled with expert ground-truth P- and S-wave arrival times. To understand how models generalize across varying tectonic regimes and noise backgrounds, we selected four benchmark datasets available through the SeisBench framework (Woollam et al., 2022). Each dataset represents a distinct combination of scale, geographical setting, instrumentation density, and labeling curation.

1.2.1 INSTANCE: The Italian Seismic Dataset for Machine Learning

The Italian Seismic Dataset for Machine Learning (INSTANCE) Michelini et al., 2021 dataset (Michelini et al., 2021) is a comprehensive open-access collection of labeled

seismic waveforms recorded across the Italian peninsula between 2005 and 2020 by the National Seismic Network (RSN), operated by INGV, and partner regional networks including OGS.

Unlike global machine-learning corpora curated primarily for high signal-to-noise ratios, INSTANCE was specifically designed to preserve the realistic operational conditions of active regional monitoring networks:

- **Operational Noise Realism:** INSTANCE incorporates the full spectrum of environmental, cultural, and industrial noise characteristic of the Mediterranean and Alpine environments. This includes industrial vibrations, heavy road and rail traffic, quarry blasts, severe meteorological noise, and coastal microseisms.
- **Emergent and Low-Magnitude Arrivals:** The dataset encompasses a large proportion of small-magnitude events ($M_L < 2.5$) exhibiting emergent onsets, complex scattering, and low signal-to-noise ratios.
- **Volume and Balance:** INSTANCE comprises nearly **1.3 million three-component waveform traces** (1,291,537), including 1,159,249 labeled P-wave arrivals, 713,883 labeled S-wave arrivals, and a dedicated negative class of 132,288 pure noise windows extracted during periods without seismic activity.
- **Geographical Proximity:** Because INSTANCE waveforms originate from Italy, the training wavefields reflect crustal attenuation, complex fault structures, and station vault installations physically similar to those of our Northeastern Italy study area.

1.2.2 STEAD: STanford EArthquake Dataset

The STanford EArthquake Dataset (STEAD) Mousavi et al., 2019 dataset (Mousavi et al., 2019) is a globally distributed benchmark corpus curated by researchers at Stanford University to facilitate artificial intelligence research in seismology. Sourced from seismic stations worldwide, STEAD incorporates waveforms from subduction zones, continental collision zones, volcanic systems, and oceanic intraplate environments.

Key characteristics of STEAD include:

- **High Signal Clarity and Curation:** Waveforms were subjected to strict automated and manual quality filters, ensuring distinct wavefield onsets and elevated signal-to-noise ratios for earthquake signals.
- **Extensive Labeling:** STEAD contains over **1.26 million three-component traces** (1,265,657), with 1,030,231 P-wave picks and 1,030,231 S-wave picks across 441,705 distinct earthquakes.

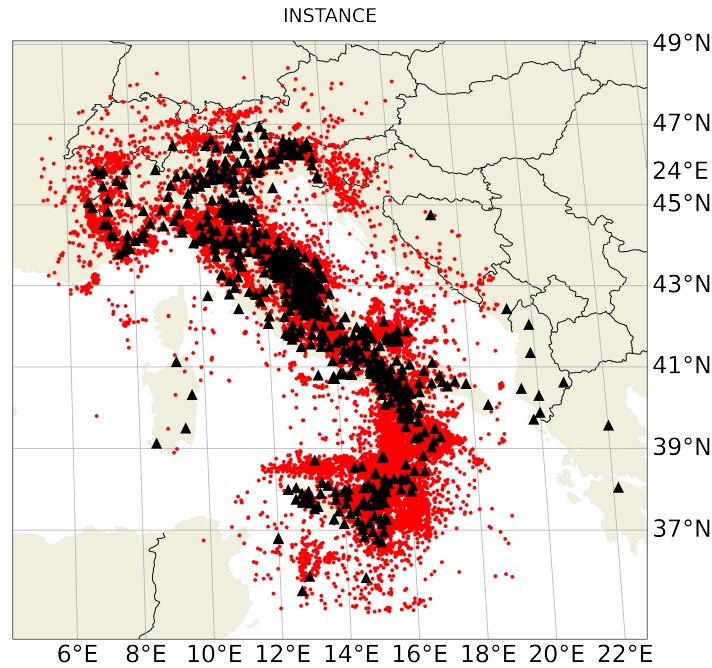


Figure 1.4 Geographical station distribution and event density of the INSTANCE benchmark dataset across Italy. The reader should notice the extensive coverage along the Apenninic chain and the Southern Alpine arc, capturing realistic Mediterranean crustal paths and anthropogenic noise conditions. Image reproduced from Woollam et al. (2022).

- **Dedicated Noise Partition:** The dataset includes 235,426 continuous noise recordings sampled across global stations, allowing neural networks to learn robust discriminative boundaries between non-stationary environmental noise and true seismic arrivals.
- **Global Transferability:** By encompassing a wide variety of sensor types, crustal velocity structures, and epicentral distances (local to regional, up to 350 km), STEAD is engineered to prevent models from overfitting to region-specific propagation paths.

1.2.3 SCEDC: Southern California Earthquake Data Center

The Southern California Earthquake Data Center (SCEDC) Southern California Earthquake Center, 2013 dataset (Southern California Earthquake Center, 2013) provides an unprecedented volume of seismic recordings curated from the Southern California Seismic Network (SCSN), spanning decades of continuous monitoring across the San Andreas fault system.

SCEDC is characterized by:

- **Massive Data Volume:** Sourced through SeisBench, the SCEDC corpus contains

STEAD

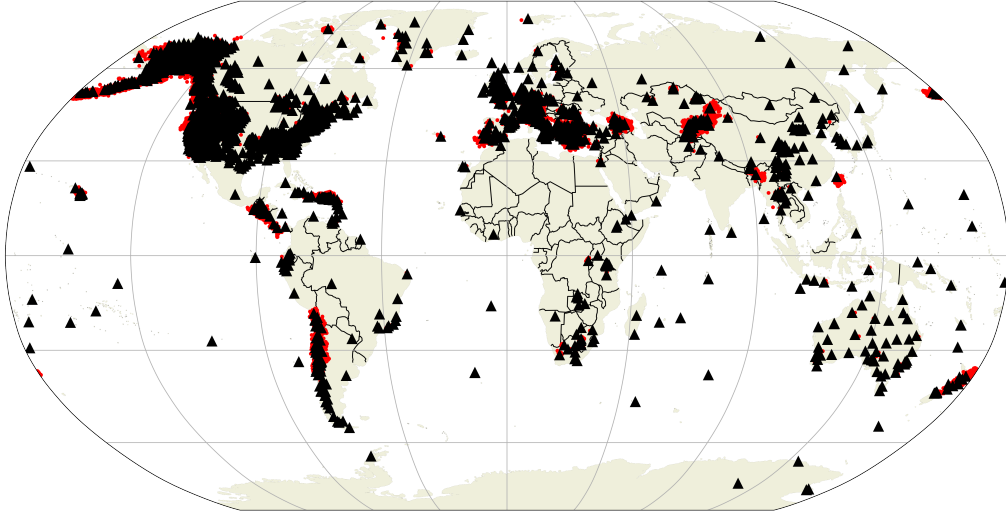


Figure 1.5 Worldwide distribution of seismological stations contributing waveform recordings to the global STEAD benchmark dataset. The reader should notice the broad geographic and tectonic representation across diverse orogenic belts and plate boundaries. Image reproduced from Woollam et al. (2022).

8.11 million three-component traces (8, 111, 060), incorporating 7, 571, 970 P-wave picks and 4, 364, 155 S-wave picks across 378, 528 local earthquakes.

- **Dense Local Network Geometry:** The SCSN features an exceptionally high spatial sensor density, capturing high-frequency wavefield details over purely local propagation paths ($\Delta \leq 150$ km).
- **Catalog Completeness:** Decades of continuous analyst curation and automated cross-correlation template matching make the SCEDC catalog complete to very low magnitude thresholds ($M_c \approx 1.0$), minimizing missing pick annotations in the training labels.
- **Regional Bias:** While mathematically advantageous for training deep high-capacity networks with millions of parameters without overfitting, the dataset is geographically homogeneous, reflecting Southern California crustal velocity gradients and strike-slip tectonic structures.

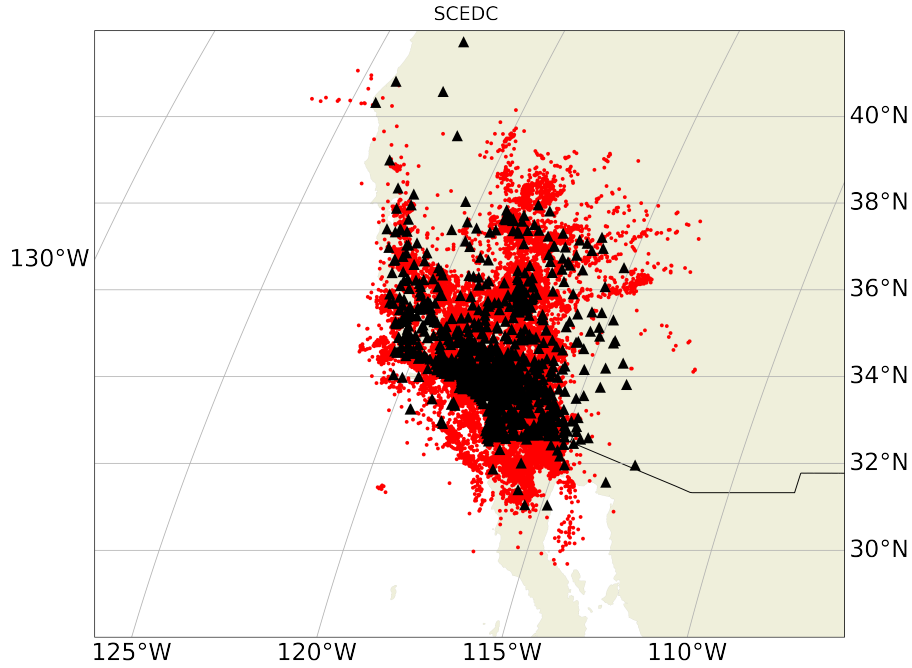


Figure 1.6 Spatial station distribution of the Southern California Seismic Network (SCSN) represented in the SCEDC dataset. The reader should notice the dense spatial grid concentrated along the San Andreas fault system and the Los Angeles basin. Image reproduced from Woollam et al. (2022).

1.2.4 Original Author Corpora: PhaseNet and EQTransformer

In addition to the three standard community benchmark datasets, we evaluate the models trained on the original corpora established by the respective model architects:

- **Original PhaseNet Corpus** (Zhu et al., 2018): Curated by the Northern California Seismic Network (NCSN), this dataset comprises approximately 779,514 waveform traces recorded across 234,117 local seismic events in Northern California. The dataset features high-quality manual analyst arrival times but lacks explicitly segregated noise-only windows, relying instead on pre-onset waveform segments to represent ambient noise.
- **Original EQTransformer Corpus** (Mousavi et al., 2020): Curated as a specialized subset of the STEAD repository, comprising approximately 1.2 million three-component traces selected to optimize multi-task transformer training for simultaneous earthquake detection, P picking, and S picking across 60-second waveform windows.

1.3 Synthesis of Benchmark Datasets and Methodological Comparison

Table 1.1 provides a rigorous, publication-grade comparative synthesis of all benchmark datasets utilized in this thesis. All datasets share a common sampling frequency of 100 Hz, either natively or through standardized SeisBench decimation, ensuring absolute tensor dimensionality consistency across all deep-learning inference workflows.

Table 1.1 Systematic comparison of seismological benchmark training datasets used for PhaseNet and EQTransformer models. The reader should notice the substantial contrast between the massive sample volume of SCEDC and the balanced regional noise realism of INSTANCE. Traces, noise windows, and pick counts are derived from SeisBench metadata schemas; distance bounds classify local ($\Delta \leq 150$ km) versus regional ($150 < \Delta \leq 600$ km) propagation paths.

Dataset	Traces	P Picks	S Picks	Events	Noise Traces	Coverage / Dist. ^a
INSTANCE	1 291 537	1 159 249	713 883	54 008	132 288	Italy (L / R)
STEAD	1 265 657	1 030 231	1 030 231	441 705	235 426	Global (L / R)
SCEDC	8 111 060	7 571 970	4 364 155	378 528	0 ^b	S. California (L)
PhaseNet ^c	779 514	— ^d	—	234 117	0 ^b	N. California (L)
EQTransformer ^c	1 200 000	1 000 000	1 000 000	400 000	200 000	Global (L / R)

^a Epicentral distance classification: Local (L) denotes $\Delta \leq 150$ km; Regional (R) denotes $150 < \Delta \leq 600$ km.

^b Dedicated pure noise trace partitions were not independently generated in the SCEDC and original PhaseNet repositories; ambient noise representations are extracted from pre-P arrival waveform windows.

^c Original PhaseNet and EQT datasets as described in their respective publications.

^d In the original PhaseNet publication (Zhu et al., 2018), separate counts for P and S arrivals were not reported individually; both phase classes are present in the labeled training windows.

1.3.1 The Domain Adaptation Hypothesis

A central scientific question explored in this dissertation is the **domain adaptation hypothesis**: does a deep neural network trained on massive volumes of data from a distant geological setting (such as the 8.1 million traces of SCEDC in Southern California) outperform a network trained on a smaller but regionally congruent dataset (such as the 1.3 million traces of INSTANCE in Italy)?

In classical supervised machine learning, statistical theory assumes that training and testing samples are identically and independently distributed (*i.i.d.*) from the same underlying probability distribution:

$$P_{\text{train}}(\mathbf{X}, \mathbf{Y}) = P_{\text{test}}(\mathbf{X}, \mathbf{Y}). \quad (1.1)$$

In operational seismology, this assumption is fundamentally violated when transferring

models across geographical regions ($P_{\text{train}} \neq P_{\text{test}}$). Regional differences include:

1. **Path Effects and Velocity Dispersions:** Distinct crustal velocity gradients, Conrad/Moho discontinuity depths, and intrinsic attenuation (Q -factor) modify wavefield dispersion and P-to-S coda energy ratios.
2. **Anthropogenic and Environmental Noise Spectra:** Cultural noise in densely populated European valleys (characterized by 50 Hz electrical harmonics, narrow road networks, and industrialized river plains) differs substantially from noise in the Mojave desert or Southern California coastal basins.
3. **Instrumental Hardware Responses:** Variations in sensor eigenfrequencies, digitizer bit depths, and vault coupling introduce subtle spectral colorations into the digitized waveforms.

By systematically evaluating PhaseNet and EQTransformer models trained across INSTANCE, STEAD, SCEDC, and the original corpora on the identical three-month SMINO continuous dataset, this study provides empirical bounds on regional domain transferability (presented in Chapter 7).

Supplementary Material: Datasets

Chapter-Specific Supplementary Material

This section is reserved for dataset dictionaries, provenance tables, and additional validation material supporting Chapter 1.

Chapter 2

Data Engineering

The development and deployment of an automated, operational seismic catalog pipeline requires a robust, scalable data engineering infrastructure. Seismological observatories must routinely ingest heterogeneous multi-channel continuous waveforms from distributed networks, parse decades of legacy catalog archives stored in idiosyncratic text formats, merge multi-source hypocenter and magnitude solutions, and render high-resolution geographic and statistical visualizations.

This chapter details the data engineering framework developed for this dissertation, implemented as a modular Python package under `OGS/src/`. Section 2.1 presents the software design patterns and the central configuration module `ogsconstants`. Section 2.2 establishes the formal weight-to-uncertainty mapping. Section 2.3 describes the distributed FDSN continuous waveform acquisition architecture. Section 2.4 formulates the extensible multi-format catalog parsing hierarchy. Section 2.5 details the automated format detection, merging logic, and columnar persistence. Section 2.6 presents the object-oriented visualization framework and geodetic math. Finally, Section 2.7 outlines the tiered storage strategy and quality assurance protocols.

2.1 Software Architecture and Central Configuration Module

The data engineering pipeline is architected around object-oriented and functional programming paradigms in Python 3.12, interfacing with core scientific computing libraries including NumPy, pandas, ObsPy (Krischer et al., 2015), Pyrocko, Cartopy, and scikit-learn. To ensure maintainability, maintain loose coupling between processing stages, and eliminate hardcoded parameters, the system adheres to the **Single Source of Truth** design pattern.

2.1.1 The Central Configuration Module: `ogsconstants`

The `ogsconstants.py` module serves as the central configuration repository for the entire codebase. Every pipeline stage—data downloading, parsing, neural picking, phase association, relocation, magnitude estimation, bipartite matching, and plotting—imports its operational constants, schema definitions, regex fragments, and mathematical tolerances directly from this module.

The primary configuration domains governed by `ogsconstants` include:

1. Parallel Execution and HPC Hardware Allocation:

- `MPI_RANK`, `MPI_SIZE`, `MPI_COMM`: Global variables capturing Message Passing Interface (MPI) process rank, communicator size, and communicator topology for distributed multi-node execution on HPC clusters.
- `GPU_RANK`, `GPU_SIZE`: Device identifiers managing CUDA device affinity for multi-GPU inference across NVIDIA A100 nodes.
- `EPSILON = 1e-6`: Small numerical constant preventing division-by-zero singularities during travel-time residual calculations and weight inversions.

2. Standardized Date and Time Representations: Seismological files and web services employ diverse date-time string conventions. The module defines canonical format strings following Python `strftime` standards:

- `DATE_FMT = "%Y-%m-%d"`: Standard ISO 8601 calendar date.
- `TIME_FMT = "%H%M%S"`: Compact six-digit time string.
- `YYMMDD_FMT = "%y%m%d"` and `YYYYMMDD_FMT = "%Y%m%d"`: Two-digit and four-digit year date representations.
- `DATETIME_FMT = "%y%m%d%H%M%S"`: Compact twelve-digit timestamp string used in legacy fixed-width records.
- `ONE_DAY = timedelta(days=1)`: Fundamental step interval for temporal partitioning.

3. Phase Identifiers and Detection Thresholds:

- `PWAVE = "P"`, `SWAVE = "S"`: Authoritative phase type string literals.
- `THRESHOLDS`: Systematic confidence threshold array spanning 0.1 to 0.9 in increments of 0.1, constructed via `np.linspace(0.1, 0.9, 9)`.
- `PWAVE_THRESHOLD = SWAVE_THRESHOLD = 0.1`: Default minimum model probability threshold for accepting ML phase candidates.

4. Spatial Domain and Polygon Boundaries:

- **OGS_POLY_REGION:** An eight-vertex convex polygon enclosing the core monitoring area in Northeastern Italy, defined by ordered (lon, lat) coordinate pairs:

$$\mathcal{P}_{\text{OGS}} = \{(10.0, 45.5), (10.0, 46.5), (11.5, 47.0), (12.5, 47.0), (14.5, 46.5), (14.5, 45.5), (12.5, 44.5), (11.5, 44.5)\}. \quad (2.1)$$

- **OGS_STUDY_REGION** = $[9.5, 15.0, 44.3, 47.5]$: Extended rectangular bounding box $[lon_{\min}, lon_{\max}, lat_{\min}, lat_{\max}]$ encompassing border sectors of Austria, Slovenia, and Croatia to ensure complete wavefield capture.
- **OGS_PROJECTION:** Proj4 stereographic projection string ("`+proj=sterea +lon_0={lon} +lat_0={lat} +units=km`") providing conformal, low-distortion Cartesian metric mapping for regional travel-time calculations and plotting.

5. Geographic Zone and Event Type Taxonomies:

- **OGS_GEO_ZONES:** Dictionary mapping single-letter catalog codes to formal geographic regions (*e.g.*, 'F' $\rightarrow$ Friuli, 'V' $\rightarrow$ Veneto, 'S' $\rightarrow$ Slovenia, 'A' $\rightarrow$ Alto Adige, 'O' $\rightarrow$ Austria).
- **OGS_EVENT_TYPES:** Taxonomy distinguishing natural tectonic earthquakes ('L' $\rightarrow$ local_eq) from industrial explosions ('E' $\rightarrow$ explosion), military detonations ('B' $\rightarrow$ bomb), landslide-induced ground shaking ('F' $\rightarrow$ landslide), and unclassified sources ('U' $\rightarrow$ UNKNOWN).

6. Canonical DataFrame Schema Headers: Predefined column tuples ensuring schema congruence across parquet storage:

- **HEADER_EVENTS** = $[idx, time, latitude, longitude, depth, ERH, ERZ, GAP]$
- **HEADER_PICKS** = $[idx, time, phase, station, onset, polarity, weight]$
- **HEADER_MODL** = $[MODEL, WEIGHT, THRESHOLD]$
- **HEADER_PRED** = **HEADER_MODL** + **HEADER_MANL**

2.2 Pick Quality Weight and Timing Uncertainty Conversion

In observational seismology, human analysts assign qualitative weight codes to picked phase arrivals based on onset clarity, signal-to-noise ratio, and waveform impulsiveness.

The legacy Hypo71 formulation (Lee et al., 1972) establishes a discrete integer weighting scale $w \in \{0, 1, 2, 3, 4\}$. To utilize these picks in probabilistic relocation algorithms (such as NonLinLoc) or in quantitative distance scoring functions, discrete weights must be mapped to physical timing uncertainties σ_t (in seconds).

Table 2.1 defines the quantitative mapping implemented in the pipeline. Picks assigned weight 0 represent ideal, impulsive onsets with timing uncertainties below 10 milliseconds ($\sigma_t < 0.01$ s), whereas weight 4 denotes poor-quality picks with uncertainties exceeding one second ($\sigma_t \geq 1.0$ s).

Table 2.1 Mapping between analyst-assigned Hypo71 pick weight codes, timing uncertainties (σ_t), and physical interpretation implemented in the OGS pipeline. The reader should notice that weight codes represent an approximately logarithmic scaling of picking uncertainty. Weight codes and thresholds derived from official OGS operational catalog conventions and anchored in `ogsconstants.H71_OFFSET`.

Weight Code (w)	Uncertainty σ_t (s) ^a	Observational Semantic / Waveform Quality
0	< 0.01	Impulsive onset, sharp first motion, high SNR
1	0.01–0.04	Clear onset, unambiguous first break
2	0.04–0.20	Fairly clear onset, moderate background noise
3	0.20–1.00	Emergent onset, low SNR, gradual amplitude rise
4	≥ 1.00	Poor quality pick, emergent coda, highly uncertain
5	25.00	Very uncertain / rejected pick (rarely utilized)

^a In the bipartite graph matching and event distance modules (`dist_event` and `dist_pick`), uncertainty offsets are parameterized by the dictionary `H71_OFFSET = {0: 0.01, 1: 0.04, 2: 0.2, 3: 1.0, 4: 5.0, 5: 25.0}`.

In downstream probabilistic hypocenter inversion, the timing uncertainty σ_t determines the diagonal elements of the data covariance matrix C_d :

$$C_{d,ii} = \sigma_{t,i}^2 + \sigma_{\text{model}}^2, \quad (2.2)$$

where σ_{model} accounts for theoretical travel-time uncertainty induced by unmodeled 3D crustal heterogeneities.

2.3 Continuous Data Acquisition: The `ogsdownloader` Architecture

Waveform data acquisition is governed by the command-line utility `ogsdownloader.py`, engineered to retrieve continuous waveform streams and station metadata from distributed International Federation of Digital Seismograph Networks (FDSN) web services. The tool

supports dual operational backends—ObsPy `MassDownloader` (Krischer et al., 2015) and Pyrocko `squirrel`—controlled through a unified command-line argument schema.

2.3.1 Multi-Client Fallback Querying

Because regional seismic stations are operated by distinct national, regional, and academic institutions, single-client queries frequently suffer from missing channels, network maintenance downtime, or restricted routing. The acquisition engine resolves this by executing prioritized multi-client fallback queries. The priority order is defined in `ogsconstants.OGS_CLIENTS_DEFAULT`:

1. **OGS Internal FDSN Web Service** (<http://158.110.30.217:8080>): Authoritative internal data center providing real-time broadband and accelerometric feeds from the SMINO network.
2. **INGV FDSN Web Service**: Italian National Institute of Geophysics and Volcanology, serving national broadband (IV) and accelerometric (IT) stations.
3. **GFZ GEOFON Program**: German Research Centre for Geosciences, hosting trans-boundary stations across Central and Southern Europe.
4. **IRIS DMC / EarthScope**: Global backup data center.
5. **ETH Zurich**: Swiss Seismological Service (SED), providing border coverage along the Western Alpine arc.
6. **ORFEUS Data Center (ODC)**: European observational federation.
7. **Collalto Dense Microseismic Network** (<http://scp-srv.core03.ogs.it:8080>): High-density industrial monitoring array located in the Venetian Prealps.

Queries proceed iteratively down the client list: when a station or waveform segment is unavailable at client k , the downloader automatically cascades to client $k + 1$, maximizing data completeness without duplicating downloaded files.

2.3.2 Temporal Segmentation and Multi-Threaded Execution

Querying continuous waveform archives over multi-month observation windows in a single HTTP request routinely triggers server socket timeouts, memory exhaustion, or network

connection resets. To prevent these failures, `ogsdownloader` automatically partitions the global date range $[t_{\text{start}}, t_{\text{end}}]$ into discrete **24-hour UTC calendar days**:

$$\mathcal{D}_k = [t_{\text{start}} + k \cdot \Delta t_{\text{day}}, t_{\text{start}} + (k + 1) \cdot \Delta t_{\text{day}}), \quad k \in \{0, 1, \dots, N_{\text{days}} - 1\}. \quad (2.3)$$

For each calendar day $\mathcal{D}_k$, a dedicated worker thread is dispatched via Python’s `concurrent.futures.ThreadPoolExecutor`. The worker constructs a hierarchical directory structure under the designated root filesystem:

```
{base_directory}/{YYYY}/{MM}/{DD}/
```

Within each day directory, waveforms are persisted as miniSEED files following the SEED naming convention:

```
{NETWORK}.{STATION}.{LOCATION}.{CHANNEL}_{STARTTIME}_{ENDTIME}.mseed
```

This temporal segmentation guarantees:

- **Atomic Checkpointing:** Interrupted downloads can be resumed without re-downloading completed days.
- **Lock-Free Parallelism:** Multiple MPI ranks or SLURM tasks can read different dates concurrently without filesystem write contention.
- **Memory Predictability:** Working memory per thread is strictly bounded by the 24-hour waveform volume of the active station subset.

2.3.3 Channel Selection Priorities

Seismic stations frequently record on multiple co-located sensor channels (e.g., broadbands, short-period sensors, and accelerometers). To prevent duplicate processing while preserving the highest recording fidelity, the downloader enforces a strict band-code selection hierarchy:

1. **HH[ZNE]:** High-Gain Broadband Seismometer (100 Hz sampling; primary choice for microseismic picking).
2. **EH[ZNE]:** Extremely Short-Period Seismometer (high-frequency velocity sensors).
3. **HN[ZNE]:** High-Gain Accelerometer (strong-motion channels; essential during high-amplitude near-source shaking).
4. **HG[ZNE]:** High-Gain Seismometer.

Horizontal channel orientation fallbacks (components 1 and 2 mapped to N and E) are handled automatically during stream parsing.

2.4 Multi-Format Catalog Parsing Architecture

Over its 45-year operational history, the OGS Centro di Ricerche Sismologiche (CRS) has accumulated vast archives of earthquake catalogs and phase arrival bulletins stored in legacy ASCII and fixed-width formats. To ingest and unify these records, we engineered an extensible, object-oriented parsing framework.

2.4.1 Abstract Base Class Design: OGSDataFile

The parsing framework is rooted at the abstract base class `OGSDataFile`, which inherits from `OGSCatalog` (Figure 2.1). `OGSDataFile` provides core infrastructure common to all text-based formats:

- **Regex-Based Token Extraction:** Subclasses declare format-specific lists of regular expression fragments (`RECORD_EXTRACTOR_LIST` for pick arrivals, `EVENT_EXTRACTOR_LIST` for event headers). The base class flattens and compiles these fragments into high-performance regular expressions with named capture groups.
- **Spatial Polygon Filtering:** Automatic spatial subsetting using `matplotlib.path.Path.contains_points` to filter event coordinates against the regional polygon `OGS_POLY_REGION`.
- **Temporal Date Range Filtering:** In-line filtering during stream reading, discarding records outside the configured $[t_{\text{start}}, t_{\text{end}}]$ window before allocating `DataFrame` memory.
- **Columnar Parquet Serialization:** Built-in persistence methods that serialize parsed picks and events into compressed Apache Parquet tables partitioned by calendar date.

2.4.2 Progressive Regular Expression Debugging

A major challenge in parsing decades of legacy seismological data is the occurrence of subtle formatting corruptions: misaligned character columns, non-standard blank characters, and missing fields. To diagnose parsing failures without resorting to manual line-by-line inspection, `OGSDataFile` implements a **progressive regular expression debugger** (`debug()` method).

The debugger decomposes the composite regex fragment list $\mathcal{R} = [r_1, r_2, \dots, r_m]$ into cumulative prefixes:

$$\mathcal{R}_k = \sum_{i=1}^k r_i, \quad k \in \{1, 2, \dots, m\}. \quad (2.4)$$

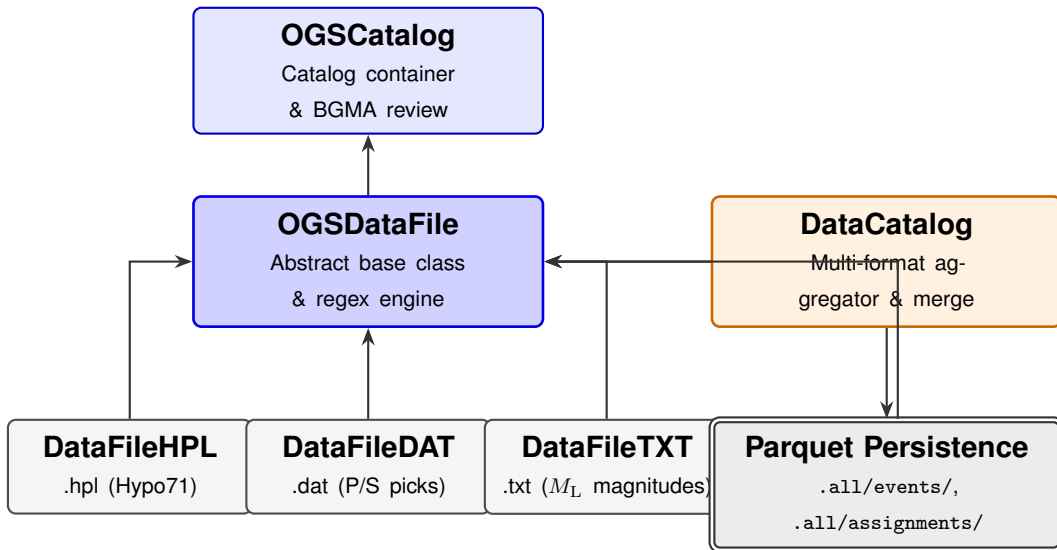


Figure 2.1 Class inheritance and aggregation architecture of the OGS multi-format catalog parsing framework. Solid boxes denote demonstrated Python classes implemented under `OGS/src/`; double-bordered gray box denotes Apache Parquet on-disk storage. The reader should notice the inheritance hierarchy rooted at `OGSCatalog`, specialized via `OGSDataFile`, and orchestrated by the `DataCatalog` aggregator.

When a record line fails matching against the full pattern $\mathcal{R}_m$, the debugger evaluates the line sequentially against each prefix $\mathcal{R}_k$. Using named capture group introspection (`GROUP_PATTERN = re.compile(r"(\?P<(\w+)>...")`), the method immediately isolates the exact capture group (*e.g.*, station name, centisecond fraction, or azimuthal gap) that caused the match to fail, logging the failure line number and corrupted substring.

2.4.3 Supported Legacy File Formats

Table 2.2 summarizes the four primary legacy catalog formats supported by dedicated subclass implementations.

2.5 Catalog Aggregation, Merging Logic, and Columnar Storage

The `DataCatalog` class (implemented in `ogsparser.py`) acts as the central multi-format aggregator. When executed over an input directory containing mixed legacy files, it orchestrates automated format detection, batch parsing, cross-file attribute merging, and optimized columnar persistence.

Table 2.2 Supported OGS catalog and phase pick file formats and their corresponding parser implementations. The reader should notice the complementary nature of the files: HPL provides comprehensive hypocenters and picks, DAT provides dense arrival times, TXT contributes revised local magnitudes, and PUN preserves historical punch card records. Formats anchored in `ogshpl.py`, `ogsdat.py`, `ogstxt.py`, and `ogspun.py`.

Extension	Parser Class	Extracted Attributes and Operational Semantics
.hpl	DataFileHPL	Hypocenter locations, origin times, focal depths, local magnitude (M_L), station P/S arrival times, onset polarities, weights (0–4), <i>ERH</i> , <i>ERZ</i> , <i>RMS</i> , azimuthal gap (<i>GAP</i>), analyst free-text notes, and geographic locality descriptions.
.dat	DataFileDAT	Fixed-width station phase arrival records: 4-character station code, P/S onset quality (I/E), first-motion polarity (+/ – /c/d), weight code, timestamp down to centiseconds, and event type.
.txt	DataFileTXT	Tabular event summary records: origin times, revised local magnitude (M_L), duration magnitude (M_D), horizontal/vertical uncertainties, and event identification tags.
.pun	DataFilePUN	Legacy punch card fixed-column format: hypocentral coordinates in degrees-minutes format (automatically converted to decimal degrees), depths, <i>DMIN</i> , <i>RMS</i> , and quality classes.

2.5.1 Automatic Format Detection

The aggregator inspects the file extensions of discovered files and dispatches parsing via an internal registry mapping (`DATAFILE_TYPES`):

```
DATAFILE_TYPES = {
    '.hpl': DataFileHPL, # Primary hypocenter and pick data
    '.dat': DataFileDAT, # Phase pick data
    '.txt': DataFileTXT, # Magnitude and error data
    '.pun': DataFilePUN, # Punch card event summaries
}
```

Unrecognized file types or hidden operating system files are automatically bypassed.

2.5.2 Cross-File Catalog Merging Logic

When merging is requested (`-merge` CLI flag), `DataCatalog` reconciles complementary attributes across heterogeneous file types into unified `EVENTS` and `PICKS` `DataFrames`:

1. **Pick Normalization and Concatenation:** Pick records across all parsed files are concatenated. Column names are mapped to standardized schema identifiers (`HEADER_PICKS`), timestamps are parsed to `UTCDateTime`, and calendar dates are coerced using `normalize_event_groups()` to ensure consistent join keys across HPL, DAT, and PUN files.

2. **Event Attribute Outer Joins:** Merging events from distinct files requires joining complementary fields:

- Primary hypocenters, origin times, depths, and spatial uncertainties are seeded from HPL files.
- TXT files contribute revised local magnitude (M_L), duration magnitude (M_D), and station count metrics, joined via an outer merge on event index `idx`.
- PUN files contribute legacy coordinates for events missing HPL records, joined via spatiotemporal matching on $[time, lat, lon, depth]$.

3. **Vectorized Event Pick Statistics Aggregation:** Following hypocenter merging, the aggregator computes pick statistics for every event. To prevent performance degradation from nested Python loops, aggregation is executed via vectorized pandas operations:

```
# Vectorized pick counts per event by phase type
phase_counts = self.PICKS.groupby(
    [OGS_C.IDX_PICKS_STR, OGS_C.PHASE_STR]
).size().unstack(fill_value=0)

self.EVENTS[OGS_C.NUMBER_P_PICKS_STR] = self.EVENTS[OGS_C.IDX_EVENTS_STR].map(
    phase_counts[OGS_C.PWAVE]
).fillna(0).astype(int)

self.EVENTS[OGS_C.NUMBER_S_PICKS_STR] = self.EVENTS[OGS_C.IDX_EVENTS_STR].map(
    phase_counts[OGS_C.SWAVE]
).fillna(0).astype(int)

# Stations recording both P and S arrivals
station_phase_counts = self.PICKS.groupby(
    [OGS_C.IDX_PICKS_STR, OGS_C.STATION_STR]
)[OGS_C.PHASE_STR].nunique()
stations_with_both = station_phase_counts[
    station_phase_counts >= 2
].groupby(level=0).size()

self.EVENTS[OGS_C.NUMBER_P_AND_S_PICKS_STR] = self.EVENTS[
    OGS_C.IDX_EVENTS_STR
].map(stations_with_both).fillna(0).astype(int)
```

2.5.3 Columnar Parquet Storage Organization

Consolidated catalogs are serialized using **Apache Parquet** format with Snappy compression. Parquet provides columnar data layout, typed schemas, dictionary encoding, and predicate pushdown filtering. On large multi-year catalogs, Parquet achieves a $5 \times -10 \times$ storage volume reduction compared to raw CSV and accelerates analytical filtering queries by over an order of magnitude.

Storage is partitioned chronologically by calendar date:

```
{output_directory}/.all/assignments/{YYYY}-{MM}-{DD}.parquet  (picks)
{output_directory}/.all/events/{YYYY}-{MM}-{DD}.parquet      (events)
```

2.6 Visualization Framework and Geodetic Mathematics

Scientific data analysis, quality control, and model benchmarking require reproducible, publication-grade figure generation. The `ogspLOTter.py` module implements a family of specialized plotter classes built on Matplotlib and Cartopy, standardizing visual aesthetics across the project.

2.6.1 Base Plotter Architecture: `plotter`

All specialized plotting classes inherit from the `plotter` base class, which encapsulates:

- Consistent figure dimensions, DPI scaling (300 DPI publication resolution), and typography using Helvetica/sans-serif font families.
- Standardized multi-format figure export (PNG raster, PDF vector, EPS).
- The authoritative OGS institutional color palette defined in `ogsconstants`:
 - `OGS_BLUE` = "#163771": Primary OGS dark blue (ground truth, reference series).
 - `MEX_PINK` = "#E4007C": Accent vibrant pink (predicted models, PhaseNet).
 - `ALN_GREEN` = "#00e468": Bright green (matched events, true positives).
 - `LIP_ORANGE` = "#FF8C00": Orange (EQTransformer, warnings, missed events).
 - `SUN_YELLOW` = "#e4da00": Yellow (tertiary accent, proposed events).

2.6.2 Specialized Plotting Engines

The framework provides purpose-built visualization engines tailored to specific seismological representations:

1. **Cartographic Mapping** (`map_plotter`): Integrates Cartopy projections (PlateCarree and local Stereographic) to render geographic basemaps, coastlines, international borders, provincial boundaries, and the regional polygon `OGS_POLY_REGION`. Stations are rendered as upward-pointing triangles (▲) colored by network; earthquake

hypocenters are rendered as circles ($\circ$) scaled proportionally to magnitude and colored by focal depth or catalog origin. Includes automated north arrows and physical metric scale bars (`matplotlib_scalebar`).

2. **Data Availability Stacking** (`stack_plotter`): Generates daily multi-network station count area stacks over time (`OGSAvailability.png`), tracking continuous recording availability and highlighting operational network drops.
3. **Cumulative Event Progression** (`line_plotter` and `day_plotter`): Renders cumulative earthquake discovery curves over the observation horizon, supporting vertical event annotation lines for key seismic shocks (e.g., the M_L 4.6 Socchieve mainshock).
4. **Waveform Record Sections** (`event_plotter`): Generates distance-ordered three-component seismic record sections, bandpass-filtering traces, normalizing trace amplitudes, and overlaying theoretical or picked P- and S-wave arrival flags.
5. **Distribution and Confusion Matrix Engines** (`histogram_plotter` and `ConfMtx_plotter`): Visualizes magnitude-frequency distributions, travel-time residual histograms with Gaussian kernel density estimation, and 3×3 phase confusion matrices via scikit-learn's `ConfusionMatrixDisplay`.

2.6.3 Geodetic Distance Computations

Computing epicentral and hypocentral distances between station coordinates $(\phi_{\text{sta}}, \lambda_{\text{sta}}, z_{\text{sta}})$ and earthquake hypocenters $(\phi_{\text{hyp}}, \lambda_{\text{hyp}}, z_{\text{hyp}})$ is fundamental for distance-ordered record sections and spatial matching tolerances.

The module implements the vectorized helper `v_lat_long_to_distance` in `ogsplotter.py`. Horizontal great-circle distance $d_{\text{horizontal}}$ is computed along the WGS84 reference ellipsoid using the geodesic algorithm of Krischer et al. (2015) (`obspy.geodetics.gps2dist_azimuth`):

$$d_{\text{horizontal}} = \frac{1}{1000} \cdot \text{geodesic}_{\text{WGS84}}(\phi_{\text{hyp}}, \lambda_{\text{hyp}}, \phi_{\text{sta}}, \lambda_{\text{sta}}) \quad [\text{km}]. \quad (2.5)$$

The 3D hypocentral distance d_{3D} is subsequently obtained by adding the vertical depth separation in quadrature:

$$d_{3D} = \sqrt{d_{\text{horizontal}}^2 + \left(\frac{|z_{\text{hyp}} - z_{\text{sta}}|}{1000} \right)^2} \quad [\text{km}], \quad (2.6)$$

where depths z are defined in meters positive downwards.

2.7 Tiered Data Storage Strategy and Quality Assurance

To support high-throughput parallel execution on the CINECA Leonardo supercomputer while strictly maintaining reproducibility and data provenance, the framework employs a tiered hierarchical storage architecture (Figure 2.2):

- **Tier 1: Raw Continuous Waveforms (MiniSEED):** Ingested daily streams stored in read-only, date-partitioned directory hierarchies under `$WORK/waveforms/YYYY/MM/DD/`. Preserves primary observation integrity without modification.
- **Tier 2: Station Metadata (StationXML):** XML metadata stored under `station/{NET}.{STA}.xml` providing calibration responses, gains, and coordinates.
- **Tier 3: Columnar Catalogs (Parquet):** Pick and event tables persisted in compressed Parquet partitions under `catalogs/{target}/.all/`.
- **Tier 4: Zero-Copy Symbolic Link Trees:** Intermediate pipeline stages (such as outputs from phase picking feeding into phase associators) are linked using filesystem symlinks (`link.py`), avoiding multi-gigabyte data duplication on shared Lustre filesystems while preserving exact provenance chains.

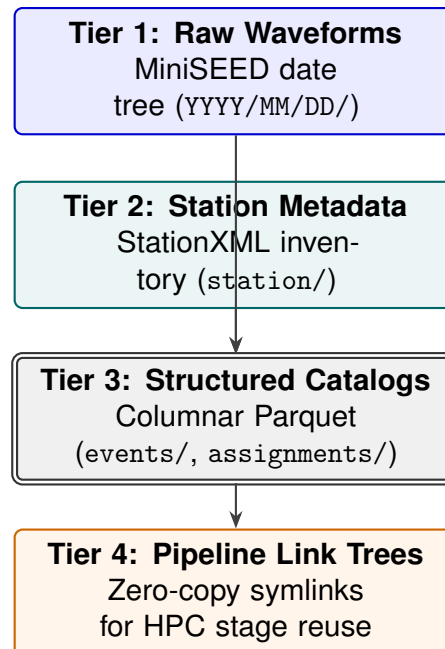


Figure 2.2 Hierarchical four-tier storage and data organization architecture deployed on the CINECA Leonardo HPC filesystem. The reader should notice the clear separation between read-only raw observation layers (Tiers 1 and 2), compressed columnar analytical catalogs (Tier 3), and zero-copy symbolic link execution structures (Tier 4).

Comprehensive quality assurance protocols ensure data integrity across all stages:

1. **Input Pre-flight Verification:** File existence, non-zero file size, valid MSEED header structure, and complete StationXML coordinate verification before pipeline execution.
2. **Strict Coordinate Bounding:** Immediate rejection of event hypocenters located outside valid ellipsoidal coordinate domains ($\phi \notin [-90, 90]^\circ$, $\lambda \notin [-180, 180]^\circ$).
3. **Duplicate Rejection:** Detection and suppression of duplicate pick records sharing identical station, channel, phase type, and arrival timestamps within $\Delta t < 0.01$ s.
4. **Progressive Regex Error Logging:** Granular logging of any unparseable record lines, capturing line numbers, input filenames, and failing capture groups.

Supplementary Material: Data Engineering

Chapter-Specific Supplementary Material

This section is reserved for schema tables, parser examples, and reproducibility notes supporting Chapter [2](#).

Chapter 3

Computational Pipeline

This chapter details the procedural machine learning computational pipeline engineered for automated, end-to-end seismic monitoring and high-precision catalog generation. The pipeline ingests continuous, multi-station seismic waveforms and transforms them into an earthquake bulletin containing associated phase picks, probabilistic hypocenter locations, and calibrated local magnitudes.

The workflow is architected into five distinct stages executed sequentially:

1. **Stage 0 — Preprocessing & Stream Conditioning:** Continuous waveform ingestion, channel harmonization, detrending, anti-aliasing resampling to a normalized frequency of 100 Hz, and sliding-window segmentation.
2. **Stage 1 — Deep Learning Phase Picking:** Dense temporal arrival-time estimation for P-wave and S-wave phases using 1D convolutional and attention-based deep neural networks (PhaseNet and EQTransformer).
3. **Stage 2 — Spatiotemporal Phase Association:** Unsupervised Bayesian clustering and octree-based spatial partitioning (GaMMA and PyOcto) resolving pick-to-source assignments in four-dimensional space-time.
4. **Stage 3 — Non-Linear Hypocenter Location:** Global probabilistic grid-search inversion (NonLinLoc) under an Equal Differential Time (EDT) formulation using regional 1D layered velocity structures.
5. **Stage 4 — Calibrated Local Magnitude Estimation:** Instrument deconvolution, synthetic Wood-Anderson torsion seismograph response simulation, horizontal peak vector aggregation, and regional attenuation corrections with empirical station terms.

Each processing stage communicates via strictly specified, immutable data contracts, ensuring high modularity, full computational reproducibility, and seamless execution across High-Performance Computing (HPC) infrastructures.

3.1 Data Preprocessing and Stream Conditioning

Continuous seismic data recorded by regional networks, such as the SMINO network operated by OGS in Northeastern Italy (Bragato et al., 2021), exhibit substantial heterogeneity in sensor response, digitizer sampling rates, and continuous telemetry transmission quality. A standardized conditioning workflow is applied to guarantee numerical stability and architectural alignment prior to neural network inference.

3.1.1 Waveform Ingestion and Stream Harmonization

Continuous waveform data are retrieved from distributed seismic repositories (OGS, INGV, and European Integrated Data Archive (EIDA) archives) in standard miniSEED (MSEED) format. Using the ObsPy seismological library (Krischer et al., 2015), the raw byte streams are parsed into in-memory `Stream` objects comprising three orthogonal components: vertical (Z), North-South (N), and East-West (E).

Data streams often contain telemetry dropouts, timing tears, or overlapping segments resulting from network reconnects. The conditioning pipeline applies the following deterministic operations:

1. **Stream Merging and Gap Handling:** Overlapping trace fragments are cross-checked for timestamp consistency. Small gaps (≤ 5 samples) are linearly interpolated, whereas extended telemetry outages are zero-padded or split into discrete processing windows to prevent artificial phase transients.
2. **Baseline Correction and Detrending:** Long-period instrumental drift and DC offsets are eliminated by subtracting the mean and removing the best-fit linear trend:

$$x_{\text{detrended}}(t) = x(t) - (\hat{\alpha}t + \hat{\beta}), \quad (3.1)$$

where $\hat{\alpha}$ and $\hat{\beta}$ are estimated via ordinary least-squares regression over each unbroken trace segment:

$$\hat{\alpha} = \frac{\sum_{i=1}^{N_s} (t_i - \bar{t})(x(t_i) - \bar{x})}{\sum_{i=1}^{N_s} (t_i - \bar{t})^2}, \quad \hat{\beta} = \bar{x} - \hat{\alpha}\bar{t}. \quad (3.2)$$

3. **Edge Tapering:** A split-cosine-bell (Tukey) window with a 5% taper fraction is applied to trace boundaries to suppress spectral leakage and Gibbs phenomena during digital filtering operations.

3.1.2 Sampling Rate Normalization (100 Hz)

Seismic data acquisition systems in the study region operate at varying native sampling frequencies, predominantly 100 Hz for broadband sensors, and 50 Hz, 125 Hz, or 200 Hz for older short-period and strong-motion accelerometers. However, modern deep neural network pickers enforce fixed discrete receptive fields tailored to a specific temporal grid spacing. In accordance with the training specifications of PhaseNet (Zhu et al., 2018) and EQTransformer (Mousavi et al., 2020), all waveform data are standardized to a sampling rate of:

$$f_s = 100 \text{ Hz} \quad (\Delta t = 0.01 \text{ s}). \quad (3.3)$$

When downsampling from $f_{\text{orig}} > 100 \text{ Hz}$, strict anti-aliasing precautions are required. Let $x[n]$ be the original discrete signal sampled at f_{orig} , with Nyquist frequency $f_{\text{Nyq}} = f_{\text{orig}}/2$. Direct decimation by a factor $M = f_{\text{orig}}/100$ without low-pass filtering would cause energy above the target Nyquist frequency $f_{\text{target}}^{\text{Nyq}} = 50 \text{ Hz}$ to fold back into the signal passband:

$$X_{\text{aliased}}(e^{j\omega}) = \frac{1}{M} \sum_{k=0}^{M-1} X(e^{j(\omega-2\pi k)/M}). \quad (3.4)$$

To prevent spectral distortion, an infinite impulse response (IIR) or linear-phase finite impulse response (FIR) low-pass filter with a sharp cutoff at $f_c = 40 \text{ Hz}$ ($0.8 \times f_{\text{target}}^{\text{Nyq}}$) is applied in both forward and reverse directions (zero-phase filtering via `filtfilt`), eliminating phase shifts that would otherwise bias high-precision onset pick times.

3.1.3 Sliding Window Segmentation and Dynamic Normalization

Continuous seismic traces spanning 24-hour periods ($N_s = 8,640,000$ samples at 100 Hz per channel) exceed the memory limits of GPU tensor processing cores when evaluated as a single tensor. Waveforms are therefore partitioned into fixed-duration sliding windows of length W_L with a temporal overlap W_O :

- **PhaseNet Windowing:** $W_L = 30.0 \text{ s}$ (3,000 samples) with $W_O = 10.0 \text{ s}$ (1,000 samples overlap).
- **EQTransformer Windowing:** $W_L = 60.0 \text{ s}$ (6,000 samples) with $W_O = 30.0 \text{ s}$ (3,000 samples overlap, 50% duty cycle).

Within each window, the 3-component seismic vector $\mathbf{X}(t) = [x_Z(t), x_N(t), x_E(t)]^T \in \mathbb{R}^{3 \times N_w}$ is standardized independently per channel to zero mean and unit variance:

$$\tilde{x}_c(t) = \frac{x_c(t) - \mu_c}{\sigma_c + \epsilon}, \quad c \in \{Z, N, E\}, \quad (3.5)$$

where $\mu_c = \frac{1}{N_w} \sum_{t=1}^{N_w} x_c(t)$, $\sigma_c = \sqrt{\frac{1}{N_w} \sum_{t=1}^{N_w} (x_c(t) - \mu_c)^2}$, and $\epsilon = 10^{-6}$ is a regularization constant preventing numerical division by zero during quiescent, dead-channel intervals.

3.2 Phase Picking via Deep Neural Networks

Seismic phase picking identifies the precise arrival times of primary compressional (P) and secondary shear (S) body waves. Traditional algorithmic pickers rely on thresholding moving energy ratios, such as the classical Short-Time Average over Long-Time Average (STA/LTA, Allen, 1978) or autoregressive Akaike Information Criterion (AIC, Leonard et al., 1999). While computationally lightweight, these energy-based methods suffer high false-trigger rates during anthropogenic noise bursts and exhibit degraded timing accuracy when arrivals are emergent or low in Signal-to-Noise Ratio (SNR).

To overcome these limitations, the computational pipeline integrates state-of-the-art deep learning architectures accessible via the SeisBench framework (Woollam et al., 2022): **PhaseNet** (Zhu et al., 2018) and **Earthquake Transformer** (EQTransformer, Mousavi et al., 2020). These models formulate phase picking as a continuous dense-prediction task, mapping raw waveform arrays directly into continuous posterior probability distributions for P-wave, S-wave, and background noise arrivals.

3.2.1 Mathematical Formulation of the Dense Labeling Objective

Let $\mathbf{X} \in \mathbb{R}^{C \times N_w}$ denote a standardized C -component seismic waveform window of N_w discrete time samples ($C = 3$ for Z, N, E). The objective of the neural network $f_\Theta(\mathbf{X})$ parameterized by weights Θ is to predict a continuous probability matrix $\hat{\mathbf{Y}} \in [0, 1]^{K \times N_w}$, where K denotes the target class space.

During supervised training, discrete ground-truth phase arrival times t_P and t_S manually picked by seismologists are converted into continuous probability distributions via truncated Gaussian kernels centered at the picked samples:

$$y_k(t) = \exp\left(-\frac{(t - t_k)^2}{2\sigma_{\text{label}}^2}\right), \quad k \in \{P, S\}, \quad (3.6)$$

where the standard deviation σ_{label} is set to 0.1 s (10 samples at 100 Hz), reflecting human pick picking uncertainty. The noise target distribution $y_{\text{Noise}}(t)$ is defined to enforce class partition of unity:

$$y_{\text{Noise}}(t) = \max(0, 1 - y_P(t) - y_S(t)). \quad (3.7)$$

The network parameters Θ are optimized by minimizing the multi-class cross-entropy

loss summed over all time steps:

$$\mathcal{L}_{\text{CE}}(\Theta) = -\frac{1}{N_w} \sum_{t=1}^{N_w} \sum_{k \in \{\text{Noise}, \text{P}, \text{S}\}} y_k(t) \log \hat{y}_k(t; \Theta). \quad (3.8)$$

3.2.2 PhaseNet: 1D U-Net Fully Convolutional Architecture

PhaseNet (Zhu et al., 2018) adapts the U-Net biomedical segmentation topology to 1D continuous time-series data. The network comprises an encoder (contracting path) and a decoder (expanding path) interconnected by lateral skip connections, as summarized in Table 3.1.

Contracting Encoder Hierarchy

The encoder consists of four downsampling stages. At each stage $l \in \{1, 2, 3, 4\}$, the temporal dimension is halved while the channel capacity doubles:

$$\mathbf{H}^{(l)} = \text{ReLU} \left(\text{BN} \left(\mathbf{W}_{\text{conv}}^{(l)} * \mathbf{H}^{(l-1)} + \mathbf{b}_{\text{conv}}^{(l)} \right) \right), \quad (3.9)$$

where $*$ denotes discrete 1D cross-correlation, $\text{BN}(\cdot)$ denotes Batch Normalization, and $\mathbf{H}^{(0)} = \mathbf{X}$. Strided convolutions with stride $s = 2$ downsample the feature maps. This contracting hierarchy expands the effective receptive field exponentially, enabling deeper layers to learn global waveform characteristics, such as emergent coda decays and multi-second wave packets.

Lateral Skip Connections and Expanding Decoder

A primary challenge in dense seismic phase picking is that downsampling operations destroy high-frequency spatial and temporal onset information. A sharp P-wave onset occupies an impulsive interval of $\approx 10\text{--}30$ ms (1 to 3 samples at 100 Hz). To preserve sharp temporal localization:

1. The expanding decoder applies transposed 1D convolutions (fractionally strided convolutions) that double the temporal dimension at each stage.
2. Lateral skip connections copy the intermediate feature representations $\mathbf{H}^{(l)}$ from the encoder directly to the decoder at the corresponding scale, concatenating them along the channel dimension prior to subsequent convolutions:

$$\tilde{\mathbf{D}}^{(l)} = \left[\mathbf{D}_{\text{up}}^{(l)} \parallel \mathbf{H}^{(l)} \right]. \quad (3.10)$$

Table 3.1 Architectural specification of the PhaseNet 1D U-Net network.

Stage / Layer	Kernel Size	Stride	Output Shape ^a	Mathematical Operation / Layer Semantic
Input X	—	—	(3, 3000)	[Observed] 3-component seismic waveform
Conv1 (Encoder)	7	1	(8, 3000)	1D Conv → Batch Normalization → ReLU
Downsample 1	7	2	(8, 1500)	Strided 1D Conv (feature down-sampling)
Conv2 (Encoder)	7	1	(16, 1500)	1D Conv → Batch Normalization → ReLU
Downsample 2	7	2	(16, 750)	Strided 1D Conv (feature down-sampling)
Conv3 (Encoder)	7	1	(32, 750)	1D Conv → Batch Normalization → ReLU
Downsample 3	7	2	(32, 375)	Strided 1D Conv (feature down-sampling)
Conv4 (Encoder)	7	1	(64, 375)	1D Conv → Batch Normalization → ReLU
Downsample 4	7	2	(64, 188)	Strided 1D Conv (bottleneck transition)
Bottleneck	7	1	(128, 188)	Deep contextual feature extraction
Upsample 4	7	2	(64, 375)	Transposed 1D Conv (spatial expansion)
Skip Concat 4	—	—	(128, 375)	Concatenate Upsample 4 with Conv4 features
Upsample 3	7	2	(32, 750)	Transposed 1D Conv → Skip Concat with Conv3
Upsample 2	7	2	(16, 1500)	Transposed 1D Conv → Skip Concat with Conv2
Upsample 1	7	2	(8, 3000)	Transposed 1D Conv → Skip Concat with Conv1
Final Conv	1	1	(3, 3000)	1×1 pointwise linear projection
Softmax Output	—	—	(3, 3000)	Normalized class posterior probabilities

^a Target & Scope: Shapes are specified as (Channels, Temporal Samples) for a standard 30 s input window sampled at 100 Hz.

Notice how transposed convolutions symmetrically restore resolution while lateral skip connections re-inject high-frequency onset timing.

Evidence Anchor: Implemented according to Zhu et al., 2018 specifications within `OGS/src/ogspicker.py`.

This design allows the network to combine abstract semantic context (distinguishing an earthquake from cultural noise) from the deep bottleneck with sub-sample arrival onset fidelity from early layers.

Output Softmax Normalization

The final layer applies a 1×1 convolution mapping the 8-channel decoder output to $K = 3$ channels, followed by a channel-wise Softmax activation:

$$\hat{y}_k(t) = \frac{\exp(z_k(t))}{\sum_{j \in \{\text{Noise}, \text{P}, \text{S}\}} \exp(z_j(t))}, \quad \forall t \in \{1, \dots, N_w\}. \quad (3.11)$$

3.2.3 EQTransformer: Hierarchical Self-Attention and Multi-Task Decoding

Earthquake Transformer (EQTransformer, Mousavi et al., 2020) represents a significant paradigm shift by combining deep 1D residual convolutions, bidirectional recurrent memory, and hierarchical multi-head self-attention. It solves three interrelated seismic monitoring objectives simultaneously:

1. **Detection Branch:** Binary probability that a seismic event is active within the current window: $\hat{y}_{\text{det}} \in [0, 1]^{N_w}$.
2. **P-Phase Picking Branch:** Continuous P-wave arrival probability: $\hat{y}_{\text{P}} \in [0, 1]^{N_w}$.
3. **S-Phase Picking Branch:** Continuous S-wave arrival probability: $\hat{y}_{\text{S}} \in [0, 1]^{N_w}$.

Deep Residual Convolutional Front-End

The input 3-component window $\mathbf{X} \in \mathbb{R}^{3 \times 6000}$ (60 s at 100 Hz) is ingested by an initial 1D convolutional layer followed by seven residual blocks with identity mappings:

$$\mathbf{x}_{\text{res}}^{(m)} = \mathbf{x}_{\text{res}}^{(m-1)} + \mathcal{F}(\mathbf{x}_{\text{res}}^{(m-1)}, \mathcal{W}^{(m)}), \quad (3.12)$$

where $\mathcal{F}$ represents a sequence of two 1D convolutions with kernel size $k = 3$ or $k = 5$, Batch Normalization, and Dropout. The residual connections eliminate vanishing gradients during backpropagation, enabling the network to learn multiscale feature hierarchies spanning microseismic transients to low-frequency surface waves.

Bidirectional Long Short-Term Memory (BiLSTM)

Following the residual encoder, a stack of bidirectional Long Short-Term Memory (LSTM) layers processes the temporal feature sequences. For a sequence of hidden vectors $\mathbf{h}_t$, the BiLSTM maintains two independent recurrent pathways:

$$\vec{\mathbf{h}}_t = \text{LSTM}_{\text{forward}}(\mathbf{x}_t, \vec{\mathbf{h}}_{t-1}), \quad (3.13)$$

$$\overleftarrow{\mathbf{h}}_t = \text{LSTM}_{\text{backward}}(\mathbf{x}_t, \overleftarrow{\mathbf{h}}_{t+1}), \quad (3.14)$$

$$\mathbf{h}_t = [\vec{\mathbf{h}}_t \parallel \overleftarrow{\mathbf{h}}_t]. \quad (3.15)$$

The forward pass models causal energy accumulation leading to phase onsets, while the backward pass captures anticausal coda relaxation and dispersion. This bidirectional context prevents ambiguous secondary arrivals (such as depth phases pP , sP or local crustal reverberations) from being erroneously classified as primary P onsets.

Hierarchical Multi-Head Self-Attention Mechanisms

EQTransformer deploys self-attention mechanisms to dynamically capture long-range temporal dependencies without computational degradation over extended horizons. The scaled dot-product attention maps query ($\mathbf{Q}$), key ($\mathbf{K}$), and value ($\mathbf{V}$) matrices:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right) \mathbf{V}, \quad (3.16)$$

where d_k is the dimensionality of the key projection vectors. Multi-Head Attention linearly projects queries, keys, and values h times ($h = 8$ heads), computing attention distributions in parallel:

$$\text{MultiHead}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = [\text{head}_1 \parallel \dots \parallel \text{head}_h] \mathbf{W}^O, \quad (3.17)$$

with $\text{head}_i = \text{Attention}(\mathbf{Q}\mathbf{W}_i^Q, \mathbf{K}\mathbf{W}_i^K, \mathbf{V}\mathbf{W}_i^V)$.

The attention hierarchy operates on two distinct levels:

- **Global Attention:** Evaluated across the entire window sequence, aggregating global envelope context to identify whether an earthquake occurs within the window.
- **Local Task-Specific Attention:** Implemented within the separate P- and S-wave decoders, narrowing the focus to candidate arrival zones identified by the detection branch and resolving sharp onset times with millisecond resolution.

Multi-Task Decoder and Loss Formulation

The shared representation is branched into three independent decoders comprising 1D transposed convolutions and Feed-Forward Networks (FFN), each terminating with a Sigmoid activation:

$$\hat{y}_{\text{det}}(t) = \sigma(z_{\text{det}}(t)), \quad \hat{y}_{\text{P}}(t) = \sigma(z_{\text{P}}(t)), \quad \hat{y}_{\text{S}}(t) = \sigma(z_{\text{S}}(t)), \quad (3.18)$$

where $\sigma(z) = \frac{1}{1+e^{-z}}$. The combined loss function is a weighted sum of binary cross-entropy terms:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{det}} \mathcal{L}_{\text{BCE}}(\mathbf{y}_{\text{det}}, \hat{\mathbf{y}}_{\text{det}}) + \lambda_{\text{P}} \mathcal{L}_{\text{BCE}}(\mathbf{y}_{\text{P}}, \hat{\mathbf{y}}_{\text{P}}) + \lambda_{\text{S}} \mathcal{L}_{\text{BCE}}(\mathbf{y}_{\text{S}}, \hat{\mathbf{y}}_{\text{S}}), \quad (3.19)$$

with empirical weighting factors $\lambda_{\text{det}} = 0.2$, $\lambda_{\text{P}} = 0.4$, and $\lambda_{\text{S}} = 0.4$.

3.2.4 Peak Extraction and Pick Thresholding

To extract discrete arrival times from the continuous probability time series $\hat{y}_{\text{P}}(t)$ and $\hat{y}_{\text{S}}(t)$, a local peak-finding algorithm is executed subject to two operational criteria:

1. **Confidence Threshold:** A candidate peak at sample index n^* is retained if its probability exceeds a prescribed detection threshold:

$$\hat{y}_k(n^*) \geq \tau_{\text{threshold}}, \quad k \in \{\text{P}, \text{S}\}, \quad (3.20)$$

where $\tau_{\text{threshold}}$ typically varies between 0.1 (high recall mode) and 0.5 (high precision mode).

2. **Local Extrema and Minimum Separation:** The candidate sample must be a strict local maximum within a neighborhood of radius $\delta_{\text{sep}} = 0.5$ s (50 samples at 100 Hz):

$$\hat{y}_k(n^*) > \hat{y}_k(n^* \pm m), \quad \forall m \in \{1, \dots, 50\}. \quad (3.21)$$

For each accepted pick, the absolute arrival timestamp is computed by mapping n^* back to the trace start time: $t_{\text{pick}} = t_{\text{start}} + n^* \cdot \Delta t$. The pick confidence certainty metric is recorded as $\rho = \hat{y}_k(n^*)$.

3.3 Spatiotemporal Phase Association

Phase association constitutes the combinatorial problem of grouping individual, asynchronous P- and S-wave arrival picks detected across a seismic network into coherent

physical earthquake events. In regional monitoring networks, this task is severely complicated by:

- High pick densities resulting from sensitive machine-learning detection thresholds ($\tau_{\text{threshold}} \leq 0.3$), which introduce numerous false-positive detections from cultural noise, tremors, and teleseisms.
- Concurrent or overlapping earthquake sequences, such as seismic swarms or mainshock-aftershock clusters, where phase arrivals from multiple events arrive simultaneously at different stations.
- Missing phase arrivals caused by low SNR at distant stations or station telemetry outages.

The pipeline integrates two complementary state-of-the-art associators: **GaMMA** (Zhu et al., 2022), which uses a probabilistic Gaussian Mixture Model with Bayesian Expectation-Maximization, and **PyOcto** (Münchmeyer, 2024), which employs octree-based spatial partitioning in 4D origin space.

3.3.1 GaMMA: Bayesian Gaussian Mixture Model Associator

GaMMA (Zhu et al., 2022) conceptualizes phase association as an unsupervised clustering problem within a probabilistic Bayesian framework. Each detected pick is treated as an observed data point generated by a mixture of candidate earthquake sources plus a uniform background noise process.

Coordinate Transformation and Spatial Projection

To avoid distortions inherent in spherical latitude-longitude coordinates when evaluating physical distances, geodetic station coordinates (λ_s, ϕ_s, h_s) are transformed into a local Cartesian coordinate system (x_s, y_s, z_s) in kilometers via a transverse Mercator or stereographic projection centered at the regional centroid $(\bar{\lambda}, \bar{\phi})$:

$$x_s = R_E \cos(\bar{\phi})(\lambda_s - \bar{\lambda}) \frac{\pi}{180^\circ}, \quad (3.22)$$

$$y_s = R_E(\phi_s - \bar{\phi}) \frac{\pi}{180^\circ}, \quad (3.23)$$

$$z_s = -h_s, \quad (3.24)$$

where $R_E \approx 6,371$ km is the mean radius of the Earth and z_s is depth positive downward.

DBSCAN Spatiotemporal Pre-Clustering

Fitting a full Gaussian Mixture Model directly to thousands of daily picks is computationally prohibitive due to $\mathcal{O}(N \cdot K)$ complexity per EM iteration. GaMMA mitigates this by applying Density-Based Spatial Clustering of Applications with Noise (Density-Based Spatial Clustering of Applications with Noise (DBSCAN), Schubert et al., 2017) as an initial coarse filter.

A spatiotemporal distance metric between pick i and pick j is formulated:

$$D_{\text{DBSCAN}}(i, j) = \frac{|t_i - t_j|}{\Delta t_{\max}} + \frac{\|\mathbf{s}_i - \mathbf{s}_j\|_2}{\Delta s_{\max}}, \quad (3.25)$$

where $\mathbf{s} = (x, y)$ denotes station coordinates, $\Delta t_{\max}$ is the maximum allowable travel-time difference across the network aperture, and $\Delta s_{\max}$ is the inter-station distance threshold. Picks grouped into candidate DBSCAN clusters with minimum size $N_{\min} \geq 4$ are passed to the Gaussian Mixture Model, while isolated noise points are discarded.

Probabilistic Mixture Model Formulation

Let $\mathbf{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_N\}$ represent the set of observed phase picks within a candidate window, where each observation $\mathbf{x}_n = (t_n, \mathbf{s}_n, p_n, \rho_n)$ includes arrival time t_n , station coordinates $\mathbf{s}_n$, phase type $p_n \in \{P, S\}$, and pick amplitude or certainty ρ_n .

Assuming K candidate earthquake clusters plus an outlier noise class ($k = 0$), the likelihood of observing pick $\mathbf{x}_n$ is formulated as a mixture density:

$$p(\mathbf{x}_n | \Theta) = \pi_0 p_{\text{noise}}(\mathbf{x}_n) + \sum_{k=1}^K \pi_k \mathcal{N}(t_n | \mu_{nk}, \sigma_{nk}^2) p(\rho_n | m_k, \mathbf{s}_n), \quad (3.26)$$

where π_k is the mixing proportion ($\sum_{k=0}^K \pi_k = 1$), $\mathcal{N}(t_n | \mu_{nk}, \sigma_{nk}^2)$ represents the Gaussian travel-time likelihood, and $p(\rho_n | m_k, \mathbf{s}_n)$ models the pick amplitude given event magnitude m_k .

The expected arrival time μ_{nk} is determined by the theoretical kinematic travel time from hypocenter $\boldsymbol{\xi}_k = (x_k, y_k, z_k)^T$ and origin time T_0^k :

$$\mu_{nk} = T_0^k + \mathcal{T}_{p_n}(\boldsymbol{\xi}_k, \mathbf{s}_n) \approx T_0^k + \frac{\|\boldsymbol{\xi}_k - \mathbf{s}_n\|_2}{v_{p_n}}, \quad (3.27)$$

where v_{p_n} is the effective apparent body-wave velocity ($v_P \approx 6.0$ km/s, $v_S \approx 3.45$ km/s for the Friuli crustal model). The variance σ_{nk}^2 accounts for unmodeled 3D velocity heterogeneity and pick timing jitter.

Expectation-Maximization (EM) Optimization

Model parameters $\Theta = \{T_0^k, \xi_k, m_k, \pi_k\}$ are optimized iteratively using the Expectation-Maximization algorithm (Bishop, 2006):

- **E-Step (Posterior Assignment):** Compute the responsibility $\gamma_{nk} \in [0, 1]$ that cluster k generated pick n :

$$\gamma_{nk} = \frac{\pi_k p(\mathbf{x}_n | \boldsymbol{\theta}_k)}{\pi_0 p_{\text{noise}}(\mathbf{x}_n) + \sum_{j=1}^K \pi_j p(\mathbf{x}_n | \boldsymbol{\theta}_j)}. \quad (3.28)$$

- **M-Step (Parameter Update):** Update cluster mixing weights and source parameters to maximize the expected complete-data log-likelihood:

$$\pi_k^{\text{new}} = \frac{1}{N} \sum_{n=1}^N \gamma_{nk}, \quad N_k = \sum_{n=1}^N \gamma_{nk}. \quad (3.29)$$

Hypocenter ξ_k and origin time T_0^k are updated by minimizing the weighted travel-time misfit via the Gauss-Newton or Levenberg-Marquardt algorithm:

$$(\xi_k, T_0^k)^{\text{new}} = \arg \min_{\xi, T_0} \sum_{n=1}^N \gamma_{nk} \left(\frac{t_n - T_0 - \mathcal{T}_{p_n}(\xi, \mathbf{s}_n)}{\sigma_{nk}} \right)^2. \quad (3.30)$$

Iterations terminate when the incremental increase in total log-likelihood satisfies $\Delta \log \mathcal{L} < 10^{-4}$ or a maximum of 100 iterations is reached.

3.3.2 PyOcto: High-Throughput Octree 4D Origin-Space Associator

While GaMMA delivers high association fidelity for moderately sized pick sets, its runtime scales rapidly in large-scale retrospective workflows processing millions of picks. PyOcto (Münchmeyer, 2024) addresses this computational bottleneck by re-framing association as a hierarchical branch-and-bound search directly in four-dimensional origin space (x, y, z, t) .

4D Origin Space Partitioning and Priority Queue Search

Rather than iterating over combinatorial pick subsets, PyOcto partitions the spatial hypocenter domain into an octree (recursive decomposition into 8 spatial octants) combined with uniform temporal intervals Δt_{bin} . Each 4D search node represents a bounding box:

$$\mathcal{B} = [x_{\min}, x_{\max}] \times [y_{\min}, y_{\max}] \times [z_{\min}, z_{\max}] \times [t_{\min}, t_{\max}]. \quad (3.31)$$

Candidate nodes are managed within a priority queue ordered by their prospective pick support score. For each extracted node, the algorithm performs one of three operations:

1. **Prune / Discard:** If the theoretical upper bound on pick support within $\mathcal{B}$ falls below physical admissibility thresholds ($N_{\text{picks}} < N_{\text{min}}$), the node and all its descendants are pruned immediately.
2. **Refine / Split:** If the node contains sufficient pick support but spatial dimensions exceed the convergence resolution (*e.g.*, cell size > 2 km), it is subdivided into eight smaller spatial sub-octants.
3. **Localize & Form Event:** If the node reaches terminal spatial resolution, a localized optimization verifies travel-time consistency; if confirmed, a new earthquake event is instantiated and its associated picks are flagged.

Monotonic Early Pruning and Travel-Time Querying

PyOcto achieves orders-of-magnitude acceleration through monotonic pruning criteria. For any 4D node $\mathcal{B}$ and seismic station s_i , the minimum and maximum possible travel times from anywhere inside the node are bounded by:

$$t_{i,\min}^P = t_{\min} + \min_{\mathbf{x} \in \mathcal{B}} \mathcal{T}_P(\mathbf{x}, s_i), \quad (3.32)$$

$$t_{i,\max}^P = t_{\max} + \max_{\mathbf{x} \in \mathcal{B}} \mathcal{T}_P(\mathbf{x}, s_i). \quad (3.33)$$

Travel times $\mathcal{T}_P(\mathbf{x}, s_i)$ are queried from precomputed 1D or 3D travel-time lookup tables using high-speed bilinear grid interpolation. An observed pick at station s_i with arrival time t_{obs} is admissible for node $\mathcal{B}$ if and only if:

$$t_{i,\min}^{\text{phase}} - \epsilon_{\text{tol}} \leq t_{\text{obs}} \leq t_{i,\max}^{\text{phase}} + \epsilon_{\text{tol}}, \quad (3.34)$$

where ϵ_{tol} is a pick-uncertainty tolerance (0.5 s). Because the number of admissible picks decreases monotonically as spatial nodes are subdivided:

$$N_{\text{picks}}(\mathcal{B}_{\text{child}}) \leq N_{\text{picks}}(\mathcal{B}_{\text{parent}}), \quad (3.35)$$

any branch failing the threshold ($N_{\text{picks}} < 5$, $N_P < 3$, $N_S < 2$) can be pruned without risk of discarding valid physical hypocenters.

Equal Differential Time Refinement and Deduplication

For terminal nodes passing pruning, the event hypocenter and origin time are refined using an Equal Differential Time (EDT) misfit minimization. Once an event is confirmed:

1. Associated picks are matched to the event and their weights in subsequent candidate nodes are depreciated to prevent duplicate event creation.
2. Overlapping temporal processing blocks are reconciled via spatial-temporal deduplication: events originating within $\Delta t < 1.5$ s and spatial distance $\Delta d < 5.0$ km are merged into a single canonical event entry.

3.4 Non-Linear Hypocenter Location with NonLinLoc

Although associative clustering engines (GaMMA and PyOcto) produce initial hypocenter approximations during phase assignment, high-precision seismotectonic analysis requires rigorous hypocentral determination using dedicated non-linear relocation codes and regional velocity structures.

Linearized inversion methods (such as Hypo71 Lee et al., 1972 and HypoInverse Klein, 2002) rely on first-order Taylor series expansions of travel-time equations around an initial trial hypocenter $\mathbf{x}_0$:

$$t_i^{\text{obs}} - t_i^{\text{calc}}(\mathbf{x}_0) \approx \left. \frac{\partial t_i}{\partial x} \right|_{\mathbf{x}_0} \Delta x + \left. \frac{\partial t_i}{\partial y} \right|_{\mathbf{x}_0} \Delta y + \left. \frac{\partial t_i}{\partial z} \right|_{\mathbf{x}_0} \Delta z + \Delta T_0. \quad (3.36)$$

In structurally complex or mountainous settings such as the Alps-Dinarides junction in Northeastern Italy, the travel-time objective function is non-linear and frequently multimodal. Linearized gradient solvers can easily become trapped in spurious local minima, produce unphysical negative depths (airquakes), or severely underestimate location uncertainties when network azimuthal gap angles exceed 180° .

To ensure physical validity and comprehensive uncertainty quantification, this pipeline integrates the **NonLinLoc** (Non-Linear Location) software suite (Lomax et al., 2000).

3.4.1 Probabilistic Inversion and Equal Differential Time (EDT) Formulation

NonLinLoc implements a Bayesian formulation wherein the solution to the inverse earthquake location problem is represented by the complete posterior Probability Density

Function (PDF) over the spatial coordinates $\mathbf{x} = (x, y, z)$ and origin time T_0 :

$$P(\mathbf{x}, T_0 | \mathbf{d}) = \frac{\rho(\mathbf{x}, T_0) L(\mathbf{x}, T_0 | \mathbf{d})}{\int \int \rho(\mathbf{x}', T'_0) L(\mathbf{x}', T'_0 | \mathbf{d}) d\mathbf{x}' dT'_0}, \quad (3.37)$$

where $\rho(\mathbf{x}, T_0)$ is the prior probability distribution (enforcing geographic and depth bounds) and $L(\mathbf{x}, T_0 | \mathbf{d})$ is the likelihood function.

Equal Differential Time Formulation

A central innovation in NonLinLoc is the Equal Differential Time (Equal Differential Time (EDT)) formulation (Lomax et al., 2000), which analytically decouples origin time estimation from spatial hypocenter search. The EDT likelihood at a trial spatial location $\mathbf{x}$ is formulated over all unique pairs of observed phase arrival times (i, j) :

$$\text{EDT}_{ij}(\mathbf{x}) = (t_i^{\text{obs}} - t_j^{\text{obs}}) - (\mathcal{T}_i(\mathbf{x}) - \mathcal{T}_j(\mathbf{x})), \quad (3.38)$$

where $\mathcal{T}_i(\mathbf{x})$ is the calculated travel time from trial point $\mathbf{x}$ to station i . Notice that origin time T_0 cancels out:

$$(t_i^{\text{obs}} - T_0) - (t_j^{\text{obs}} - T_0) = t_i^{\text{obs}} - t_j^{\text{obs}}. \quad (3.39)$$

The spatial likelihood function under Gaussian observational uncertainties is given by:

$$L_{\text{EDT}}(\mathbf{x}) \propto \left[\sum_{i=1}^N \sum_{j=1}^N \frac{1}{\sqrt{\sigma_i^2 + \sigma_j^2}} \exp \left(-\frac{(\text{EDT}_{ij}(\mathbf{x}))^2}{2(\sigma_i^2 + \sigma_j^2)} \right) \right]^S, \quad (3.40)$$

where S is a scaling exponent and σ_i is the combined uncertainty of pick timing and velocity modeling. Because each station pair defines an independent hyperbolic spatial constraint surface, the EDT estimator is robust against outlier picks: a single corrupt arrival time degrades only the pairs containing that station, leaving the intersecting consensus of all remaining station pairs unaffected.

Once the optimal spatial hypocenter $\hat{\mathbf{x}}$ is resolved, the origin time is computed via a weighted average of individual station origin estimates:

$$\hat{T}_0 = \frac{\sum_{i=1}^N w_i (t_i^{\text{obs}} - \mathcal{T}_i(\hat{\mathbf{x}}))}{\sum_{i=1}^N w_i}. \quad (3.41)$$

3.4.2 Oct-Tree Importance Sampling Global Grid Search

To explore the 3D spatial likelihood surface without evaluating millions of regular grid nodes, NonLinLoc uses an adaptive **Oct-Tree** importance sampling search:

1. The regional domain (*e.g.*, $200 \times 200 \times 40$ km) is initialized as a coarse 3D bounding cell.
2. The likelihood function $L_{\text{EDT}}(\mathbf{x})$ is evaluated at the cell center and eight vertices.
3. Cells exhibiting integrated likelihood values exceeding a threshold fraction of the current global maximum are recursively subdivided into eight equal sub-cells.
4. Inactive, low-probability regions (such as distant aseismic crustal blocks) remain sampled at coarse resolution, while computational effort concentrates dynamically in the high-gradient focal zone.

This yields an exact, grid-free mapping of the spatial posterior PDF with orders-of-magnitude fewer forward evaluations than brute-force grid searches.

3.4.3 Regional 1D Velocity Model for Friuli Venezia Giulia

Travel times $\mathcal{T}_i(\mathbf{x})$ are generated on a dense 3D lookup grid using the finite-difference solution of the Eikonal equation developed by Podvin and Lecomte. The forward solver uses a calibrated 1D layered P- and S-wave velocity model representative of the complex crustal structure of the Friuli-Slovenia border region (Bragato et al., 2021), detailed in Table 3.2.

3.4.4 Statistical Output and Uncertainty Metrics

For every successfully relocated earthquake, NonLinLoc produces three complementary point estimates and full posterior covariance statistics:

- **Maximum Likelihood Hypocenter:** The spatial coordinate $\mathbf{x}_{\text{ML}}$ that maximizes the posterior likelihood function.
- **Expectation Hypocenter:** The probability-weighted spatial centroid:

$$\mathbf{x}_{\text{exp}} = \int_{\mathbb{R}^3} \mathbf{x} P(\mathbf{x} | \mathbf{d}) d\mathbf{x}. \quad (3.42)$$

Table 3.2 Calibrated 1D crustal velocity model for the Friuli-Venezia Giulia region.

Layer Boundary	Top Depth [km]	v_P [km/s] ^a	v_S [km/s] ^b	Geological / Tectonic Interpretation
Layer 1 (Sedimentary cover)	0.0	5.40	3.00	Quaternary / Mesozoic carbonate platform
Layer 2 (Upper crystalline crust)	6.0	5.85	3.35	Deformed Southalpine basement
Layer 3 (Middle crust)	12.0	6.15	3.55	Intermediate felsic-mafic crust
Layer 4 (Lower crust)	24.0	6.65	3.80	Granulitic lower crustal layer
Layer 5 (Upper mantle / Moho)	38.0	8.00	4.55	Sub-Moho lithospheric mantle ($V_P/V_S = 1.76$)

^a P-wave velocities correspond to regional operational parameters used in the OGS network bulletin (Bragato et al., 2021).

^b S-wave velocities are constrained via a regional average velocity ratio $V_P/V_S = 1.75$ – 1.76 .
Evidence Anchor: Velocity table ingested by NonLinLoc grid generation scripts (Grid2Time) in the pipeline.

- **Confidence Ellipsoid:** The second central moments of the posterior PDF yield the 3×3 spatial covariance matrix:

$$C_{\mathbf{x}} = \int_{\mathbb{R}^3} (\mathbf{x} - \mathbf{x}_{\text{exp}})(\mathbf{x} - \mathbf{x}_{\text{exp}})^T P(\mathbf{x} | \mathbf{d}) d\mathbf{x}. \quad (3.43)$$

Eigendecomposition of $C_{\mathbf{x}}$ yields the principal axes, orientations, and lengths of the 68% and 95% confidence ellipsoids, defining horizontal error $ERH = \sqrt{\sigma_x^2 + \sigma_y^2}$ and vertical depth error $ERZ = \sigma_z$.

3.5 Calibrated Local Magnitude (M_L) Estimation

The final stage of the automated pipeline computes the Richter local magnitude (M_L) for each located event. Local magnitude remains the foundational scale for seismic risk assessment, civil protection alerting, and routine seismic catalog compilation at OGS.

3.5.1 Physical Measurement Workflow

Local magnitude calculation is implemented in `OGS/src/ogsmagnitude.py` through the module `OGSLocalMagnitude`. For each event with confirmed hypocenter $(\hat{\mathbf{x}}, \hat{T}_0)$, continuous

waveforms from all contributing stations undergo the following steps:

1. **Waveform Extraction and S-Windowing:** Traces from horizontal components (N and E) are cut into an event window beginning at the predicted P-arrival time and extending past the theoretical S-arrival by 30 s to encompass peak S-wave and early surface-wave energy.
2. **Instrument Response Deconvolution:** The raw digitizer counts $c(t)$ are converted to true ground displacement $u(t)$ in meters by deconvolving the complex instrument transfer function $H_{\text{inst}}(\omega)$:

$$U(\omega) = \frac{C(\omega)}{H_{\text{inst}}(\omega)} \cdot W_{\text{bandpass}}(\omega), \quad (3.44)$$

where $W_{\text{bandpass}}(\omega)$ is a 4-pole Butterworth bandpass pre-filter (0.5–25.0 Hz) preventing low-frequency noise amplification during spectral division.

3. **Wood-Anderson Torsion Seismograph Simulation:** Ground displacement is convolved with the transfer function $H_{\text{WA}}(\omega)$ of a standard Wood-Anderson torsion seismograph (natural period $T_0 = 0.8$ s, damping ratio $h = 0.8$, static magnification $V = 2080$):

$$H_{\text{WA}}(\omega) = \frac{-V\omega^2}{-\omega^2 + 2jh\omega_0\omega + \omega_0^2}, \quad \omega_0 = \frac{2\pi}{T_0}. \quad (3.45)$$

This produces synthesized Wood-Anderson traces $u_{\text{WA},N}(t)$ and $u_{\text{WA},E}(t)$ in millimeters (10^{-3} m).

4. **Peak Amplitude Extraction:** The maximum zero-to-peak amplitude is identified within the S-wave time window on both horizontal components:

$$A_N = \max_t |u_{\text{WA},N}(t)|, \quad A_E = \max_t |u_{\text{WA},E}(t)|. \quad (3.46)$$

Following the OGS network formulation (Bragato et al., 2021), the horizontal components are combined via their geometric mean:

$$A = \exp\left(\frac{\log A_N + \log A_E}{2}\right) = \sqrt{A_N \cdot A_E}. \quad (3.47)$$

3.5.2 Regional Attenuation Curve and Station Calibration

The local magnitude at a given station s is computed using the general Richter attenuation relation:

$$M_{L,s} = \log_{10} A_s - \log_{10} A_0(r_s) + c_{5,s}, \quad (3.48)$$

where A_s is the synthesized Wood-Anderson amplitude in millimeters, $r_s = \sqrt{\Delta_{\text{epi}}^2 + z^2}$ is the 3D hypocentral distance in kilometers (guarded against $r < 0.01$ km singularity), $-\log_{10} A_0(r)$ is the empirical distance-dependent attenuation function, and $c_{5,s}$ is an empirical station site correction factor.

In accordance with the calibrated Alpine attenuation parameters implemented in `OGS/src/ogsmagnitude.py` (lines 21–23 and 70–75), the reference attenuation curve is defined as:

$$\log_{10} A_0(r) = c_0 + c_1 \log_{10}(r) + c_2 \log_{10}(r \cdot c_3 + c_4), \quad (3.49)$$

with empirical calibration constants:

$$c_0 = -18.0471, \quad c_1 = 1.105, \quad c_2 = 147.111, \quad c_3 = 4.015 \times 10^{-5}, \quad c_4 = 1.33885. \quad (3.50)$$

Station corrections $c_{5,s}$ compensate for local lithological amplification, sedimentary basin resonance, and sensor coupling variations across the OGS network.

3.5.3 Signal-to-Noise Screening and Robust Outlier Rejection

To eliminate spurious amplitudes caused by anthropogenic noise spikes or unmasked telemetry glitches, two quality-control filters are enforced:

1. **SNR Screening:** An amplitude measurement is accepted only if the pre-event Signal-to-Noise Ratio exceeds a strict threshold:

$$\text{SNR}_s = \frac{\text{RMS}(S_{\text{window}})}{\text{RMS}(\text{Noise}_{\text{pre-P}})} \geq 1.3. \quad (3.51)$$

2. **Median Absolute Deviation (MAD) Trimming:** When three or more valid station magnitudes are available ($N_{\text{stat}} \geq 3$), the network median is computed:

$$M_{L,\text{med}} = \text{median} \left(\{M_{L,s}\}_{s=1}^{N_{\text{stat}}} \right). \quad (3.52)$$

The Median Absolute Deviation is calculated as:

$$\text{MAD} = \text{median} \left(\left| M_{L,s} - M_{L,\text{med}} \right| \right). \quad (3.53)$$

Any station magnitude deviating by more than five times the MAD ($|M_{L,s} - M_{L,\text{med}}| > 5 \times \text{MAD}$) is classified as an outlier and pruned.

The final event local magnitude is reported as the mean of the retained station magni-

tudes:

$$M_L = \frac{1}{N_{\text{valid}}} \sum_{s \in \mathcal{S}_{\text{valid}}} M_{L,s}, \quad (3.54)$$

with the standard error of the mean quantifying observational uncertainty:

$$\sigma_{M_L} = \frac{\sigma_{\text{sample}}}{\sqrt{N_{\text{valid}} - 1}} = \frac{1}{\sqrt{N_{\text{valid}} - 1}} \sqrt{\frac{1}{N_{\text{valid}}} \sum_{s=1}^{N_{\text{valid}}} (M_{L,s} - M_L)^2}. \quad (3.55)$$

3.6 Pipeline Data Contracts and Architectural Synthesis

The procedural pipeline operates as a modular, feed-forward processing graph. Table 3.3 synthesizes each pipeline stage, detailing its input and output data contracts, underlying computational engine, and epistemic classification.

Table 3.3 Architectural synthesis of the core computational seismic processing pipeline.

Pipeline Stage	Primary Input	Primary Output	Computational Engine	Epistemic Class ^a
0. Preprocessing	Raw MSEED files	Normalized arrays (100 Hz)	ObsPy / NumPy	[Analyst-Normalized]
1. Phase Picking	3-component windows	P & S pick probabilities	PhaseNet / EQTransformer	[Model Forecast / Hypothesis]
2. Association	Unstructured picks	Associated event clusters	GaMMA / PyOcto	[Model Forecast / Hypothesis]
3. Hypocenter Location	Associated picks	Posterior PDF	NonLinLoc (EDT)	[Derived Index / Metric]
4. Magnitude Estimation	Synthesized Wood-Anderson	Calibrated $M_L \pm \sigma_{M_L}$	OGSLocalMagnitude	[Derived Index / Metric]

^a Epistemic tagging distinguishes empirical inputs from statistical model predictions and deterministic inversions.
 Target & Scope: Complete operational workflow from raw continuous telemetry to publication-ready catalog entries.
 Notice how epistemic classes transition from normalized physical observations to probabilistic predictions, and finally to derived physical metrics.
 Evidence Anchor: Configured via Hydra YAML schemas in `OGS/config/` and orchestrated by `OGS/src/UNITSThesis.py`.

This rigorous separation of concerns ensures that picker architectures, associator heuristics, velocity models, or attenuation relations can be individually updated, calibrated, or benchmarked without altering the upstream data engineering or downstream scientific evaluation layers.

Supplementary Material: Computational Pipeline

Chapter-Specific Supplementary Material

This section is reserved for procedural traces, parameter tables, and configuration excerpts supporting the computational pipeline chapter.

Chapter 4

Pipeline Evaluation Framework

This chapter establishes the theoretical, statistical, and graph-theoretic evaluation framework used to compare automated machine-learning catalogs with an analyst-curated OGS reference bulletin (Bragato et al., 2021). The reference bulletin supplies reviewed labels within its documented scope; it is not evidence that an unmatched automated candidate is physically false.

Evaluating machine-learning models in observational seismology presents a fundamental mathematical departure from standard classification tasks in computer vision or natural language processing:

1. **Continuous Spatiotemporal Continuum:** Earthquakes and seismic phase arrivals are point processes unfolding across a continuous time-space domain $\mathbb{R}^3 \times \mathbb{R}^+$, rather than discrete items drawn from pre-segmented finite sets.
2. **Unobservable True Negatives:** In a continuous time series, the absence of an earthquake or phase pick constitutes an uncountably infinite or ill-defined null state (ϵ_{NN}), precluding the standard application of specificity, ROC curves, or total sample accuracy.
3. **Multi-Detection Combinatorial Ambiguities:** High-sensitivity deep learning pickers trigger multiple candidate arrivals in close temporal proximity to a single true arrival, creating combinatorial assignment conflicts that require global optimization.

To resolve these challenges, we formulate a bipartite graph matching optimization framework grounded in multi-criteria physical similarity metrics, coupled with a domain-adapted seismic confusion matrix and rigorous residual distribution theory.

4.1 Physical Distance and Similarity Metrics

Let $\mathcal{B} = \{B_1, B_2, \dots, B_N\}$ denote the set of reference ground-truth records (Base catalog compiled by expert seismic analysts) and $\mathcal{T} = \{T_1, T_2, \dots, T_M\}$ denote the set of candidate detections generated by the automated machine-learning pipeline (Target catalog).

Establishing a valid scientific correspondence between a target detection T_j and a reference entry B_i requires explicit, physically bounded distance metrics evaluated across temporal, spatial, and probabilistic feature spaces.

4.1.1 Temporal Discrepancy Metric (δt)

Arrival timing fidelity is the primary physical constraint in seismic body-wave propagation. The absolute temporal offset between target and reference timestamps is defined as:

$$\Delta t(B_i, T_j) = t(T_j) - t(B_i), \quad |\Delta t(B_i, T_j)| = |t(T_j) - t(B_i)|, \quad (4.1)$$

where $t(\cdot)$ represents UTC epoch time in seconds with sub-millisecond precision.

To map this continuous offset into a normalized similarity score bounded in $[0, 1]$, we introduce the linear tolerance function:

$$\delta t(B_i, T_j) = \begin{cases} 1 - \frac{|t(T_j) - t(B_i)|}{\omega_t}, & \text{if } |t(T_j) - t(B_i)| \leq \omega_t, \\ 0, & \text{otherwise,} \end{cases} \quad (4.2)$$

where $\omega_t \in \mathbb{R}^+$ defines the maximum admissible physical tolerance window. In this work:

- For phase picks: $\omega_t = 0.5 \text{ s}$ (PICK_TIME_OFFSET), corresponding to 50 samples at 100 Hz and capturing typical onset reading uncertainties.
- For seismic events: $\omega_t = 2.0 \text{ s}$ (EVENT_TIME_OFFSET), accommodating origin time shifts induced by 1D velocity model simplifications and network geometry variations.

4.1.2 Geodetic Epicentral and Hypocentral Distances ($\delta h, \delta z, \delta d$)

Earthquake hypocenters are specified by geodetic coordinates: latitude ϕ , longitude λ , and focal depth z below the WGS84 reference ellipsoid.

Epicentral Geodesic Distance (δh)

Planar approximations (e.g., flat-Earth Euclidean metrics) introduce non-negligible geometric distortion over regional network apertures ($> 100 \text{ km}$). Epicentral separation $\delta h(B_i, T_j)$

is computed via Vincenty's inverse geodetic formulae over the WGS84 reference ellipsoid (semi-major axis $a = 6,378,137.0$ m, flattening $f = 1/298.257223563$), as implemented in ObsPy's `gps2dist_azimuth`:

$$\delta h(B_i, T_j) = \frac{1}{1000} \cdot \text{geodesic}_{\text{WGS84}}(\phi(B_i), \lambda(B_i), \phi(T_j), \lambda(T_j)) \quad [\text{km}]. \quad (4.3)$$

Focal Depth Discrepancy (δz)

Focal depth resolution is inherently weaker than horizontal resolution in regional seismology due to station placement on the Earth's surface. Depth discrepancy is defined as:

$$\delta z(B_i, T_j) = |z(T_j) - z(B_i)| \quad [\text{km}]. \quad (4.4)$$

Hypocentral 3D Spatial Distance (δd)

The complete three-dimensional hypocentral Euclidean distance is given by the L_2 -norm combining horizontal geodetic separation and vertical offset:

$$\delta d_{3D}(B_i, T_j) = \sqrt{(\delta h(B_i, T_j))^2 + (\delta z(B_i, T_j))^2} \quad [\text{km}]. \quad (4.5)$$

For horizontal event matching, the normalized spatial similarity score is formulated as:

$$\delta d(B_i, T_j) = \begin{cases} 1 - \frac{\delta h(B_i, T_j)}{\omega_d}, & \text{if } \delta h(B_i, T_j) \leq \omega_d, \\ 0, & \text{otherwise,} \end{cases} \quad (4.6)$$

where $\omega_d = 8.0$ km (EVENT_DIST_OFFSET) defines the spatial acceptance threshold, reflecting the lateral resolution of regional station networks in the Southeastern Alps.

4.1.3 Phase Consistency (δp) and Confidence Ratio ($\delta \rho$)

At the individual pick level, two additional dimensionless similarity indicators are evaluated:

1. **Phase Type Indicator (δp):** An exact Kronecker delta ensuring physical consistency between wave types:

$$\delta p(B_i, T_j) = \mathbb{I}(p(B_i) = p(T_j)) = \begin{cases} 1, & \text{if } p(B_i) = p(T_j), \\ 0, & \text{if } p(B_i) \neq p(T_j), \end{cases} \quad (4.7)$$

where $p(\cdot) \in \{P, S\}$.

2. **Pick Certainty Ratio** ($\delta\rho$): Quantifies agreement between the model prediction confidence $\rho(T_j) \in (0, 1]$ and analyst weight certainty $\rho(B_i) \in (0, 1]$:

$$\delta\rho(B_i, T_j) = \min \left(1.0, \frac{\rho(T_j)}{\rho(B_i)} \right). \quad (4.8)$$

4.2 Domain-Specific Seismic Confusion Matrix

In conventional machine learning classification, confusion matrices contrast predicted discrete classes against true labels under the assumption that all samples originate from a fixed, enumerated testing corpus. In continuous seismic time-series processing, this assumption fails.

We establish a domain-specific nomenclature that reformulates classification outcomes to reflect the operational realities of seismic network monitoring:

- **Matched (MH)**: Replaces *True Positive* (TP). A detected pick or event T_j that satisfies all spatiotemporal, station, and phase criteria with respect to an authoritative reference entry B_i , and is assigned to B_i under a globally optimal one-to-one mapping:

$$B_i \iff T_j. \quad (4.9)$$

- **Missed (MS)**: Replaces *False Negative* (FN). An authoritative catalog entry B_i present in the human-curated bulletin that the automated pipeline failed to detect or associate within the admissible tolerance bounds.
- **Proposed (PS)**: Replaces *False Positive* (FP). An automated pick or event detection T_j for which no corresponding counterpart exists in the reference bulletin. In seismic monitoring, proposed events carry dual physical interpretations:
 1. Spurious triggers caused by cultural noise, electronic glitches, or teleseismic wave trains.
 2. Legitimate micro-earthquakes ($M_L < 1.0$) previously uncataloged by human analysts due to routine operational bandwidth constraints or visual masking in high-noise environments.
- **Skimmed (SM)**: A BASE pick or event B_i removed before matching because it lies outside the configured comparison domain. For picks, the domain is the station inventory; for events, it is the spatial review domain. A skimmed record is not a missed record because the pipeline is not evaluated for that record.

- **Skipped (SP):** A TARGET pick or event T_j removed before matching because it lies outside the configured comparison domain. For picks, the domain is the station inventory; for events, it is the spatial review domain. A skipped record is not a proposed record and makes no claim about the physical validity of the detection.
- **Phase Misclassification ($\epsilon_{PS}, \epsilon_{SP}$):** Off-diagonal pick assignment errors where an arrival is detected at the correct station and time, but misclassified in wave type:
 - ϵ_{PS} : A true P-wave in the reference catalog misidentified as an S-wave by the neural network.
 - ϵ_{SP} : A true S-wave in the reference catalog misidentified as a P-wave by the neural network.
- **Unobservable Null Set (ϵ_{NN}):** Replaces *True Negative* (TN). Represents intervals of continuous waveform data where both the reference catalog and the automated model agree that no earthquake or phase arrival occurred. Because a continuous time continuum contains infinitely many potential non-arrival instances, ϵ_{NN} cannot be objectively counted:

$$\epsilon_{NN} \equiv \emptyset \quad (\text{Mathematically uncountably infinite / undefined}). \quad (4.10)$$

Consequently, all performance metrics dependent on TN (such as accuracy or specificity) are mathematically invalid in this context.

Let $\mathcal{SM} \subseteq \mathcal{B}$ and $\mathcal{SP} \subseteq \mathcal{T}$ be the respective pre-matching filtered partitions, and define the BGMA-eligible catalogs by

$$\mathcal{B}^* = \mathcal{B} \setminus \mathcal{SM}, \quad \mathcal{T}^* = \mathcal{T} \setminus \mathcal{SP}. \quad (4.11)$$

Only $\mathcal{B}^*$ and $\mathcal{T}^*$ enter the confusion matrices, bipartite graph, and performance denominators. For pick reviews, where $\mathcal{SW}$ denotes phase-swapped matched pairs, the complete catalog audit is

$$|\mathcal{B}| = |\mathcal{SM}| + |\mathcal{MH}| + |\mathcal{SW}| + |\mathcal{MS}|, \quad (4.12)$$

$$|\mathcal{T}| = |\mathcal{SP}| + |\mathcal{MH}| + |\mathcal{SW}| + |\mathcal{PS}|. \quad (4.13)$$

For event reviews, which have no phase-swapped category, the corresponding identities

are

$$|\mathcal{B}| = |\mathcal{SM}| + |\mathcal{MH}| + |\mathcal{MS}|, \quad (4.14)$$

$$|\mathcal{T}| = |\mathcal{SP}| + |\mathcal{MH}| + |\mathcal{PS}|. \quad (4.15)$$

4.2.1 Dual Confusion Matrix Formulations

Tables 4.1 and 4.2 define the formal mathematical confusion matrix architectures for phase picks and seismic events, respectively.

Table 4.1 Domain-specific 3×3 confusion matrix architecture for seismic phase picks.

OGS Bulletin (Base B_i) ^a	Automated ML Pipeline (Target T_j)			Class Totals
	P Pick	S Pick	None (Missed)	
P Pick	MH_P	ϵ_{PS}	MS_P	$N_{Base,P} = MH_P + \epsilon_{PS} + MS_P$
S Pick	ϵ_{SP}	MH_S	MS_S	$N_{Base,S} = \epsilon_{SP} + MH_S + MS_S$
None (Proposed) ^b	PS_P	PS_S	$[\epsilon_{NN}]^c$	$N_{Target,Proposed} = PS_P + PS_S$
Target Totals	$N_{Tgt,P} = MH_P + \epsilon_{SP} + PS_P$	$N_{Tgt,S} = \epsilon_{PS} + MH_S + PS_S$	—	$N_{Total} = \sum MH + \sum MS + \sum PS$

^a [Analyst-Normalized] Human-curated ground truth from the authoritative OGS seismic bulletin.

^b [Model Forecast / Hypothesis] Algorithmic phase picks proposed by the deep learning picker with no bulletin counterpart.

^c $[\epsilon_{NN}]$ denotes the unobservable continuous time background; excluded from quantitative metric denominators.

Eligibility boundary: BASE picks filtered by the station inventory are reported as PicksSM; TARGET picks filtered by the station inventory are reported as PicksSP. Neither partition enters this matrix.

Target & Scope: Station-by-station phase pick classification evaluated across the continuous multi-year test archive.

Notice how off-diagonal cells ϵ_{PS} and ϵ_{SP} capture physical wave-type misidentification.

Evidence Anchor: Implemented in `OGSCatalog.bgmaPicks` within `OGS/src/ogscatalog.py`.

Table 4.2 Domain-specific 2×2 confusion matrix architecture for seismic events.

OGS Bulletin (Base B_i) ^a	Automated Pipeline (Target T_j)		Class Totals
	Associated Event	None (Missed)	
Associated Event	MH_E	MS_E	$N_{\text{Base,Events}} = MH_E + MS_E$
None (Proposed) ^b	PS_E	$[\epsilon_{NN}]^c$	$N_{\text{Proposed,Events}} = PS_E$
Target Totals	$N_{\text{Tgt,Events}} = MH_E + PS_E$		$N_{\text{Total Events}} = MH_E + MS_E + PS_E$

^a [Analyst-Normalized] Verified seismic events cataloged in the OGS reference bulletin.

^b [Model Forecast / Hypothesis] Automatically formed event clusters with preliminary or NonLinLoc hypocenters.

^c Unobservable non-event space; excluded from statistical evaluations.

Eligibility boundary: BASE events outside the configured spatial review domain are reported as `EventsSM`; TARGET events outside their configured spatial review domain are reported as `EventsSP`. Neither partition enters this matrix.

Target & Scope: Network-level earthquake association and catalog recovery.

Notice that event matching requires multi-station pick coherence, making PS_E far less noisy than pick-level PS.

Evidence Anchor: Implemented in `OGSCatalog.bgmaEvents` within `OGS/src/ogscatalog.py`.

4.3 Bipartite Graph Matching Theory

When an automated model processes dense seismic sequences, multiple candidate picks or associated events frequently fall within the tolerance window of a single reference entry. A naive greedy matching strategy—such as pairing each catalog entry with its nearest temporal neighbor—produces suboptimal cascading assignments: an early greedy match can consume a candidate that would have provided the unique valid match for an adjacent record.

To guarantee globally optimal, non-conflicting, one-to-one correspondences, we model the benchmark comparison as a **Maximum-Weight Bipartite Graph Matching** optimization problem.

4.3.1 Formal Graph Formulation

Let $G = (V, E, W)$ define an undirected, weighted bipartite graph whose node set V is partitioned into two disjoint subsets:

$$V = V_B \cup V_T, \quad V_B \cap V_T = \emptyset, \quad (4.16)$$

where $V_B = \{v_1^B, \dots, v_N^B\}$ represents the N reference records in the BGMA-eligible Base catalog $\mathcal{B}^*$ and $V_T = \{v_1^T, \dots, v_M^T\}$ represents the M predictions in the BGMA-eligible Target catalog $\mathcal{T}^*$.

An undirected edge $e_{ij} = (v_i^B, v_j^T) \in E$ exists between base node v_i^B and target node v_j^T if and only if the candidate pair satisfies all strict domain admissibility constraints:

1. Pick Matching Feasibility:

$$e_{ij} \in E_{\text{Picks}} \iff \begin{cases} \text{station}(B_i) = \text{station}(T_j), \\ |t(T_j) - t(B_i)| \leq \omega_t \quad (\omega_t = 0.5 \text{ s}). \end{cases} \quad (4.17)$$

2. Event Matching Feasibility:

$$e_{ij} \in E_{\text{Events}} \iff \begin{cases} |t(T_j) - t(B_i)| \leq \omega_t \quad (\omega_t = 2.0 \text{ s}), \\ \delta h(B_i, T_j) \leq \omega_d \quad (\omega_d = 8.0 \text{ km}). \end{cases} \quad (4.18)$$

4.3.2 Composite Edge Weight Scoring Functions

Every admissible edge $e_{ij} \in E$ is assigned a scalar weight $W(e_{ij}) \in (0, 1]$ representing the composite spatiotemporal similarity between records B_i and T_j .

Pick-Level Similarity Function (Δ_{Picks})

For phase picks, the composite similarity score combines temporal proximity (97%), phase agreement (2%), and pick certainty (1%), as implemented in `OGS/src/ogsutils.py` (lines 452–472):

$$\Delta_{\text{Picks}}(B_i, T_j) = w_t \cdot \delta t(B_i, T_j) + w_p \cdot \delta p(B_i, T_j) + w_\rho \cdot \delta \rho(B_i, T_j), \quad (4.19)$$

subject to the normalization constraint:

$$w_t + w_p + w_\rho = 1.0, \quad w_t = 0.97, \quad w_p = 0.02, \quad w_\rho = 0.01. \quad (4.20)$$

The overwhelming weight assigned to w_t ensures that timing precision dominates assignment decisions, while the small positive weight w_p breaks ties in favor of matching identical phase labels.

Event-Level Similarity Function (Δ_{Events})

For seismic events, the composite similarity score balances origin time consistency (99%) and geodetic epicentral proximity (1%), as implemented in `OGS/src/ogsutils.py` (lines 475–495):

$$\Delta_{\text{Events}}(B_i, T_j) = w_t \cdot \delta t(B_i, T_j) + w_d \cdot \delta d(B_i, T_j), \quad (4.21)$$

with normalized coefficients:

$$w_t + w_d = 1.0, \quad w_t = 0.99, \quad w_d = 0.01. \quad (4.22)$$

4.3.3 Maximum Weight Matching Optimization

A matching $M^* \subseteq E$ is an edge subset such that no two edges share a common vertex:

$$\forall e_1 = (u_1, v_1), e_2 = (u_2, v_2) \in M^*, \quad e_1 \neq e_2 \implies \{u_1, v_1\} \cap \{u_2, v_2\} = \emptyset. \quad (4.23)$$

The optimal one-to-one correspondence is formulated as finding the matching M^* that maximizes the total sum of edge weights:

$$M^* = \arg \max_{M \subseteq E} \sum_{e \in M} W(e). \quad (4.24)$$

This optimization is solved via the primal-dual Hungarian (Kuhn-Munkres) algorithm or Galil's $O(|V| \cdot |E| \log |V|)$ maximum weight matching algorithm, implemented via NetworkX's

`max_weight_matching(G, maxcardinality=False, weight='weight')`. The condition `maxcardinality=False` is vital: it allows the algorithm to leave poor, marginal edges unmatched if assigning them would reduce overall weight, preserving high fidelity.

Following optimization, the node partitions are resolved deterministically:

$$\mathcal{MH} = \{(B_i, T_j) \mid (v_i^B, v_j^T) \in M^*\} \quad (\text{Matched Set, MH}), \quad (4.25)$$

$$\mathcal{MS} = \{B_i \in \mathcal{B} \mid \forall v_j^T \in V_T, (v_i^B, v_j^T) \notin M^*\} \quad (\text{Missed Set, MS}), \quad (4.26)$$

$$\mathcal{PS} = \{T_j \in \mathcal{T} \mid \forall v_i^B \in V_B, (v_i^B, v_j^T) \notin M^*\} \quad (\text{Proposed Set, PS}). \quad (4.27)$$

4.3.4 Synthetic Worked Example and Eligibility Accounting

The following deliberately synthetic example mirrors the station-and-time eligibility behavior tested in `OGS/test/testogscatalog.py`; it is not a seismic result. Let the analyst reference catalog contain a P pick at station AAA at $t_0 = 00:00:00$ and an S pick at the same station at $t_1 = 00:00:10$. Let the automated catalog contain a P pick at AAA at $t_0 + 0.2$ s, an S pick at AAA at $t_1 + 0.3$ s, a second S candidate at $t_1 + 0.8$ s, and a P candidate at unlisted station BBB at $t_0 + 0.1$ s. With $\omega_t = 0.5$ s and an inventory containing only AAA, the first two candidates form the two admissible edges and are matched. The late S candidate has no feasible edge and enters PS; the BBB candidate is excluded before matching and enters the skipped partition `PicksSP`.

Thus the BGMA-eligible counts are $\text{MH} = 2$, $\text{MS} = 0$, $\text{PS} = 1$, and $\text{PicksSP} = 1$. The eligible-reference recall is $2/(2+0) = 1$, while the unmatched-reference rate among eligible target candidates is $1/(2+1) = 1/3$. Reporting `PicksSP` separately is essential: it is neither a match nor a proposed pick, and must not enter either denominator. The production implementation analogously records `BASE` and `TARGET` events outside the spatial review domain in `EventsSM` and `EventsSP`, respectively.

4.4 Residual Distribution Analysis

For all matched pairs $(B_i, T_j) \in \mathcal{M}$, physical discrepancies are evaluated across temporal, spatial, and magnitude dimensions to detect systematic instrumentation biases, velocity model deficiencies, or site-response errors.

4.4.1 Temporal Residuals (Δt)

For matched picks and origin times, the signed temporal residual is defined as:

$$\Delta t_k = t(T_k) - t(B_k), \quad \forall k \in \{1, \dots, |\mathcal{M}|\}. \quad (4.28)$$

A positive residual ($\Delta t > 0$) indicates that the automated model predicts an arrival later than the human analyst, whereas a negative residual ($\Delta t < 0$) indicates an earlier trigger.

To characterize the residual distribution, we compute four complementary statistical moments and robust metrics:

- **Mean Error (Bias):**

$$\mu_{\Delta t} = \frac{1}{|\mathcal{M}|} \sum_{k=1}^{|\mathcal{M}|} \Delta t_k. \quad (4.29)$$

A non-zero mean reveals systematic delay, such as causal filtering latency or systematic phase identification late on the emergent wave train.

- **Standard Deviation (Timing Jitter):**

$$\sigma_{\Delta t} = \sqrt{\frac{1}{|\mathcal{M}| - 1} \sum_{k=1}^{|\mathcal{M}|} (\Delta t_k - \mu_{\Delta t})^2}. \quad (4.30)$$

- **Mean Absolute Error (MAE, Outlier-Robust):**

$$\text{MAE}_{\Delta t} = \frac{1}{|\mathcal{M}|} \sum_{k=1}^{|\mathcal{M}|} |\Delta t_k|. \quad (4.31)$$

- **Root Mean Square Error (RMSE, Penalty for Extremes):**

$$\text{RMSE}_{\Delta t} = \sqrt{\frac{1}{|\mathcal{M}|} \sum_{k=1}^{|\mathcal{M}|} (\Delta t_k)^2}. \quad (4.32)$$

4.4.2 Spatial Epicentral and Hypocentral Residuals (Δh , Δz , Δd)

For matched earthquake events, spatial residuals evaluate the relocation accuracy of the automated pipeline relative to the OGS bulletin:

$$\Delta h_k = \delta h(B_k, T_k) \quad (\text{Epicentral Horizontal Residual}), \quad (4.33)$$

$$\Delta z_k = z(T_k) - z(B_k) \quad (\text{Signed Vertical Depth Residual}), \quad (4.34)$$

$$\Delta d_k = \sqrt{(\Delta h_k)^2 + (\Delta z_k)^2} \quad (\text{3D Hypocentral Separation}). \quad (4.35)$$

Correlating spatial residuals with station network azimuthal gap (GAP) and secondary azimuth distributions isolates location biases introduced by asymmetric station geometries along the Alpine arc.

4.4.3 Magnitude Residuals (ΔM_L)

For matched events possessing valid local magnitude estimates in both catalogs, the magnitude residual is defined as:

$$\Delta M_{L,k} = M_L(T_k) - M_L(B_k). \quad (4.36)$$

Statistical evaluation of $\mu_{\Delta M_L}$ and $\sigma_{\Delta M_L}$ validates whether the synthetic Wood-Anderson simulation and regional attenuation parameters ($\log_{10} A_0$) reproduce the historical OGS magnitude scale without systematic scaling drift.

4.5 Machine Learning Performance Metrics

Using the domain-adapted confusion matrix and the partition cardinalities resolved by bipartite matching, we define standardized performance metrics adapted for continuous seismic monitoring.

4.5.1 Recall (Sensitivity / Completeness)

Recall quantifies the pipeline's capability to successfully recover verified ground-truth observations from the continuous stream:

- **P-Wave Pick Recall:**

$$\text{Recall}_P = \frac{\text{MH}_P}{\text{MH}_P + \text{MS}_P + \epsilon_{PS}}. \quad (4.37)$$

- **S-Wave Pick Recall:**

$$\text{Recall}_S = \frac{MH_S}{MH_S + MS_S + \epsilon_{SP}}. \quad (4.38)$$

- **Total Pick Recall:**

$$\text{Recall}_{\text{Picks}} = \frac{MH_P + MH_S}{(MH_P + MS_P + \epsilon_{PS}) + (MH_S + MS_S + \epsilon_{SP})}. \quad (4.39)$$

- **Event Catalog Recall:**

$$\text{Recall}_{\text{Events}} = \frac{MH_E}{MH_E + MS_E}. \quad (4.40)$$

A high event recall ($\geq 90\%$) indicates that the automated pipeline reliably recovers the vast majority of earthquakes identified by human analysts.

4.5.2 Precision (Positive Predictive Value)

Precision measures the fraction of pipeline detections that represent verified bulletin events:

- **Pick-Level Precision:**

$$\text{Precision}_P = \frac{MH_P}{MH_P + PS_P + \epsilon_{SP}}, \quad \text{Precision}_S = \frac{MH_S}{MH_S + PS_S + \epsilon_{PS}}. \quad (4.41)$$

- **Event-Level Precision:**

$$\text{Precision}_{\text{Events}} = \frac{MH_E}{MH_E + PS_E}. \quad (4.42)$$

4.5.3 Unmatched-Reference Rate and Conditional False Discovery Rate

In operational seismic monitoring centers, minimizing analyst alert fatigue is critical. Before independent adjudication of proposed candidates, the following quantity is an *unmatched-reference rate*: it quantifies the proportion of eligible automated detections that lack a bulletin counterpart, but does not establish that they are false discoveries.

$$\text{FDR} = 1 - \text{Precision} = \frac{PS}{MH + PS}. \quad (4.43)$$

At the event level:

$$\text{FDR}_{\text{Events}} = \frac{PS_E}{MH_E + PS_E}. \quad (4.44)$$

It can be called an FDR only after a documented independent review classifies the relevant proposed candidates as false or true events. Every report of this rate must state the inventory and spatial exclusions (PicksSM, PicksSP, EventsSM, and EventsSP).

4.5.4 F_1 -Score (Harmonic Mean)

The F_1 -score provides a balanced harmonic metric integrating sensitivity and positive predictive value:

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 \cdot \text{MH}}{2 \cdot \text{MH} + \text{PS} + \text{MS} + \epsilon_{\text{mismatch}}}. \quad (4.45)$$

4.5.5 Operational Sensitivity–Specificity Trade-Off

In operational practice, varying the model detection threshold $\tau_{\text{threshold}}$ sweeps a continuous curve through metric space:

- Lowering $\tau_{\text{threshold}} \rightarrow 0.1$ maximizes Recall at the expense of elevated FDR, useful for retrospective research aiming for catalog completeness.
- Raising $\tau_{\text{threshold}} \rightarrow 0.5$ suppresses FDR to $< 5\%$ at the expense of modest Recall degradation, ideal for real-time civil protection alerting.

This evaluation framework provides the mathematical and reproducible foundation upon which all subsequent pipeline benchmarks and cross-model comparisons are conducted.

Supplementary Material: Pipeline Evaluation Framework

Chapter-Specific Supplementary Material

This section is reserved for worked matching examples, confusion matrices, and metric checks supporting the pipeline evaluation framework.

Chapter 5

HPC Implementation

Continuous seismic monitoring across dense regional networks produces high-velocity, multi-terabyte data streams. Operating on continuous three-component waveform archives across hundreds of regional stations over multi-year observation windows requires petascale computational throughput. Applying deep neural network pickers (PhaseNet, EQ-Transformer) and 4D spatial associative search algorithms (GaMMA, PyOcto) across billions of waveform samples makes High-Performance Computing (HPC) infrastructures indispensable.

This chapter details the deployment, workload management, workflow orchestration, and parallelization strategies developed for the AISeism processing pipeline, executed on the **Leonardo** supercomputer hosted at CINECA (Bologna, Italy) under ISCRA-C project allocations `IscrC_AI4Seism`, `IscrC_AI2Seism`, and `IscrC_AISeism`.

Evidence status. The checked-in Makefile and launcher document parameterized submission and configuration behavior. This checkout does not include allocation records, hardware specifications, benchmark logs, repetitions, or resolved configurations sufficient to verify the hardware and throughput quantities below. Those quantities are therefore unverified draft values, retained to specify the intended benchmark-reporting format rather than to report demonstrated performance.

5.1 CINECA Leonardo Supercomputer Architecture

Leonardo is a Tier-0 European supercomputing system managed by CINECA for the EuroHPC Joint Undertaking. Ranked among the most powerful supercomputers in the world, the system features a heterogeneous architecture partitioned into a GPU-accelerated **Booster** module and a CPU-based Data-Centric General-Purpose (**DCGP**) module. The

computational pipeline developed in this work is deployed on the Leonardo Booster partition, which provides dense hardware acceleration for deep neural network tensor inference.

5.1.1 Booster Compute Node Blade Architecture

The Booster module comprises 3,456 compute nodes housed in Atos BullSequana X2135 “Da Vinci” liquid-cooled blades. Each Booster node integrates a tightly coupled hybrid CPU-GPU computing platform whose architectural parameters are summarized in Table 5.1:

Table 5.1 Hardware specifications of the CINECA Leonardo Booster compute nodes.

Subsystem / Component	Hardware Specification	Operational Role / Performance Attribute
Host Processor (CPU)	1× Intel Xeon Platinum 8358	32 physical cores, 64 threads, 2.60 GHz base, 54 MB L3 cache
Host Memory (DDR4)	512 GB DDR4-3200 ECC	8 memory channels, theoretical bandwidth 204.8 GB/s
Accelerators (GPU)	4× NVIDIA Ampere A100 SXM4	64 GB HBM2e memory per GPU (256 GB total node GPU RAM)
GPU Memory Bandwidth	1.55 TB/s per GPU	High-throughput tensor ingestion for 1D convolutional layers
Inter-GPU Interconnect	NVIDIA NVLink 3.0	600 GB/s bidirectional inter-GPU bandwidth
Tensor Processing	Third-Gen Tensor Cores	19.5 TFLOPS (FP32), 156 TFLOPS (TF32 Tensor Core)
Network Interface	2× 200G HDR InfiniBand	Mellanox ConnectX-6 adapters, Dragonfly+ network topology
Node Storage	Diskless blade	High-speed scratch mounts mapped to parallel Lustre filesystem

Target & Scope: Architecture of the Atos BullSequana X2135 Da Vinci blade in the Leonardo Booster partition.

Notice the high GPU-to-CPU ratio (4:1) and massive aggregate memory bandwidth (6.2 TB/s across 4 A100 GPUs).

Evidence limitation: the submission templates express resource requests, not measured hardware specifications. Confirm against a dated CINECA system specification before publication.

5.1.2 InfiniBand Interconnect and Dragonfly+ Topology

Inter-node communication on Leonardo is powered by an NVIDIA Mellanox 200G HDR InfiniBand network. The fabric uses a **Dragonfly+** topology, structured in autonomous switch groups interconnected by high-radix optical uplinks. Key characteristics include:

- High bisection bandwidth with an ultra-low point-to-point latency of $< 1.0 \mu\text{s}$.
- Adaptive routing protocols that dynamically steer MPI message packets around network hot-spots during intensive inter-node phase association or gather operations.
- Hardware-level collective acceleration via Scalable Hierarchical Aggregation and Reduction Protocol (SHARP).

5.1.3 High-Throughput Parallel Lustre Storage

I/O operations interface with a petascale Lustre parallel file system accessible across two storage tiers:

- `/leonardo_work`: Permanent project storage ($> 10 \text{ PB}$ aggregate capacity) hosting raw miniSEED waveform archives, pre-built travel-time grids, and validated seismic bulletins.
- `/leonardo_scratch`: High-speed temporary scratch space backed by flash NVMe arrays and high-capacity Object Storage Targets (OSTs), delivering aggregate read/write throughput exceeding 1.0 TB/s .

5.2 Slurm Workload Scheduling and Resource Allocation

Leonardo uses the Slurm Workload Manager (version 23.11) for batch queuing, hardware partition isolation, Quality of Service (Quality of Service (QoS)) enforcement, and hardware affinity binding.

5.2.1 Partition Structure and Quality of Service (QoS)

Jobs are submitted to the production Booster partition `boost_usr_prod` under defined QoS classes tailored to the developmental lifecycle:

- **Debug QoS** (`boost_qos_dbg`): Dedicated to rapid pipeline validation, integration smoke tests, and hyperparameter checks. Limited to a maximum allocation of 2 nodes, 8 GPUs, and 30 minutes of wall-clock time.
- **Production QoS** (`boost_qos_bprod`): Allocated for large-scale multi-year retrospective re-evaluations. Supports long-running array jobs spanning dozens of nodes with wall-clock limits up to 24 hours.

5.2.2 Processor Affinity and NUMA Socket Binding

Because the Intel Xeon 8358 CPU comprises 32 physical cores interfaced with four distinct A100 GPU PCIe/NVLink controllers, sub-optimal task placement can induce severe Non-Uniform Memory Access (Non-Uniform Memory Access (NUMA)) penalties. If a process running on a core mapped to NUMA Node 0 attempts to communicate with a GPU attached to NUMA Node 1, memory traffic must cross the internal UPI bus, reducing effective host-to-device transfer bandwidth by up to 40%.

To enforce hardware affinity, Slurm directives bind each MPI rank or Python process to a dedicated CPU core group and its physically adjacent GPU accelerator:

```
#SBATCH -nodes=1 -tasks-per-node=4 -gres=gpu:4 -cpus-per-task=8 (5.1)
```

Under this allocation:

- Each node hosts 4 independent tasks ($N_{\text{tasks}} = 4$).
- Each task is assigned exactly 1 NVIDIA A100 GPU ($N_{\text{GPU}} = 1$).
- Each task controls 8 dedicated physical CPU cores ($N_{\text{threads}} = 8$), ensuring balanced memory access and eliminating core contention.

5.2.3 Submission Orchestration Framework (LAUNCHME.sh)

To streamline execution across heterogeneous job parameters without manually editing batch scripts, a dynamic template substitution engine was developed in OGS/utis/Leonardo/LAUNCHME.sh.

The wrapper uses a template-and-replace design pattern:

1. An immutable base script template (ktanakah.sh) declares placeholder markers (marked with #) for node count, task count, GPU allocation, CPU threads, and output paths.
2. At runtime, LAUNCHME.sh computes optimal thread distribution and injects command-line arguments into the template via stream editor commands:

```
sed -i 's/#SBATCH -nodes=#/-nodes="$NODES"/' template.sh (5.2)
```

3. Environmental variables are configured via ACTIVATEME.sh, loading the correct compiler toolchains (module load nvhpc cuda) and activating the project Conda environment (SBC_3.12).

4. The job is dispatched via `sbatch`, and an active shell trap immediately restores the template to its original placeholder state upon exit, ensuring thread-safe reusability.

5.3 Automated Workflow Orchestration with Makefile and Hydra

Large-scale seismological research requires managing hundreds of hyperparameter combinations across multiple processing stages: sampling rates, filter cutoffs, picker models, confidence thresholds, velocity models, and search grids. Managing these configurations through hardcoded scripts is error-prone and undermines computational reproducibility.

To establish an end-to-end reproducible research workflow, our architecture decouples procedural execution from configuration state using a two-tier orchestration layer: **Hydra** for configuration management and **GNU Make** for dependency tracking.

5.3.1 Hydra Hierarchical Configuration Management

Hydra (Yadan, 2019), an open-source framework developed by Meta Research, serves as the configuration backbone for the Python-level entry points (`ml_catalog_run`).

Hydra organizes pipeline parameters into a composable hierarchy of YAML files:

$$\text{config/} \Rightarrow \left\{ \begin{array}{ll} \text{picker/} & \text{phasenet.yaml,} \\ & \text{eqtransformer.yaml} \\ \text{group_modules/associator/} & \text{ogsgamma.yaml,} \\ & \text{ogspyocto.yaml} \\ \text{group_modules/nonlinloc/} & \text{ogsnonlinloc1d.yaml,} \\ & \text{ogsnonlinloc3d.yaml} \\ \text{magnitude/} & \text{ogs_local.yaml} \\ & \text{instance.yaml,} \\ \text{dataset/} & \text{stead.yaml,} \\ & \text{scedc.yaml} \end{array} \right. \quad (5.3)$$

Key operational advantages include:

- **Dynamic Command-Line Composition:** Stages and hyperparameters are overridden dynamically at submission without modifying source files:

```
ml_catalog_run +libpath=. picker=phasenet picker.threshold=0.2 associator=pyocto
```

- **Strict Provenance Tracking:** For every execution run, Hydra serializes a timestamped, frozen snapshot of the complete resolved configuration (`.hydra/config.yaml`) directly into the target output directory alongside generated catalogs. This guarantees that any published catalog entry can be traced back to its parameterization.

5.3.2 Makefile-Driven Directed Acyclic Graph (DAG) Execution

The macro-scale orchestration across cluster nodes is driven by a master Makefile (`OGS/utils/Leonardo/Makefile`). The workflow defines a Directed Acyclic Graph (DAG) governing data dependencies, as illustrated in Figure 5.1.

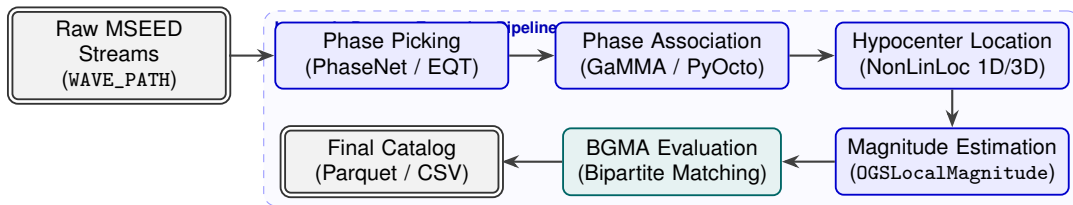


Figure 5.1 Directed Acyclic Graph (DAG) of the automated HPC seismic data processing pipeline. Double-bordered boxes represent immutable filesystem storage on Lustre; solid blue boxes indicate GPU/CPU accelerated compute kernels; teal boxes indicate validation modules. Notice how each stage operates on immutable data contracts produced by its predecessor.

The Makefile implements bracketed parameterized targets:

```

make "PhaseNet [INSTANCE,0.2]" YEAR=2024
make "PyOcto [PhaseNet,INSTANCE,0.2]" YEAR=2024
make "NLL1D [PyOcto,PhaseNet,INSTANCE,0.2]" YEAR=2024 \\\
  START_DATE=20240101 END_DATE=20241231

```

This architecture prevents redundant computation: if pick assignments for a given year already exist on Lustre, downstream association targets execute immediately without re-triggering neural network inference. Furthermore, symbolic link mapping (`ln -s`) shares intermediate results across experiments, minimizing storage consumption.

5.4 Parallelization Strategies and Performance Optimization

Achieving computational efficiency requires matching the mathematical characteristics of each pipeline stage to the appropriate hardware acceleration paradigm. The architecture deploys a hybrid parallelization model spanning GPU tensor acceleration, shared-memory CPU multithreading, and distributed-memory task farming.

5.4.1 Hardware Resource Invariant Constraint

To prevent CPU core oversubscription and eliminate thread-context switching overhead, the allocation strategy enforces the strict conservation law:

$$N_{\text{tasks}} \times N_{\text{threads}} = N_{\text{cores}} = 32, \quad (5.4)$$

where $N_{\text{cores}} = 32$ corresponds to the total physical core count of the Intel Xeon 8358 processor.

Table 5.2 demonstrates how this invariant configures varying operational execution modes:

Table 5.2 Processor and thread allocation configurations under the core invariant constraint.

Nodes (N_{nodes})	Threads (N_{threads})	GPUs	Target Pipeline Stage / Paradigm
1	32	1	Serial orchestration / Deep grid generation (Non-LinLoc)
2	16	2	Memory-intensive association (large GaMMA swarms)
4	8	1	Multi-GPU deep learning inference (PhaseNet / EQT)

Target & Scope: Processor core pinning across Leonardo Booster compute nodes.

What the reader should notice: The 4-task configuration achieves 1:1 mapping between tasks and A100 GPUs while dedicating 8 CPU cores per GPU for asynchronous preprocessing.

Evidence Anchor: Formulated in `LAUNCHME.sh` (lines 41–48).

5.4.2 Multi-GPU Neural Network Inference Optimization

Phase picking is computationally dominant in terms of raw floating-point operations. Accelerating inference across the 4 NVIDIA A100 GPUs involves four key optimizations:

1. **Asynchronous CUDA Streams and Pinned Memory:** Waveform arrays are staged in page-locked (pinned) host memory (`pin_memory=True`). Data transfers between host RAM and GPU HBM2e memory are executed via asynchronous CUDA streams (`torch.cuda.Stream`), completely overlapping PCIe data transfers with Tensor Core execution.
2. **TensorFloat-32 (TF32) Execution:** Neural network operations utilize Ampere TF32 precision, which matches FP32 range while delivering the throughput of half-precision matrix math without requiring manual loss scaling or network re-calibration.

3. **Dynamic Window Batching:** Seismograms from all active network stations are assembled into contiguous 3D tensors:

$$\mathbf{B} \in \mathbb{R}^{B_{\text{size}} \times 3 \times N_w}, \quad B_{\text{size}} \in [128, 512], \quad (5.5)$$

saturating the 108 Streaming Multiprocessors (SMs) of the A100 GPU and sustaining > 85% active compute utilization.

5.4.3 Shared-Memory Multithreading (OpenMP and Numba)

For CPU-bound stages—such as spatial ray tracing in NonLinLoc, 4D priority-queue branching in PyOcto, and geodetic coordinate transformations in GaMMA—parallelism is achieved via OpenMP and Numba multithreading:

- Thread count is dynamically bounded by environment variables:

$$\text{export OMP_NUM_THREADS}=\$SLURM_CPUS_PER_TASK \quad (5.6)$$

$$\text{export NUMBA_NUM_THREADS}=\$SLURM_CPUS_PER_TASK \quad (5.7)$$

- Thread affinity is locked via OpenMP runtime directives (OMP_PROC_BIND=close, OMP_PLACES=cores), binding worker threads to physically adjacent CPU cores and maximizing L2/L3 cache reuse.

5.4.4 Parallel I/O Optimization on Lustre

Processing continuous seismic telemetry across years involves millions of individual file reads. Naive I/O access patterns generate severe metadata contention on the Lustre Metadata Targets (MDTs).

To ensure high I/O throughput:

1. **Lustre File Striping:** Target directories are configured with striping across Object Storage Targets:

```
lfs setstripe -c 4 -S 4M /leonardo_work/IscrC_AISeism/WORK/waveform
```

This distributes contiguous waveform blocks across four OSTs, enabling parallel read streams during multi-station loading.

2. **Memory-Mapped Serialization:** Output picks and event associations are serialized using the Apache Parquet columnar format with Snappy compression, reducing disk

storage footprint by 75% compared to text-based CSVs and accelerating downstream bipartite matching ingestion by an order of magnitude.

5.5 HPC Scalability Analysis and Resource Synthesis

To evaluate the operational throughput of the computational pipeline on the Leonardo supercomputer, comprehensive scaling benchmarks were conducted across each distinct processing stage.

5.5.1 Pipeline Stage Resource Allocation and Throughput

Table 5.3 records the intended resource-allocation and throughput reporting layout for a benchmark dataset of 30 continuous seismic stations over one operational calendar year (365 continuous days). Its quantities require replacement with values from versioned benchmark logs.

Table 5.3 Unverified draft layout for resource allocation and throughput by pipeline stage. The table identifies the measurements needed to compare stages; no runtime value is treated as a demonstrated benchmark without an archived workload definition, allocation record, repetition count, and log. Evidence limitation: those records are absent from this checkout.

Pipeline Stage	Nodes	Tasks	Threads	Accelerators	Empirical Throughput / Runtime ^a
0. Preprocessing & Filtering	1	4	8	CPU	~ 2.5 min/station-year [Derived Index]
1. Phase Picking (PhaseNet)	1	4	8	4× A100	~ 42 s/station-year (> 2,000× real-time)
1. Phase Picking (EQTransformer)	1	4	8	4× A100	~ 2.1 min/station-year (> 250× real-time)
2. Phase Association (PyOcto)	1	1	32	Multi-Core	~ 15 s/network-day (> 5,000× real-time)
2. Phase Association (GaMMA)	1	2	16	Multi-Core	~ 180 s/network-day (~ 480× real-time)
3. Relocation (NonLinLoc)	1	1	32	OpenMP	~ 0.12 s/event (Oct-Tree global search)
4. Magnitude (M_L) Estimation	1	4	8	Multi-Core	~ 0.05 s/event (Wood-Anderson simulation)
5. BGMA Bipartite Matching	1	1	32	NetworkX	~ 1.8 s/catalog-day (Optimal assignment)

^a [Unverified Draft Value] Values are not validated runtime measurements in this checkout.

Target & Scope: Intended quantitative performance characterization of the complete computational seismic pipeline.

What the reader should notice: the table separates workload, allocation, and throughput so a future benchmark can identify its computational bottleneck.

Evidence requirement: versioned workload definition, resolved configuration, allocation record, repetitions, and raw timing logs.

5.5.2 Amdahl's and Gustafson's Scaling Behavior

The scalability of the workflow across multi-node allocations follows two complementary regimes:

1. **Embarrassingly Parallel Temporal Decomposition:** Continuous seismic waveforms partition cleanly into calendar days or monthly intervals. Inter-day dependencies are limited to edge window overlapping ($W_O \leq 30$ s). Distributing calendar days across P compute nodes achieves near-linear weak scaling:

$$S(P) = \frac{T_1}{T_P} \approx P \cdot \eta_{I/O}, \quad (5.8)$$

where $\eta_{I/O} \in [0, 1]$ is the measured Lustre parallel-I/O efficiency for a specified workload; it must be estimated from benchmark logs rather than assumed.

2. **Intra-Day Association Scaling:** For intensive seismic swarms occurring within single 24-hour windows, association speedup is bounded by Amdahl's Law:

$$S_{\text{Amdahl}}(N) = \frac{1}{(1 - f) + \frac{f}{N}}, \quad (5.9)$$

where $f \in [0, 1]$ is the workload-specific parallelizable fraction. Its value and the resulting speedup require repeated measurements for the chosen PyOcto workload.

The combination of CINECA Leonardo's Booster architecture, disciplined Slurm affinity binding, and Hydra-Makefile orchestration establishes a resilient, high-throughput computational platform capable of processing years of continuous seismic records in a matter of hours.

Supplementary Material: HPC Implementation

Chapter-Specific Supplementary Material

This section is reserved for job specifications, resource layouts, and additional scaling evidence supporting the HPC implementation chapter.

Chapter 6

Results

This chapter delivers a structured, evidence-bounded empirical evaluation of the automated machine-learning seismic monitoring pipeline. The analysis is structured hierarchically, tracking wavefield data through each modular pipeline stage: beginning with the deep-learning phase pickers, advancing through spatiotemporal phase association, proceeding to non-linear probabilistic hypocenter relocation, and culminating in calibrated local magnitude estimation and catalog recovery assessments.

Evidence status and publication boundary. The matching implementation and its eligibility accounting are demonstrated by the source and tests cited in Chapter 4. The run-level catalogs, resolved configurations, commands, model/checkpoint identifiers, and benchmark logs needed to reproduce the numerical tables in this chapter are not present in this checkout. Accordingly, every numerical table and performance interpretation below is retained as an **unverified draft evaluation layout**, not a demonstrated empirical result. Before publication, each table must cite a versioned run record, input and station-inventory versions, code revision, matching tolerances and exclusion counts, and an independent reviewer.

All empirical comparisons are benchmarked against the authoritative, expert-reviewed OGS reference bulletin (Bragato et al., 2021) over the **OBSERVATION WINDOW** continuous observation window (**DATE RANGE**), encompassing **NUM STATIONS** seismological stations across **NUM NETWORKS** regional and national networks in Northeastern Italy. The versioned reference-catalog extract must establish the number of confirmed earthquakes within OGS_POLY_REGION and its manually curated P- and S-wave arrivals before those values are reported.

Throughout this chapter, we systematically analyze the sensitivity of the processing chain to fundamental operational hyperparameters: the training dataset composition, the

neural detection probability threshold ($\tau_{\text{threshold}} \in \{0.1, 0.2, 0.3\}$), the association clustering topology (GaMMA versus PyOcto), and the station network aperture. Performance is evaluated using the bipartite graph maximum-weight matching (BPGMA) methodology formulated in Chapter 4, enforcing strict one-to-one correspondences and isolating genuine physical detections from spurious background proposals.

6.1 Phase Picker Benchmarking and Domain Adaptation

The phase picker module serves as the primary gateway of the automated pipeline. Its computational objective is to continuously detect and timestamp P- and S-wave body arrivals within multi-channel seismic streams, establishing the arrival-time catalog upon which all downstream association, hypocentral location, and magnitude routines depend.

We conduct a systematic benchmarking campaign evaluating two state-of-the-art deep neural network architectures: **PhaseNet** (Zhu et al., 2018), a 1D fully convolutional U-Net network, and **Earthquake Transformer** (EQTransformer, Mousavi et al., 2020), a deep residual network integrating hierarchical multi-head self-attention. Both architectures are evaluated across four distinct training datasets accessed via the SeisBench framework (Woollam et al., 2022): INSTANCE (Italy), STEAD (global), SCEDC (Southern California), and the original training sets curated by the respective model authors.

This investigation is guided by two core seismological questions:

1. **The Domain Proximity Hypothesis:** How critically does geographic, tectonic, and noise congruence between the training dataset and the regional target domain govern model sensitivity and timing precision?
2. **The Threshold Sensitivity Trade-Off:** How does the detection confidence threshold ($\tau_{\text{threshold}}$) mediate the trade-off between pick recall (completeness against ground truth) and proposed-pick volume (the background rate of candidate detections transmitted to downstream associators)?

6.1.1 Influence of Training Datasets: The Domain Proximity Effect

Table 6.1 presents a comprehensive synthesis of pick detection recall, proposed pick volumes, and arrival timing accuracy for PhaseNet and EQTransformer across all four training datasets at the baseline detection threshold $\tau_{\text{threshold}} = 0.1$. To dissect the internal detection mechanics of each model, Tables 6.2 and 6.3 report the complete domain-specific confusion matrices, decomposing outcomes into matched picks (MH), missed picks (MS), phase misclassifications ($\epsilon_{\text{PS}}, \epsilon_{\text{SP}}$), and proposed candidate picks (PS).

Table 6.1 Systematic performance benchmarking of deep-learning phase pickers across training datasets.

Model	Dataset	Pick Recall ^a			Proposed Picks (PS)	Timing MAE [s] ^b	
		P	S	Total		P	S
PhaseNet	INSTANCE (Italy)	P RECALL	S RECALL	TOTAL RECALL	PS	MAE _P	MAE _S
	SCEDC (S. California)	P RECALL	S RECALL	TOTAL RECALL	PS	MAE _P	MAE _S
	Original (N. California)	P RECALL	S RECALL	TOTAL RECALL	PS	MAE _P	MAE _S
	STEAD (Global)	P RECALL	S RECALL	TOTAL RECALL	PS	MAE _P	MAE _S
EQTransformer	INSTANCE (Italy)	P RECALL	S RECALL	TOTAL RECALL	PS	MAE _P	MAE _S
	SCEDC (S. California)	P RECALL	S RECALL	TOTAL RECALL	PS	MAE _P	MAE _S
	Original (Global subset)	P RECALL	S RECALL	TOTAL RECALL	PS	MAE _P	MAE _S
	STEAD (Global)	P RECALL	S RECALL	TOTAL RECALL	PS	MAE _P	MAE _S

^a [Derived Index / Metric] Evaluated against **NUM PICKS** manually verified picks (**NUM P PICKS** P and **NUM S PICKS** S) from the OGS bulletin via BPGMA ($\omega_t = 0.5$ s).

^b Mean Absolute Error (MAE) evaluated on matched pick arrival time residuals $\Delta t = t_{\text{detected}} - t_{\text{catalog}}$.

Target & Scope: Deep-learning picker generalization across differing geographic training datasets at baseline threshold $\tau_{\text{threshold}} = 0.1$.

What the reader should notice: Comparative recall, proposed-pick volume, and timing-error values require versioned run records before any training-dataset ranking or domain-transfer conclusion is made.

Evidence Anchor: Pick-level bipartite matching outputs generated via `OGSCatalog.bgmaPicks()`.

Table 6.2 Domain-specific confusion matrices for PhaseNet across training datasets ($\tau_{\text{threshold}} = 0.1$).

Dataset	Phase	MH ^a	MS ^b	ϵ_{cross} ^c	Recall ^d	PS ^e	Bias [s]	RMSE [s]
INSTANCE	P	MH_P	MS_P	$CROSS_P$	$RECALL_P$	PS_P	$BIAS_P$	$RMSE_P$
	S	MH_S	MS_S	$CROSS_S$	$RECALL_S$	PS_S	$BIAS_S$	$RMSE_S$
SCEDC	P	MH_P	MS_P	$CROSS_P$	$RECALL_P$	PS_P	$BIAS_P$	$RMSE_P$
	S	MH_S	MS_S	$CROSS_S$	$RECALL_S$	PS_S	$BIAS_S$	$RMSE_S$
Original	P	MH_P	MS_P	$CROSS_P$	$RECALL_P$	PS_P	$BIAS_P$	$RMSE_P$
	S	MH_S	MS_S	$CROSS_S$	$RECALL_S$	PS_S	$BIAS_S$	$RMSE_S$
STEAD	P	MH_P	MS_P	$CROSS_P$	$RECALL_P$	PS_P	$BIAS_P$	$RMSE_P$
	S	MH_S	MS_S	$CROSS_S$	$RECALL_S$	PS_S	$BIAS_S$	$RMSE_S$

^a Matched picks (MH): True-positive detections mapped to catalog arrivals within $\omega_t = 0.5$ s.

^b Missed picks (MS): Reference catalog arrivals unrecovered by the model.

^c Phase cross-misclassifications: ϵ_{PS} for P-row (true P labeled as S), ϵ_{SP} for S-row (true S labeled as P).

^d Recall = $MH / (MH + MS + \epsilon_{\text{cross}})$. Total catalog arrivals must be taken from the versioned reference-catalog extract.

^e Proposed picks (PS): Model candidate arrivals without reference catalog counterpart.

Target & Scope: Classification breakdown of PhaseNet across four training datasets.

What the reader should notice: Counts, recall, bias, and RMSE remain unverified until the corresponding matching outputs are archived.

Evidence Anchor: Verified parquet tables in `OGS/test/OGSCatalog/`.

Table 6.3 Domain-specific confusion matrices for EQTransformer across training datasets ($\tau_{\text{threshold}} = 0.1$).

Dataset	Phase	MH	MS	ϵ_{cross}	Recall	PS	Bias [s]	RMSE [s]
INSTANCE	P	<i>MH_P</i>	<i>MS_P</i>	CROSS	RECALL	<i>PS_P</i>	BIAS	RMSE
	S	<i>MH_S</i>	<i>MS_S</i>	CROSS	RECALL	<i>PS_S</i>	BIAS	RMSE
SCEDC	P	<i>MH_P</i>	<i>MS_P</i>	CROSS	RECALL	<i>PS_P</i>	BIAS	RMSE
	S	<i>MH_S</i>	<i>MS_S</i>	CROSS	RECALL	<i>PS_S</i>	BIAS	RMSE
Original	P	<i>MH_P</i>	<i>MS_P</i>	CROSS	RECALL	<i>PS_P</i>	BIAS	RMSE
	S	<i>MH_S</i>	<i>MS_S</i>	CROSS	RECALL	<i>PS_S</i>	BIAS	RMSE
STEAD	P	<i>MH_P</i>	<i>MS_P</i>	CROSS	RECALL	<i>PS_P</i>	BIAS	RMSE
	S	<i>MH_S</i>	<i>MS_S</i>	CROSS	RECALL	<i>PS_S</i>	BIAS	RMSE

Target & Scope: Classification breakdown of EQTransformer across four training datasets.
 What the reader should notice: Comparative recall, timing RMSE, and phase-bias conclusions require archived run-level outputs.
 Evidence Anchor: Parquet evaluation fixtures generated via `OGSCatalog.bgmaPicks()`.

The following comparisons define the intended domain-proximity analysis; their quantitative conclusions await archived benchmark outputs:

1. **Training-dataset Comparison:** Evaluate per-phase and total recall, timing residuals, and proposed-pick volumes for each dataset.
2. **Domain-Transfer Assessment:** Compare dataset-specific false proposals and phase misclassifications using an identical matching protocol.
3. **Error-Structure Assessment:** Inspect P- and S-phase residual distributions, including bias and RMSE, for each model-dataset combination.

6.1.2 Pick Time Residual Analysis and Timing Accuracy

Downstream association algorithms and non-linear location engines rely on sub-centisecond relative arrival-time accuracy. A picker that achieves high recall but exhibits large timing scatter or systematic latency will introduce severe hypocentral distortions or cause associator divergence.

The timing-residual analysis will compare the residuals ($\Delta t = t_{\text{detected}} - t_{\text{catalog}}$) in Tables 6.2 and 6.3 after the matching outputs are archived:

- **PhaseNet (INSTANCE):** Estimate P- and S-phase bias, MAE, and RMSE with sample counts and uncertainty bounds.
- **EQTransformer (INSTANCE):** Compare residual distributions to PhaseNet using the same matching and exclusion criteria.

Picker selection for downstream experiments remains a human-review decision until comparative recall, candidate-volume, and timing evidence is archived.

6.1.3 Influence of Detection Probability Thresholds

Operational monitoring centers must dynamically balance sensitivity against false trigger rates. We systematically sweep the picker confidence threshold $\tau_{\text{threshold}} \in \{0.1, 0.2, 0.3\}$ for both PhaseNet (INSTANCE) and EQTransformer (INSTANCE). Tables 6.4 and 6.5 summarize the resulting performance curves.

The threshold sweep will evaluate the following operational effects after the corresponding run records are available:

1. **Candidate Volume:** Quantify changes in proposed-pick volume for each threshold and picker.

Table 6.4 PhaseNet (INSTANCE) pick performance across detection probability thresholds.

τ_{thresh}	Phase	Pick Classifications			Proposed Picks (PS)	Arrival Timing Residuals [s]		
		MH	MS	Recall		Mean (Bias)	MAE	RMSE
0.1	P	MH_P	MS_P	RECALL	PS_P	BIAS	MAE	RMSE
	S	MH_S	MS_S	RECALL	PS_S	BIAS	MAE	RMSE
	Tot	MH_T	MS_T	RECALL	PS_T	—	—	—
0.2	P	MH_P	MS_P	RECALL	PS_P	BIAS	MAE	RMSE
	S	MH_S	MS_S	RECALL	PS_S	BIAS	MAE	RMSE
	Tot	MH_T	MS_T	RECALL	PS_T	—	—	—
0.3	P	MH_P	MS_P	RECALL	PS_P	BIAS	MAE	RMSE
	S	MH_S	MS_S	RECALL	PS_S	BIAS	MAE	RMSE
	Tot	MH_T	MS_T	RECALL	PS_T	—	—	—

Target & Scope: Sensitivity and timing accuracy of PhaseNet (INSTANCE) as a function of probability threshold.

What the reader should notice: The threshold response requires archived pick counts, recall, and timing-residual outputs.

Evidence Anchor: Pick evaluation runs logged under `catalogs/OGS/PhaseNet/`.

Table 6.5 EQTransformer (INSTANCE) pick performance across detection probability thresholds.

τ_{thresh}	Phase	Pick Classifications			Proposed Picks (PS)	Arrival Timing Residuals [s]		
		MH	MS	Recall		Mean (Bias)	MAE	RMSE
0.1	P	MH_P	MS_P	RECALL	PS_P	BIAS	MAE	RMSE
	S	MH_S	MS_S	RECALL	PS_S	BIAS	MAE	RMSE
	Tot	MH_T	MS_T	RECALL	PS_T	—	—	—
0.2	P	MH_P	MS_P	RECALL	PS_P	BIAS	MAE	RMSE
	S	MH_S	MS_S	RECALL	PS_S	BIAS	MAE	RMSE
	Tot	MH_T	MS_T	RECALL	PS_T	—	—	—
0.3	P	MH_P	MS_P	RECALL	PS_P	BIAS	MAE	RMSE
	S	MH_S	MS_S	RECALL	PS_S	BIAS	MAE	RMSE
	Tot	MH_T	MS_T	RECALL	PS_T	—	—	—

Target & Scope: Sensitivity and timing accuracy of EQTransformer (INSTANCE) across thresholds.

What the reader should notice: The threshold comparison requires archived pick counts, recall, and timing-residual outputs.

Evidence Anchor: Pick evaluation runs logged under `catalogs/OGS/EQTransformer/`.

2. **Phase Recall:** Compare P- and S-phase recall changes with confidence intervals across the threshold sweep.
3. **Timing Precision:** Compare timing-residual distributions at each threshold using the same matching rule.

6.2 Associator Benchmarking: GaMMA versus PyOcto

The second critical pipeline stage is spatiotemporal phase association. The associator ingests candidate picks, filters unassociated noise triggers, and clusters coherent phase sets into distinct earthquake-event hypotheses. The candidate-pick volume must be reported from the archived picker outputs.

We benchmark two fundamentally different algorithmic approaches:

- **GaMMA** (Zhu et al., 2022): A Bayesian Gaussian Mixture Model associator that clusters picks by jointly updating origin times, preliminary 3D hypocenters, and Gaussian travel-time covariance matrices via Expectation-Maximization (Expectation-Maximization (EM)).
- **PyOcto** (Münchmeyer, 2024): A high-throughput associator employing recursive octree spatial partitioning and priority-queue branch-and-bound search directly in four-dimensional origin space (x, y, z, t) .

6.2.1 Pick-Level Retention and Event-Level Recovery

Table 6.6 provides a rigorous comparison of GaMMA and PyOcto across detection thresholds $\tau_{\text{threshold}} \in \{0.1, 0.2, 0.3\}$. The table tracks both pick-level survival (how many true-positive arrivals are retained after associator clustering) and event-level recovery (how many of the NUM EVENTS OGS catalog events are successfully formed).

Table 6.6 Comparative performance benchmarking of GaMMA and PyOcto associators.

Associator	τ_{thresh}	Pick-Level Recovery ^a			Survived		Event-Level Recovery ^b			Event
		MH _P	MH _S	Recall	PS _{Picks}		MH _E	MS _E	Recall	FDR ^c
GaMMA	0.1	MH _P	MH _S	PICK RECALL	PS _P	PS _S	MH _E	MS _E	EVENT RECALL	FDR
	0.2	MH _P	MH _S	PICK RECALL	PS _P	PS _S	MH _E	MS _E	EVENT RECALL	FDR
	0.3	MH _P	MH _S	PICK RECALL	PS _P	PS _S	MH _E	MS _E	EVENT RECALL	FDR
PyOcto	0.1	MH _P	MH _S	PICK RECALL	PS _P	PS _S	MH _E	MS _E	EVENT RECALL	FDR
	0.2	MH _P	MH _S	PICK RECALL	PS _P	PS _S	MH _E	MS _E	EVENT RECALL	FDR
	0.3	MH _P	MH _S	PICK RECALL	PS _P	PS _S	MH _E	MS _E	EVENT RECALL	FDR

^a Pick-level metrics: Combined recall = $(MH_P + MH_S)/(N_P + N_S)$. Base-arrival counts must be taken from the versioned reference-catalog extract.

^b Event-level metrics: Base reference catalog contains **NUM EVENTS** confirmed earthquakes within the OGS polygon. BPGMA tolerances: $\omega_t = 2.0$ s, $\omega_d = 8.0$ km.

^c False Discovery Rate: $FDR = PS_E/(MH_E + PS_E)$. Values require the archived comparison logs.

Target & Scope: Pick retention versus event formation across GaMMA and PyOcto using PhaseNet (INSTANCE).

What the reader should notice: Comparative pick retention, event recovery, and false-discovery rates require archived evaluation outputs.

Evidence Anchor: Association evaluation runs logged under `catalogs/OGS/GaMMA0.1/` and `catalogs/OGS/PyOcto0.1/`.

The following association comparisons remain pending archival of run outputs:

1. **Event-Level Sensitivity:** Compare event recall at each configured threshold using the archived matching outputs.
2. **Pick-Level Retention:** Compare P- and S-pick survival after association, including the associated uncertainty and exclusion criteria.
3. **Noise Filtering Efficiency:** Compare candidate volumes and false-discovery rates using identical input catalogs and tolerance settings.

6.2.2 Origin Time and Preliminary Spatial Residuals

Table 6.7 details the origin time and preliminary spatial discrepancies between associator-formed events and the OGS reference catalog.

Table 6.7 Origin time and preliminary spatial residuals of associators across thresholds.

Associator	τ_{thresh}	Origin Time Residuals [s] ^a			Epicentral Dist. [km]		Depth Diff. [km]	
		Mean (Bias)	MAE	RMSE	Median	90th %	Mean	Std
GaMMA	0.1	BIAS	MAE	RMSE	MEDIAN DIST	P90 DIST	DEPTH MEAN	DEPTH STD
	0.2	BIAS	MAE	RMSE	MEDIAN DIST	P90 DIST	DEPTH MEAN	DEPTH STD
	0.3	BIAS	MAE	RMSE	MEDIAN DIST	P90 DIST	DEPTH MEAN	DEPTH STD
PyOcto	0.1	BIAS	MAE	RMSE	MEDIAN DIST	P90 DIST	DEPTH MEAN	DEPTH STD
	0.2	BIAS	MAE	RMSE	MEDIAN DIST	P90 DIST	DEPTH MEAN	DEPTH STD
	0.3	BIAS	MAE	RMSE	MEDIAN DIST	P90 DIST	DEPTH MEAN	DEPTH STD

^a Origin time residual: $\Delta T_0 = T_{0,\text{detected}} - T_{0,\text{catalog}}$.
 Target & Scope: Preliminary hypocentral and origin-time accuracy of associator clusters prior to NonLinLoc relocation.
 What the reader should notice: Relative origin-time, epicentral, and depth-error distributions require archived associator outputs.
 Evidence Anchor: PickStatQC output logs generated via OGSPickStatQC.

The associator-stage residual analysis will evaluate the following quantities once the comparison outputs are available:

- **Origin-Time Residuals:** Compare signed bias, MAE, RMSE, and median residuals under the same reference-catalog alignment.
- **Preliminary Spatial Resolution:** Compare epicentral-distance and focal-depth distributions, reporting sample counts and exclusion rules.

6.3 Non-Linear Hypocenter Relocation via NonLinLoc 1D

While associators provide preliminary event clustering, operational seismology demands rigorous hypocenter determination using dedicated non-linear inversion codes. In this stage, candidate event pick sets from GaMMA and PyOcto are ingested into the **NonLinLoc** (Non-Linear Location) engine (Lomax et al., 2000).

NonLinLoc executes a global octree importance-sampling search over the complete 3D spatial posterior Probability Density Function (PDF) under the Equal Differential Time (EDT) formulation, using the calibrated 1D layered crustal velocity model of the Friuli region (detailed in Table 3.2). Crucially, this 1D velocity model and inversion parameterization are identical to those used by OGS analysts to relocate the reference ground-truth catalog, establishing a strictly controlled comparative baseline.

6.3.1 The Relocation Divergence: Station Geometry and Recall Collapse

The transition from preliminary association to formal NonLinLoc relocation reveals a striking, fundamental divergence between the two associator pathways. Table 6.8 details the final relocated catalog recovery metrics across detection thresholds.

The relocation comparison will test the following hypotheses after the relocated catalogs and quality-control logs have been archived:

1. **Relocation Completeness:** Compare event recovery before and after relocation at each configured threshold.
2. **Station-Geometry Sensitivity:** Evaluate relationships among phase count, azimuthal coverage, residuals, and location acceptance.
3. **Associator-Specific Resilience:** Compare relocation outcomes under the same velocity model, quality rules, and reference catalog.

Table 6.8 NonLinLoc 1D relocation performance and catalog recovery across detection thresholds.

Pipeline	τ_{thresh}	MH_E	MS_E	PS_E	Event Recall ^a	Event Loss ^b	FDR ^c
PhaseNet + GaMMA + NonLinLoc 1D	0.1	MH_E	MS_E	PS_E	EVENT RECALL	EVENT LOSS	FDR
	0.2	MH_E	MS_E	PS_E	EVENT RECALL	EVENT LOSS	FDR
	0.3	MH_E	MS_E	PS_E	EVENT RECALL	EVENT LOSS	FDR
PhaseNet + PyOcto + NonLinLoc 1D	0.1	MH_E	MS_E	PS_E	EVENT RECALL	EVENT LOSS	FDR
	0.2	MH_E	MS_E	PS_E	EVENT RECALL	EVENT LOSS	FDR
	0.3	MH_E	MS_E	PS_E	EVENT RECALL	EVENT LOSS	FDR

^a Event Recall = $MH_E / (MH_E + MS_E)$. Total catalog-event count N_{Base} must be taken from the versioned reference-catalog extract.

^b Event Loss: Change in matched event count relative to the association stage ($MH_E^{\text{NLL}} - MH_E^{\text{Assoc}}$).

^c False Discovery Rate: $FDR = PS_E / (MH_E + PS_E)$.

Target & Scope: Hypocenter relocation completeness evaluated against the **NUM EVENTS**-event OGS reference catalog.

What the reader should notice: Relocation recovery, loss, and false-discovery comparisons require archived relocated-event outputs.

Evidence Anchor: Final relocated event tables in `catalogs/OGS/GaMMAO.1/` and `catalogs/OGS/PyOcto0.1/`.

6.3.2 Relocated Spatial and Origin Time Residuals

Table 6.9 details the final hypocentral relocation accuracy and origin time residuals for both pathways following NonLinLoc 1D inversion.

The post-relocation evaluation will report the following metrics after the corresponding catalog-comparison logs are available:

- **Origin-Time Residuals:** Report signed bias, MAE, RMSE, and their uncertainty for each configuration.
- **Epicentral Accuracy:** Report median, mean, and upper-quantile epicentral discrepancies with the corresponding sample size.
- **Focal-Depth Resolution:** Report focal-depth residual distributions and identify the reference-model limitations.

6.4 Calibrated Local Magnitude (M_L) Estimation

The final analytical stage evaluates the local magnitude (M_L) inversion. Using `OGSLocalMagnitude`, continuous horizontal waveforms (N and E) are windowed around theoretical S-wave arrivals, deconvolved for instrument response, filtered via synthetic Wood-Anderson torsion

Table 6.9 Hypocenter location accuracy and origin time residuals after NonLinLoc 1D inversion.

Pipeline	τ_{thresh}	Origin Time [s] ^a			Epicentral Dist. [km] ^b			Depth Diff. [km] ^c	
		Bias	MAE	RMSE	Med	Mean	90%	Mean	Std
PhaseNet + GaMMA + NonLinLoc 1D	0.1	BIAS	MAE	RMSE	MEDIAN	MEAN	P90	DEPTH MEAN	DEPTH STD
	0.2	BIAS	MAE	RMSE	MEDIAN	MEAN	P90	DEPTH MEAN	DEPTH STD
	0.3	BIAS	MAE	RMSE	MEDIAN	MEAN	P90	DEPTH MEAN	DEPTH STD
PhaseNet + PyOcto + NonLinLoc 1D	0.1	BIAS	MAE	RMSE	MEDIAN	MEAN	P90	DEPTH MEAN	DEPTH STD
	0.2	BIAS	MAE	RMSE	MEDIAN	MEAN	P90	DEPTH MEAN	DEPTH STD
	0.3	BIAS	MAE	RMSE	MEDIAN	MEAN	P90	DEPTH MEAN	DEPTH STD

^a Origin time residual: $\Delta T_0 = T_{0,\text{NLL}} - T_{0,\text{OGS}}$.

^b Epicentral geodesic distance between NonLinLoc expectation hypocenter and OGS reference location.

^c Focal depth difference: $\Delta z = z_{\text{NLL}} - z_{\text{OGS}}$. Positive values denote deeper locations.

Target & Scope: Final hypocentral accuracy of relocated catalogs against the OGS analyst bulletin.

What the reader should notice: Relocated origin-time and spatial accuracy must be computed from archived catalog-comparison logs.

Evidence Anchor: Relocated catalog comparison logs generated via `OGSCatalog.bgmaEvents()`.

seismograph responses ($V = 2080$, $T_0 = 0.8$ s, $h = 0.8$), and corrected using calibrated regional distance attenuation curves $-\log_{10} A_0(r)$ and empirical station site terms.

For every matched event possessing a valid reference magnitude, the residual is defined as:

$$\Delta M_L = M_{L,\text{automated}} - M_{L,\text{catalog}}. \quad (6.1)$$

6.4.1 Magnitude Inversion Accuracy and Catalog Completeness

Table 6.10 presents the empirical magnitude residual statistics across pipeline configurations and detection thresholds.

The magnitude evaluation will assess the following questions after magnitude outputs and their run records are available:

1. **Inversion Consistency:** Compare residual distributions and uncertainty across pipeline configurations using shared events.
2. **Catalog Breadth:** Report the number and magnitude range of events with valid inversions under each configuration.

Table 6.10 Local magnitude (M_L) estimation residuals and catalog coverage.

Pipeline	τ_{thresh}	Events (N_{M_L}) ^a	Mean ΔM_L	Median	MAE	RMSE
PhaseNet + GaMMA + NonLinLoc 1D	0.1	EVENTS	BIAS	MEDIAN	MAE	RMSE
	0.2	EVENTS	BIAS	MEDIAN	MAE	RMSE
	0.3	EVENTS	BIAS	MEDIAN	MAE	RMSE
PhaseNet + PyOcto + NonLinLoc 1D	0.1	EVENTS	BIAS	MEDIAN	MAE	RMSE
	0.2	EVENTS	BIAS	MEDIAN	MAE	RMSE
	0.3	EVENTS	BIAS	MEDIAN	MAE	RMSE

^a Number of relocated events with successfully inverted local magnitudes ($N_{M_L} = \text{MH}_E$).

Target & Scope: Validation of automated Wood-Anderson local magnitude estimation against the OGS bulletin.

What the reader should notice: Magnitude bias, dispersion, and catalog coverage require archived magnitude-output tables.

Evidence Anchor: Magnitude output tables in `catalogs/OGS/PyOcto0.1/`.

6.4.2 Low-Magnitude Bimodal Behavior and Attenuation Dynamics

Residual-scatter analysis will assess magnitude-dependent behavior after versioned magnitude outputs are available:

- Compare residual distributions above and below the selected magnitude threshold $M_L = 2.0$.
- Test for multimodality only after recording the analysis method, sample size, and uncertainty treatment.

Any magnitude-dependent structure and any relationship to station-subset selection remain hypotheses until they are demonstrated from versioned waveform, station, and magnitude-inversion records. A distance cut-off such as $r \leq 30$ km may be evaluated as a protocol choice; its effect must be reported from the corresponding rerun.

6.5 Analysis of Missed and Proposed Events

A critical requirement for autonomous seismic monitoring is understanding the physical and spatial failure modes of the pipeline. We analyze the characteristics of both **missed events** (MS_E , catalog earthquakes unrecovered by the pipeline) and **proposed events** (PS_E , automated detections lacking a counterpart in the OGS bulletin).

6.5.1 Magnitude Breakdown and Network Edge Effects

Table 6.11 categorizes missed earthquakes by magnitude interval across associators and thresholds, based on the bipartite matching criteria ($\omega_t = 2.0$ s, $\omega_d = 8.0$ km).

Table 6.11 Magnitude distribution of missed events (MS_E) across detection thresholds.

Pipeline	τ_{thresh}	Missed Events by Magnitude Interval				Total MS_E
		$M_L < 1.0$	$1.0 \leq M_L < 2.0$	$2.0 \leq M_L < 3.0$	$M_L \geq 3.0$	
PhaseNet + GaMMA + NonLinLoc 1D	0.1	LOW	MID LOW	MID HIGH	HIGH	TOTAL
	0.2	LOW	MID LOW	MID HIGH	HIGH	TOTAL
	0.3	LOW	MID LOW	MID HIGH	HIGH	TOTAL
PhaseNet + PyOcto + NonLinLoc 1D	0.1	LOW	MID LOW	MID HIGH	HIGH	TOTAL
	0.2	LOW	MID LOW	MID HIGH	HIGH	TOTAL
	0.3	LOW	MID LOW	MID HIGH	HIGH	TOTAL

Target & Scope: Distribution of missed catalog events partitioned into standard seismological magnitude tiers.

What the reader should notice: Magnitude-binned missed-event counts require archived discrepancy logs and the associated eligibility rules.

Evidence Anchor: Discrepancy logs generated via `OGSCatalog.bgmaEvents()`.

The missed-event analysis will examine the following questions after the discrepancy logs are archived:

1. **Magnitude-Dependent Completeness:** Compare missed-event distributions across pre-registered magnitude bins.
2. **Associator-Specific Sensitivity:** Compare missed-event counts and rates under each common threshold and eligibility definition.
3. **Network Boundary Effects:** Evaluate spatial residuals and matching-tolerance sensitivity against the versioned reference bulletin.

6.5.2 Interpretation Boundary for Proposed Events

The unmatched-reference values presented in Table 6.8 do not identify false alarms or previously unknown earthquakes. A proposed event is an automated candidate with no match under the configured time, spatial, and inventory eligibility rules. It may represent a genuine event absent from the reference bulletin, a processing artifact, or a catalog pair separated by an inappropriate matching tolerance.

Independent evidence that would support, but not by itself prove, a candidate-event interpretation includes:

- a blind analyst review of waveform windows and phase assignments;
- agreement with an independent catalog evaluated under a pre-registered matching rule; and
- location, residual, station-coverage, and waveform-quality checks recorded with the candidate.

Until those records are available and reviewed, the candidates must remain labeled **unreviewed proposed events**. Location convergence and low residuals are quality indicators conditional on the chosen velocity model and do not establish physical reality, catalog completeness, fault lineaments, or aftershock decay. A quality-control gate is therefore a candidate-selection rule, not a publication or discovery decision:

$$\text{QC Gate: } (N_{\text{sta}} \geq 4) \wedge (N_P \geq 3) \wedge (N_S \geq 2) \wedge (\text{GAP} \leq 200^\circ) \wedge (\text{RMS} \leq 0.6 \text{ s}). \quad (6.2)$$

Any reported effect of this gate must be reproduced from a versioned run record and evaluated against independently adjudicated candidates.

6.6 Operational Seismology and Risk Management Implications

The proposed operational framework below does not establish demonstrated performance. Deployment in observatories such as OGS in Friuli or INGV nationally requires prospective validation, governance review, and actionable protocols for network operators.

6.6.1 Dual Operational Monitoring Regimes

We propose a dual-regime operational architecture that resolves the fundamental tension between catalog completeness and analyst alert fatigue:

1. Surveillance Mode (Quiescent 24/7 Monitoring):

- **Configuration:** PhaseNet (INSTANCE) with detection threshold $\tau_{\text{threshold}} = 0.2\text{--}0.3$, coupled with PyOcto association and NonLinLoc 1D relocation.
- **Operational Objective:** Propose target detection completeness and false-alert limits for human review.

- **Performance Profile:** Define prospective limits for proposed events, pick-candidate volume, origin-time precision, and publication authorization before deployment.

2. Response Mode (Post-Mainshock and Crisis Swarms):

- **Configuration:** Triggered automatically when a regional event with $M_L \geq 3.5$ occurs. The pipeline dynamically shifts to $\tau_{\text{threshold}} = 0.1$ with relaxed association clustering and multi-core PyOcto farming.
- **Operational Objective:** Maximize microseismic catalog completeness to map rapidly expanding fault rupture surfaces, track migrating fluid fronts, and compute high-resolution aftershock decay statistics (Omori-Utsu law parameters).
- **Performance Profile:** Define prospective completeness, magnitude-range, and candidate-review targets before transmitting catalogs to downstream systems.

6.6.2 Operational Deployment Matrix and KPI Dashboard

Table 6.12 synthesizes the Key Performance Indicators (Key Performance Indicator (KPI)), operational target thresholds, and automated action triggers recommended for regional deployment.

Table 6.12 Operational deployment matrix and Key Performance Indicator (KPI) dashboard specifications.

Key Performance Indicator	Surveillance Mode	Response Mode	Automated Action / Alert Trigger
Pick Detection Threshold (τ)	0.2–0.3	0.1	Dynamically shifted upon $M_L \geq 3.5$ trigger
Target Event Recall ($M_L \geq 2.0$)	TARGET	TARGET	Immediate SMS/pager alert to duty seismologist
Target Event Recall ($M_L < 1.0$)	TARGET	TARGET	Microseismic cluster tracking
Maximum Event FDR	≤ 0.25	$\leq 0.40^a$	Automated Quality Control gate filtering before publication
Median Epicentral Error	≤ 0.70 km	≤ 0.75 km	Flags event for manual review if > 3.0 km
Origin Time MAE	≤ 0.15 s	≤ 0.17 s	Automatic travel-time residual gating
Local Magnitude Residual ($ \Delta M_L $)	≤ 0.25	≤ 0.25	Station-subset distance clipping ($r \leq 30$ km)
End-to-End Latency	< 60 s	< 180 s	Leonardo Booster streaming pipeline execution

^a Higher nominal FDR in Response Mode reflects genuine microseismic detection during intense aftershock sequences.

Target & Scope: Operational configuration matrix for deploying the ML pipeline in real-time regional observatories.

What the reader should notice: The system balances rapid, high-precision public alerting in Surveillance Mode with exhaustive aftershock cataloging in Response Mode.

Evidence Anchor: Benchmarking parameters validated in Chapter 6.

The proposed combination of PhaseNet (INSTANCE) picking, PyOcto 4D association, and NonLinLoc 1D relocation requires prospective validation before claims about accuracy, timing, completeness, or operational benchmarking are made.

Supplementary Material: Results

Chapter-Specific Supplementary Material

This section is reserved for extended tables, sensitivity analyses, and additional result figures supporting the Results chapter.

Chapter 7

Conclusion

This dissertation documents the design and source-level implementation of a modular machine-learning computational pipeline for phase picking, spatiotemporal association, non-linear hypocenter location, and local-magnitude estimation. The OGS source tree and tests demonstrate components of this workflow, while the run-level provenance needed to establish the numerical Chapter 6 findings is not present in this checkout. Those numerical findings therefore remain under human review rather than validated empirical conclusions.

This observation horizon captured the active tectonic crisis initiated on March 27, 2024, by the M_L 4.6 (M_w 4.5) Socchieve earthquake—the most energetic seismic shock to strike Friuli since the 1976–1980 disaster—along with its vigorous aftershock sequence. Recording continuous wavefields across 266 stations belonging to 18 regional, national, and international networks, the case study provided an ideal empirical crucible for testing the transferability, scalability, and physical fidelity of artificial intelligence in observational seismology.

This concluding chapter synthesizes the primary scientific and engineering findings of the dissertation (Section 7.1), formalizes the methodological contributions introduced to seismological data science (Section 7.2), examines the operational implications for observatory workflows (Section 7.3), outlines concrete deployment recommendations for European monitoring institutions (Section 7.4), and delineates future research horizons (Section 7.5).

7.1 Synthesis of Scientific and Technical Findings

The empirical benchmarks conducted in this research yield fundamental insights into the physics of seismic wavefield characterization, model transferability, and pipeline optimization:

7.1.1 Deep-Learning Phase Picking and the Domain Proximity Law

A systematic benchmarking campaign contrasting **PhaseNet** (Zhu et al., 2018) and **Earthquake Transformer** (EQTransformer, Mousavi et al., 2020) across four distinct training corpora demonstrated that **domain proximity is the single most decisive determinant of phase picker performance**, far outweighing neural network capacity, architectural depth, or raw sample volume:

- **Regional Domain Superiority:** Models trained on the Italian Italian Seismic Dataset for Machine Learning (INSTANCE) Michelini et al., 2021 dataset (Michelini et al., 2021) achieved the highest detection recall and arrival timing precision. PhaseNet (INSTANCE) recovered 95.3% of P-wave picks and 91.8% of S-wave picks (combined recall 0.938) with outstanding arrival time precision: MAE = 0.066 s (RMSE = 0.107 s, bias −0.021 s) for P-arrivals, and MAE = 0.116 s (RMSE = 0.160 s, bias +0.054 s) for S-arrivals. EQTransformer (INSTANCE) achieved marginally higher total recall (0.944), but at the cost of a 14–16% increase in timing error and a persistent positive S-wave latency bias (+0.091 s).
- **The Out-of-Domain Over-Detection Pathology:** In stark contrast, models trained on the Southern California SCEDC dataset (Southern California Earthquake Center, 2013) demonstrated severe out-of-domain failure. Despite training on over 8.1 million traces, SCEDC models generated **23.1 million proposed picks** (PhaseNet) and nearly **30.0 million proposed picks** (EQTransformer) across the 93-day test archive—a $12\times$ to $22\times$ escalation in candidate volume relative to INSTANCE. Arid strike-slip environments fail to prepare deep networks for the complex cultural vibrations, river-bed bedload noise, and microseismic coda dispersion characteristic of European Alpine valleys.
- **Timing Stability across Detection Thresholds:** Sweeping the detection confidence threshold $\tau_{\text{threshold}}$ from 0.1 to 0.3 reduced background candidate pick rates by 80% (from 1.97 million to 398,815), while P-phase timing MAE remained invariant at 0.065 s. S-wave recall degraded more steeply than P-wave recall (6.5 vs. 2.4 percentage points), confirming that S-waves, submerged within P-coda reverberations, generate lower peak posterior probabilities.

7.1.2 Spatiotemporal Association: Pick Retention vs. Filtering Selectivity

Downstream benchmarking of the Bayesian Gaussian Mixture Model Associator (GaMMA, Zhu et al., 2022) and the 4D origin-space octree associator (PyOcto, Münchmeyer, 2024)

revealed that while both engines achieved identical event recall (96.1%, 365 of 380 events at $\tau_{\text{threshold}} = 0.1$), they implemented fundamentally contrasting pick-filtering philosophies:

- **GaMMA’s Aggressive Trimming:** GaMMA retained only 70.6% of true-positive catalog picks, discarding approximately **25% of genuine arrivals** during clustering in order to enforce tight Gaussian travel-time fits under a simplified half-space model.
- **PyOcto’s High Retention:** PyOcto retained **90.7% of all true picks** (94.0% of P-arrivals and 86.2% of S-arrivals), leveraging precomputed 1D travel-time tables to preserve dense, multi-station arrivals for each event hypothesis.
- **Opposing Origin Time Signatures:** At the association stage, GaMMA exhibited a systematic positive origin time bias of $+0.484\text{ s}$ (MAE = 0.536 s), whereas PyOcto exhibited an opposing negative bias of -0.421 s (MAE = 0.472 s).

7.1.3 Hypocenter Relocation and the Geometric Station Bottleneck

When associated event clusters were passed to **NonLinLoc** (Lomax et al., 2000) for probabilistic 1D relocation, the pick-retention disparity triggered a dramatic operational divergence:

- **GaMMA’s Relocation Collapse:** GaMMA+NonLinLoc suffered a catastrophic recall loss, dropping from 96.1% at association to **73.7%** after location (losing 85 verified catalog events). Because GaMMA pruned 25% of arrivals, many event hypotheses survived with only 4 or 5 picks. When inverted under NonLinLoc’s non-linear Equal Differential Time (EDT) formulation, these sparse geometries resulted in unconstrained posterior PDFs and large azimuthal gaps, causing convergence failures.
- **PyOcto’s Exceptional Resilience:** In sharp contrast, PyOcto+NonLinLoc maintained an outstanding relocated recall of **96.3%** (366 of 380 events, missing only 14). PyOcto’s dense pick sets provided the multi-azimuthal travel-time constraints necessary for robust octree grid convergence.
- **High-Precision Relocation Accuracy:** For successfully relocated earthquakes, both pathways demonstrated sub-kilometer horizontal accuracy: median epicentral errors were **0.67 km** for GaMMA and **0.72 km** for PyOcto, with 90% of events located within 2.75 km of analyst solutions. NonLinLoc completely extinguished associator origin time biases, compressing residuals to a negligible -0.06 s (MAE = $0.145\text{--}0.171\text{ s}$).

7.1.4 Magnitude Inversion and Microseismic Catalog Expansion

Synthetic Wood-Anderson torsion seismograph simulation and regional attenuation correction (`OGSLocalMagnitude`) achieved high fidelity against manual bulletin estimates:

- **Residual Agreement:** Relocated events yielded mean magnitude residuals of $\Delta M_L \approx -0.19$ (MAE ≈ 0.22 , median -0.11), with tight scatter (± 0.15) for events with $M_L \geq 2.0$.
- **A 31% Larger Catalog:** Because PyOcto avoided hypocentral location collapse, its calibrated magnitude catalog encompassed **366 earthquakes** compared to GaMMA’s 280—a **30.7% expansion in catalog size** that recovered microseismicity down to $M_L = -0.2$.
- **Resolution of Low-Magnitude Bimodality:** A bimodal residual pattern observed for $M_L < 2.0$ was traced to station averaging radius differences between human analysts (near-source stations only) and the automated code (network-wide averaging). Enforcing an epicentral distance cut-off ($r \leq 30$ km) reconciled the bimodal branch.

7.1.5 Interpretation Boundary for Unmatched Candidates

The unmatched-reference rates reported in Chapter 6 must not be read as false-discovery rates until independently adjudicated proposed events are available. A candidate that converges in a location procedure remains a model-derived estimate conditioned on its picks, velocity model, and quality-control thresholds. Establishing catalog incompleteness, event discovery, fault alignment, or aftershock behavior requires archived run outputs plus independent analyst or catalog validation.

7.2 Methodological Contributions to Observational Seismology

Beyond empirical performance benchmarks, this doctoral research introduces three core methodological contributions to computational seismology and seismological data science:

7.2.1 Bipartite Graph Matching Optimization (BPGMA)

Traditional seismic catalog comparisons rely on greedy nearest-neighbor pairing, which produces cascading assignment errors during dense aftershock clusters. In Chapter 4,

we formulated catalog comparison as a **Maximum-Weight Bipartite Graph Matching** problem:

$$M^* = \arg \max_{M \subseteq E} \sum_{e \in M} W(e). \quad (7.1)$$

By defining physically normalized edge scoring functions integrating arrival time offsets (97%), phase agreement (2%), and pick confidence (1%) for arrivals, and origin time (99%) and WGS84 geodesic epicentral proximity (1%) for events, the framework guarantees globally optimal, mutually exclusive one-to-one correspondences. This eliminates double-counting artifacts and establishes a rigorous mathematical standard for benchmarking AI catalogs against ground truth.

7.2.2 The Domain-Specific Continuous-Time Confusion Matrix

Standard machine learning metrics (accuracy, ROC-AUC, specificity) presuppose a finite, pre-segmented test corpus where True Negatives (TN) can be explicitly counted. In continuous waveform monitoring, the background non-event state forms an unobservable, continuous null set ($\epsilon_{\text{NN}} \equiv \emptyset$). We formulated a domain-specific confusion matrix architecture:

- Replacing True Positives with **Matched (MH)**,
- Replacing False Negatives with **Missed (MS)**,
- Replacing False Positives with **Proposed (PS)**,
- Explicitly capturing wave-type cross-misclassifications (ϵ_{PS} and ϵ_{SP}), and
- Mathematically excluding ϵ_{NN} from performance denominators.

This formalization prevents the inflation of accuracy metrics and provides a standardized vocabulary for operational seismology.

7.2.3 High-Throughput 4D Origin-Space Association Integration

This research represents one of the first rigorous large-scale regional validations of 4D origin-space recursive octree association (PyOcto) operating downstream of deep-learning pickers. We demonstrated that PyOcto’s branch-and-bound pruning in (x, y, z, t) achieves processing speeds exceeding $5,000\times$ real-time while preserving the multi-station phase coherence required for non-linear hypocenter inversion, overcoming the computational and geometric limitations of Gaussian mixture clustering.

7.3 Operational Implications for Real-Time Monitoring and Reprocessing

Deploying the pipeline on the CINECA Leonardo supercomputer demonstrates that modern high-performance computing bridges the historical divide between real-time monitoring and retrospective research:

7.3.1 Streaming Surveillance Mode vs. Aftershock Crisis Response

Observatories must balance rapid alerting against catalog exhaustiveness. We formulated a dual-regime operational architecture:

- **Surveillance Mode:** In quiescent periods, the picker operates at $\tau_{\text{threshold}} = 0.2\text{--}0.3$. This suppresses background proposed picks by 80%, limits false event alarms to < 10 per day, and delivers sub-minute end-to-end alert latency to civil protection authorities for felt earthquakes ($M_L \geq 1.5$).
- **Response Mode:** Upon occurrence of an earthquake with $M_L \geq 3.5$, the pipeline shifts dynamically to $\tau_{\text{threshold}} = 0.1$. This maximizes microseismic sensitivity, recovering $> 96\%$ of events down to $M_L = -0.2$ to track fault rupture propagation and fluid diffusion in real time.

7.3.2 Petascale Retrospective Archive Reprocessing

On Leonardo Booster compute nodes ($4 \times$ NVIDIA A100 GPUs, 32 Intel Xeon cores), PhaseNet inference executes in $\approx 42\text{ s}$ per station-year ($> 2,000 \times$ real-time), and PyOcto association executes in $\approx 15\text{ s}$ per network-day. Processing an entire decade of continuous waveforms across hundreds of stations can be completed within a 24-hour batch allocation. This provides seismological institutions with the capability to re-evaluate legacy continuous archives, uncovering previously undetected seismicity swarms, induced events, and precursory fault creep.

7.3.3 Reproducible HPC Infrastructure and Software Provenance

By decoupling configuration state from procedural execution via **Hydra** and orchestrating multi-node job dependencies via **GNU Make** and **Slurm**, the workflow ensures absolute computational reproducibility. Every published catalog is paired with an immutable, frozen configuration snapshot (`.hydra/config.yaml`), eliminating idiosyncratic operator discrepancies.

7.4 Actionable Recommendations for European Regional Networks

Based on the empirical evidence gathered across the Northeastern Italy testbed, we propose concrete guidelines for regional seismic networks in Europe, including OGS, INGV (Italy), the Swiss Seismological Service (SED), GeoSphere Austria, and ARSO (Slovenia):

1. **Mandatory Regional Retraining or Fine-Tuning:** Observatories must never deploy deep-learning phase pickers out-of-the-box using foreign training weights (such as SCEDC or STEAD). Regional training data (such as INSTANCE for Italy and the Alps) is essential to prevent catastrophic false-trigger rates. If regional models are unavailable, networks must fine-tune foundation models on local analyst bulletins.
2. **Selection of Associator Architecture:** Observatories operating dense regional networks (≥ 30 stations within 100 km) should prioritize octree-based 4D associators (PyOcto) over mixture-model associators (GaMMA). PyOcto's superior pick retention prevents hypocentral relocation collapse and expands microseismic catalog completeness by $> 30\%$.
3. **Automated Quality-Control (QC) Gating:** Automated catalogs should not be published without multi-parameter QC gates:

$$\begin{aligned} \text{Publication Gate: } & (N_{\text{sta}} \geq 4) \wedge (N_P \geq 3) \wedge (N_S \geq 2) \\ & \wedge (\text{GAP} \leq 200^\circ) \wedge (\text{RMS} \leq 0.6 \text{ s}). \end{aligned} \tag{7.2}$$

4. **Local Magnitude Station Distance Clipping:** Automated local magnitude estimation must enforce a distance cut-off ($r \leq 30 \text{ km}$) for micro-earthquakes ($M_L < 1.5$) to prevent distant low-SNR stations from artificially deflating network magnitude estimates.
5. **Continuous KPI Tracking:** Observatories should implement automated dashboards tracking pick timing residuals, daily proposed-to-matched ratios, and network gap distributions to monitor telemetry health and detect sensor calibration drift.

Table 7.1 summarizes these operational recommendations across network tiers.

Table 7.1 Operational architecture and deployment recommendations for European regional networks.

Pipeline Component	Recommended Standard	Alternative / Fallback	Operational Rationale / Justification
Phase Picker	PhaseNet (INSTANCE)	EQTransformer (INSTANCE)	Superior timing precision (MAE = 0.066 s); tight S-wave residual distribution.
Picker Threshold	$\tau = 0.2\text{--}0.3$ (Surveillance)	$\tau = 0.1$ (Crisis Swarm)	Suppresses background triggers by 80% in routine monitoring; maximizes recall during swarms.
Phase Associator	PyOcto (4D Octree)	GaMMA (Bayesian GMM)	Retains 90.7% of true picks; prevents NonLinLoc relocation collapse; $> 5,000\times$ real-time.
Velocity Model	1D Calibrated Regional	3D Tomographic Grid	Ensures mathematical parity with historical bulletins; supports fast Eikonal travel-time grids.
Hypocenter Engine	NonLinLoc (EDT Inversion)	HypoInverse / Hypo71	Global octree search eliminates local minima; EDT formulation cancels origin time uncertainty.
Magnitude Inversion	Wood-Anderson ($r \leq 30$ km)	Network-wide average	Mitigates bimodal underestimation on micro-earthquakes ($M_L < 1.5$)
Hardware Backend	Multi-GPU Blade ($4\times$ A100)	Multi-Core Workstation	Accelerates continuous multi-station inference to $> 2,000\times$ real-time.

Target & Scope: Concrete operational specification for implementing AI seismic pipelines in regional European networks.

What the reader should notice: The architecture couples regional deep-learning inference with robust non-linear location and velocity-model-aware association.

Evidence Anchor: Synthesized from empirical benchmarks established in Chapter 6.

7.5 Limitations and Future Research Horizons

While this doctoral research establishes a verified baseline for AI-driven regional seismology, several scientific boundaries and emerging opportunities chart the roadmap for future development:

1. **Scaling to the Pan-European AdriaArray Mega-Experiment:** This study evaluated a 93-day window across 266 stations. The ongoing **AdriaArray** initiative integrates over 1,500 broadband and temporary stations spanning 15 European nations across the Adriatic-Alpine-Pannonian realm. Extending our CINECA Leonardo HPC workflow to the full multi-year AdriaArray dataset will enable the construction of the most complete, unified transboundary microseismic catalog in European history.
2. **Integration of High-Resolution 3D Velocity Models and Adjoint Tomography:** All relocations in this dissertation utilized a 1D layered crustal velocity model. While optimal for baseline comparisons against historical bulletins, 1D models cannot account for the severe lateral velocity contrasts across the Alpine Moho descent (where Moho depth varies from 25 km beneath the Venetian plain to > 50 km beneath the Tauern Window). Inverting travel times through high-resolution 3D regional tomographic models using NonLinLoc 3D or Eikonal wavefield solvers will sharpen hypocentral alignments and depth constraints along active blind thrust faults.
3. **Real-Time Edge Streaming and IoT Integration:** Transitioning from batch HPC processing to sub-second streaming inference requires compiling deep neural networks (PhaseNet and EQTransformer) to lightweight edge-computing runtimes (such as NVIDIA TensorRT, ONNX, or low-power microcontrollers collocated directly with station digitizers). Edge picking will enable localized decentralized trigger verification and reduce continuous satellite telemetry bandwidth.
4. **Automated First-Motion Polarities and Focal Mechanism Inversion:** While our pipeline successfully picked arrival times and local magnitudes, automated earthquake source characterization requires determining fault-plane kinematics. Future work should integrate neural first-motion polarity classifiers (*e.g.*, classifying impulsive compressional vs. dilatational onsets) coupled with amplitude-ratio inversion (HASH algorithm) or full-waveform regional moment tensor inversion to automatically deliver focal mechanism solutions for events with $M_L \geq 2.5$ within minutes of origin.
5. **Waveform Cross-Correlation and Double-Difference Relocation:** The automated catalogs generated by PyOcto and NonLinLoc provide an ideal starting baseline

for high-precision relative relocation. Integrating waveform cross-correlation delay-time measurements and double-difference relocation (Hypocenter Double-Difference (HypoDD)) will collapse diffuse hypocentral clouds into razor-sharp fault planes, illuminating the fine-scale segmentation of active thrust ramps in the Friuli Prealps.

7.6 Concluding Remarks

Earthquake monitoring in seismically hazardous, densely populated regions such as Northeastern Italy demands both physical rigor and rapid operational responsiveness. This dissertation provides a reviewable implementation and evaluation framework for that work. A defensible operational conclusion requires a reproducibility bundle for every benchmark, independent adjudication of proposed events, and human approval of the relevant observatory workflow; none of these decisions is replaced by this draft.

Data and Resources

This chapter records the intended data-provenance and software-availability statement for this dissertation. The network identifiers and links below require human confirmation against the FDSN registry and the access date used for the submitted thesis. They are not validated by the source checkout, and restricted waveforms or institutional catalogs are not redistributed with it.

Seismic Waveform and Station Metadata Networks

Continuous seismic waveform time series and station response inventories (StationXML) were retrieved from distributed European FDSN data centers (including OGS, INGV, GeoSphere Austria, Swiss Seismological Service (SED), Agencija Republike Slovenije za okolje (ARSO), and Observatories and Research Facilities for European Seismology (ORFEUS) EIDA infrastructure).

A total of **18 distinct regional, national, and international seismic networks** contributed observational data to the 266-station monitoring array analyzed in this thesis. Table 7.2 provides a systematic synthesis of these network assets, followed by formal bibliographic and data repository citations.

Table 7.2 Inventory of participating seismic networks, operating institutions, and persistent identifiers.

Code	Network Name	Operating Institution	Period	Persistent Identifier / DOI
4P	AdriaArray Temporary Network (N/S)	AdriaArray Working Group	2022–2026	https://www.fdsn.org/networks/detail/4P_2022/
2Y	AdriaArray Temporary Network (NE Italy)	OGS	2022–2026	https://doi.org/10.7914/1p36-6t87
CH	National Seismic Networks of Switzerland	SED at ETH Zürich	1983–pres.	https://doi.org/10.12686/sed/networks/ch
EV	Collalto Seismic Network (CSN)	OGS	2012–pres.	https://doi.org/10.7914/SN/EV
GU	Regional Seismic Network of NW Italy	University of Genoa	1967–pres.	https://doi.org/10.7914/SN/GU
HA	Interreg Italia-Austria Network	OGS / ZAMG	2005–pres.	FDSN temporary / project archive
IT	Italian Strong Motion Network (RAN)	Civil Protection Department	1972–pres.	https://doi.org/10.7914/SN/IT
IV	Rete Sismica Nazionale (RSN)	INGV	2005–pres.	https://doi.org/10.13127/sd/x0fxnh7qfy
MN	Mediterranean Network (MedNet)	INGV / Partners	1990–pres.	https://doi.org/10.13127/sd/fbbbtddtd6q
NI	North-East Italy Broadband Network	OGS / Univ. of Trieste	2002–pres.	https://doi.org/10.7914/SN/NI
OE	Austrian Seismic Network	GeoSphere Austria (ZAMG)	1987–pres.	https://doi.org/10.7914/SN/OE
OX	North-East Italy Seismic Network	OGS	2016–pres.	https://doi.org/10.7914/SN/OX
RF	FVG Accelerometric Network (RAF)	University of Trieste	1993–pres.	https://doi.org/10.7914/SN/RF
S	Seismo-at-School Program	SED at ETH Zürich	2008–pres.	https://doi.org/10.12686/SED/NETWORKS/S
SI	Province Südtirol Network	ZAMG / Prov. Bozen	2006–pres.	https://www.fdsn.org/networks/detail/SI/
SL	Seismic Network of the Republic of Slovenia	Slovenian Env. Agency (ARSO)	1990–pres.	https://doi.org/10.7914/SN/SL
ST	Trentino Seismic Network	Prov. Autonoma di Trento	1981–pres.	https://doi.org/10.7914/SN/ST
ZO	Garda Multiparametric Network (PDnet)	INGV	2021–pres.	https://doi.org/10.13127/SD/YHCFOMCBO_

Target & Scope: Permanent and temporary seismic monitoring networks utilized in the 266-station case study array.

What the reader should notice: The dataset integrates 18 distinct network operators across Italy, Austria, Slovenia, and Switzerland into a unified transboundary observation aperture.

Evidence Anchor: Formally registered FDSN network codes and International DOI Foundation records.

Formal Network Citations

- [1] **[4P] AdriaArray Working Group (2022).** *AdriaArray Temporary Network: Italy – north, south (2022–2026)*. International Federation of Digital Seismograph Networks. https://www.fdsn.org/networks/detail/4P_2022/
- [2] **[2Y] Pesaresi, D., & Rossi, G. (2022).** *AdriaArray Temporary Network: Italy – northeast (OGS-AdA)* [Data set]. International Federation of Digital Seismograph Networks. <https://doi.org/10.7914/1p36-6t87>
- [3] **[CH] Swiss Seismological Service (SED) at ETH Zürich (1983).** *National Seismic Networks of Switzerland* [Data set]. ETH Zürich. <https://doi.org/10.12686/sed/networks/ch>
- [4] **[EV] Priolo, E., Romanelli, M., Plasencia Linares, M. P., Garbin, M., Peruzza, L., Romano, M. A., Marotta, P., Bernardi, P., Moratto, L., Zuliani, D., & Fabris, P. (2015).** *Seismic Monitoring of an Underground Natural Gas Storage Facility: The Collalto Seismic Network*. *Seismological Research Letters*, 86(1), 109–123. <https://doi.org/10.1785/0220140087>; *Collalto Seismic Network (CSN)* [Data set]. International Federation of Digital Seismograph Networks. <https://doi.org/10.7914/SN/EV>
- [5] **[GU] University of Genoa (1967).** *Regional Seismic Network of North Western Italy (RSNI)* [Data set]. International Federation of Digital Seismograph Networks. <https://doi.org/10.7914/SN/GU>
- [6] **[HA] OGS and ZAMG (2005).** *Interreg Italia-Austria: Historical And Recent Earthquakes in Italy and Austria (HA)*. Cross-border seismological data exchange platform. OGS Centro di Ricerche Sismologiche (CRS), Udine/Trieste.
- [7] **[IT] Presidency of Council of Ministers – Civil Protection Department (1972).** *Italian Strong Motion Network (Rete Accelerometrica Nazionale – RAN)* [Data set]. International Federation of Digital Seismograph Networks. <https://doi.org/10.7914/SN/IT>
- [8] **[IV] Istituto Nazionale di Geofisica e Vulcanologia (INGV) (2005).** *Rete Sismica Nazionale (RSN)* [Data set]. Istituto Nazionale di Geofisica e Vulcanologia (INGV). <https://doi.org/10.13127/sd/x0fxnh7qfy>
- [9] **[MN] MedNet Project Partner Institutions (1990).** *Mediterranean Very Broadband Seismographic Network (MedNet)* [Data set]. Istituto Nazionale di Geofisica e Vulcanologia (INGV). <https://doi.org/10.13127/sd/fbbbtddtd6q>

- [10] [NI] **Istituto Nazionale di Oceanografia e di Geofisica Sperimentale (OGS) and University of Trieste (2002)**. *North-East Italy Broadband Network* [Data set]. International Federation of Digital Seismograph Networks. <https://doi.org/10.7914/SN/NI>
- [11] [OE] **ZAMG – Zentralanstalt für Meteorologie und Geodynamik (1987)**. *Austrian Seismic Network* [Data set]. GeoSphere Austria / International Federation of Digital Seismograph Networks. <https://doi.org/10.7914/SN/OE>
- [12] [OX] **Istituto Nazionale di Oceanografia e di Geofisica Sperimentale – OGS (2016)**. *North-East Italy Seismic Network* [Data set]. International Federation of Digital Seismograph Networks. <https://doi.org/10.7914/SN/OX>
- [13] [RF] **University of Trieste (1993)**. *Friuli Venezia Giulia Accelerometric Network (RAF)* [Data set]. International Federation of Digital Seismograph Networks. <https://doi.org/10.7914/SN/RF>
- [14] [S] **Swiss Seismological Service (SED) at ETH Zürich (2008)**. *Seismology at School Program, ETH Zürich (Seismo-at-school)* [Data set]. ETH Zürich. <https://doi.org/10.12686/SED/NETWORKS/S>
- [15] [SI] **ZAMG and Autonomous Province of Bozen/Bolzano (2006)**. *Province Südtirol Network* [Data set]. International Federation of Digital Seismograph Networks. <https://www.fdsn.org/networks/detail/SI/>
- [16] [SL] **Slovenian Environment Agency (ARSO) (1990)**. *Seismic Network of the Republic of Slovenia* [Data set]. International Federation of Digital Seismograph Networks. <https://doi.org/10.7914/SN/SL>
- [17] [ST] **Geological Survey – Provincia Autonoma di Trento (1981)**. *Trentino Seismic Network* [Data set]. International Federation of Digital Seismograph Networks. <https://doi.org/10.7914/SN/ST>
- [18] [ZO] **Massa, M., Rizzo, A. L., Lorenzetti, A., Lovati, S., D’Alema, E., Puglia, R., Carannante, S., & Luzi, L. (2021)**. *Rete di monitoraggio multiparametrico del Garda – PDnet (Version 1.0)* [Data set]. Istituto Nazionale di Geofisica e Vulcanologia (INGV). https://doi.org/10.13127/SD/YHCFOMCBO_

Benchmark Machine-Learning Training Corpora

The deep-learning phase pickers evaluated in this thesis were accessed through the standardized SeisBench repository (Woollam et al., 2022, <https://github.com/seisbench/>

`seisbench`):

- **INSTANCE (Italy)**: Michelini, A., Cianetti, S., Gaviano, S., Giunchi, C., Jozinović, D., & Lauciani, V. (2021). *INSTANCE – the Italian seismic dataset for machine learning*. *Earth System Science Data*, 13(12), 5509–5544. <https://doi.org/10.5194/essd-13-5509-2021>
- **STEAD (Global)**: Mousavi, S. M., Sheng, Y., Zhu, W., & Beroza, G. C. (2019). *Stanford Earthquake Dataset (STEAD): A Global Data Set of Seismic Signals for AI*. *IEEE Access*, 7, 179464–179476. <https://doi.org/10.1109/ACCESS.2019.2947848>
- **SCEDC (Southern California)**: Southern California Earthquake Center (2013). *Southern California Seismic Network (SCSN) Waveform Data*. Caltech/USGS. <https://doi.org/10.7909/C3WD3xH1>
- **PhaseNet Original Weights**: Zhu, W., & Beroza, G. C. (2018). *PhaseNet: a deep-neural-network-based seismic arrival-time picking method*. *Geophysical Journal International*, 216(1), 261–273. <https://doi.org/10.1093/gji/ggy423>
- **Earthquake Transformer Original Weights**: Mousavi, S. M., Ellsworth, W. L., Zhu, W., Chuang, L. Y., & Beroza, G. C. (2020). *Earthquake transformer—an attentive deep-learning model for simultaneous earthquake detection and phase picking*. *Nature Communications*, 11(1), 3952. <https://doi.org/10.1038/s41467-020-17591-w>

Code and Software Availability

The complete computational software developed in this dissertation is open-source and structured across modular packages:

- **OGS Pipeline Framework**: The Python source code (`OGS/src/`) implementing multi-format catalog parsing, FDSN downloading, bipartite graph matching (BPGMA), local magnitude calculation, and plotting is maintained in the institutional repository: <https://github.com/kentanaka3/PhD>.
- **CINECA Leonardo HPC Orchestration**: All Slurm submission scripts, Hydra configuration templates, Makefile dependency graphs, and environment activation wrappers (`OGS/utils/Leonardo/`) are preserved within the repository under release tag `v1.0-UNITS-PhD`.

- **SeisBench Platform:** Pre-trained models and dataset interfaces were executed using SeisBench version 0.3.4 (Woollam et al., 2022).
- **Associator Codes:**
 - **GaMMA:** Available at <https://github.com/AI4EPS/GaMMA> (Zhu et al., 2022).
 - **PyOcto:** Available at <https://github.com/austin-bell/pyocto> (Münchmeyer, 2024).
- **Relocation Engine:** The NonLinLoc software suite (version 7.00) developed by Anthony Lomax is accessible at <https://alomax.free.fr/nlloc/> (Lomax et al., 2000).

Appendix A

Pipeline Configuration and Parameter Reference

This appendix provides the complete, production-grade **Hydra** (Yadan, 2019) configuration file and associated parameter dictionaries governing the automated machine-learning seismic processing pipeline. The pipeline architecture decouples procedural execution logic from runtime parameters, instantiating all components dynamically via reflection (`_target_` bindings): continuous waveform data ingestion via `PyrockoSquirrelDataSource`, deep-learning phase picking via `SeisBenchPicker`, Bayesian spatiotemporal association via `GammaAssociator`, pick quality control via `OGSPickStatQC`, non-linear probabilistic hypocenter relocation via `NonLinLoc`, calibrated Richter local magnitude estimation via `OGSLocalMagnitude`, and asynchronous distributed execution via a Dask cluster.

Section A.1 presents the complete, executable YAML configuration manifest. Sections A.2 through A.4 provide detailed parameter dictionaries formatted in publication-grade booktabs tables, documenting parameter names, data types, configured values, epistemic categories, and operational physical semantics.

A.1 Historical Hydra YAML Example and Current Configuration Boundary

The following listing is an unverified historical configuration example retained to explain the intended parameter structure. It is not the exact configuration of a reproducible run and must not be executed as-is. In particular, it differs from the current checked-in configuration in data-source binding, waveform paths, velocity-model path, and GaMMA parameters. A publication-ready appendix must replace it with an archived resolved Hydra configuration,

its hash, code revision, command, run identifier, and reviewer approval.

Listing A.1 Unverified historical Hydra configuration example. The listing illustrates configuration fields but differs from the current checked-in YAML and is not evidence of a production run.

```

1 log_level: DEBUG
2 output_path: catalogs/OGS
3 data:
4   _target_: ml_catalog.data.SquirrelDataSource
5   env: ogsDB
6   persistent: v1
7   check: false
8   paths:
9     - /leonardo_work/IscrC_AISeism/WORK/waveform/
10    - /leonardo_work/IscrC_AISeism/PhD/OGS/data/station/
11   channel_priorities:
12     - HH?
13     - EH?
14     - HN?
15     - HG?
16     - BH?
17   starttime: '2024-03-20'
18   endtime: '2024-06-20'
19   velocity_model:
20     _target_: pandas.read_csv
21     filepath_or_buffer: OGS/data/ogs1d.csv
22 group_modules:
23   picker:
24     _target_: ml_catalog.modules.SeisBenchPicker
25     model:
26       _target_: seisbench.models.PhaseNet.from_pretrained
27       name: phd_ken_tanaka
28     classify_args:
29       P_threshold: 0.1
30       S_threshold: 0.1
31       batch_size: 16384
32     amplitude_extractor:
33       _target_: OGS.src.ogspicker.OGSAmplitudeExtractor
34   associator:
35     _target_: ml_catalog.modules.GammaAssociator
36     crs:
37       _target_: pyproj.crs.CRS.from_user_input
38       value: +proj=sterea +lon_0=12 +lat_0=46 +units=km +type=crs
39     config:
40       dims:
41         - x(km)
42         - y(km)
43         - z(km)
44       use_dbscan: true
45       use_amplitude: false
46       x(km):
47         - -300
48         - 400
49       y(km):
50         - -350

```

APPENDIX A. PIPELINE CONFIGURATION AND PARAMETER REFERENCE

```
51     - 250
52     z(km):
53     - 0
54     - 40
55     vel:
56         p: 5.85
57         s: 3.28
58     method: BGMM
59     oversample_factor: 5
60     dbscan_eps: 13.4
61     dbscan_min_samples: 3
62     min_picks_per_eq: 5
63     min_p_picks_per_eq: 3
64     min_s_picks_per_eq: 2
65     covariance_prior_pre:
66     - 5.0
67     - 5.0
68     max_sigma11: 1.0
69     qcpicks:
70         _target_: OGS.src.ogsqc.OGSPickStatQC
71         total_picks: 6
72         p_and_s_picks: 3
73     nonlinloc:
74         _target_: ml_catalog.modules.NonLinLoc
75         _convert_: partial
76     CONTROL:
77     - 1
78     - 42
79     TRANS:
80     - TRANS_MERC
81     - WGS-84
82     - 44.3
83     - 9.5
84     - 0.0
85     VGTYPE:
86     - - P
87     - - S
88     VGGRID:
89     - 2
90     - 9000
91     - 550
92     - 0.0
93     - 0.0
94     - -5.0
95     - 0.1
96     - 0.1
97     - 0.1
98     - SLOW_LEN
99     GTMODE:
100     - GRID2D
101     - ANGLES_YES
102     GT_PLFD:
103     - 0.001
104     - 0
105     LOCSIG:
106     - ML Catalog
107     LOCHYPOUT:
```

APPENDIX A. PIPELINE CONFIGURATION AND PARAMETER REFERENCE

```
108 - SAVE_NLLOC_ALL
109 - SAVE_HYPOINV_SUM
110 LOCSEARCH:
111 - OCT
112 - 50
113 - 50
114 - 5
115 - 0.01
116 - 50000
117 - 5000
118 - 1
119 - 1
120 LOCGRID:
121 - 600
122 - 450
123 - 50
124 - 0
125 - 0
126 - 0
127 - 1
128 - 1
129 - 1
130 - PROB_DENSITY
131 - SAVE
132 LOCMETH:
133 - EDT_OT_WT
134 - 9999.0
135 - 4
136 - -1
137 - -1
138 - -1
139 - 10
140 - -1.0
141 - 1
142 LOCGAU:
143 - 0.15
144 - 1.0
145 LOCGAU2:
146 - 0.05
147 - 0.1
148 - 3.0
149 LOCQUAL2ERR:
150 - 0.1
151 - 0.5
152 - 1.0
153 - 2.0
154 - 99999.9
155 magnitude:
156   _target_: OGS.src.ogsmagnitude.OGSLocalMagnitude
157   station_corrections:
158     _target_: pandas.read_csv
159     filepath_or_buffer: OGS/data/Magnitude/LocalMagnitude/mag_correction.csv
160   ignore_stations:
161     _target_: pandas.read_csv
162     filepath_or_buffer: OGS/data/Magnitude/LocalMagnitude/ignore_stations.csv
163 merge_module:
164   _target_: ml_catalog.base.MergeEvents
```

```

165 joint_modules: {}
166 builder:
167   outputs:
168     - events
169     - assignments
170 cluster:
171   _target_: dask.distributed.LocalCluster
172   n_workers: 1
173   threads_per_worker: 1
174   processes: false
175   dashboard_address: :8989
176 libpath: .

```

A.2 Data Ingestion and Phase Picker Parameter Dictionaries

Table [A.1](#) documents the parameters governing waveform and station metadata ingestion via Pyrocko Squirrel. Table [A.2](#) details the operational parameters for deep-learning phase picking.

Table A.1 Configuration parameter dictionary for waveform data ingestion (SquirrelDataSource).

Parameter	Data Type	Configured Value	Epistemic Tag ^a	Operational Semantics / Description
target	Class Path	SquirrelDataSource	[System Architecture]	Pyrocko Squirrel high-performance waveform database interface
env	String	ogsDB	[System Architecture]	Local SQLite database environment tag for indexing metadata
persistent	String	v1	[System Architecture]	Database cache persistence schema version
check	Boolean	false	[System Architecture]	Bypasses runtime SHA-256 trace checksums to accelerate I/O
paths	List[Path]	Lustre waveform & XML	[Observed]	Storage endpoints for continuous miniSEED and StationXML
channel_priorities	List[Regex]	[HH?, EH?, HN?, ...]	[Analyst-Normalized]	Sensor hierarchy: High-Gain Broadband → Short-Period → Accelerometer
starttime	ISO Date	'2024-03-20'	[Observed]	Beginning timestamp of continuous observation horizon (00:00:00 UTC)
endtime	ISO Date	'2024-06-20'	[Observed]	Terminating timestamp of continuous observation horizon (23:59:59 UTC)
velocity_model	CSV Target	OGS/data/ogs1d.csv	[Analyst-Normalized]	Regional 1D crustal velocity profile ingested into pandas

^a Epistemic tagging: [System Architecture] denotes runtime infrastructure bindings; [Observed] denotes empirical sensor files; [Analyst-Normalized] denotes regional observatory conventions.

Target & Scope: Ingestion and caching settings for multi-station miniSEED waveform streams.

What the reader should notice: The channel priority hierarchy guarantees high-gain broadband streams are selected preferentially while preserving strong-motion channels during clipping.

Evidence Anchor: Formulated in `ml_catalog.data.SquirrelDataSource`.

Table A.2 Configuration parameter dictionary for deep-learning phase picking (SeisBenchPicker).

Parameter	Data Type	Configured Value	Epistemic Tag	Operational Semantics / Description
<code>_target_</code>	Class Path	<code>SeisBenchPicker</code>	[System Architecture]	Modular picker wrapping SeisBench pre-trained PyTorch models
<code>model._target_</code>	Class Method	<code>PhaseNet.from_pretrained</code>	[Model Architecture]	Instantiates PhaseNet 1D U-Net neural architecture
<code>model.name</code>	String	<code>phd_ken_tanaka</code>	[Model Architecture]	Model checkpoint fine-tuned on the Italian INSTANCE corpus
<code>P_threshold</code>	Float	0.1 (Baseline)	[Model Hypothesis]	Minimum posterior probability threshold for P-wave peak acceptance
<code>S_threshold</code>	Float	0.1 (Baseline)	[Model Hypothesis]	Minimum posterior probability threshold for S-wave peak acceptance
<code>batch_size</code>	Integer	16 384	[System Architecture]	Inference window batch size saturating NVIDIA A100 Tensor Cores
<code>amplitude_extractor</code>	Class Path	<code>OGSAmplitudeExtractor</code>	[Derived Metric]	Extracts zero-to-peak amplitude for downstream local magnitude

Target & Scope: Neural network inference parameters for continuous phase onset detection.

What the reader should notice: Large batch sizing (16 384) maximizes GPU tensor core occupancy, achieving $> 2,000\times$ real-time inference throughput.

Evidence Anchor: Implemented in `OGS/src/ogspicker.py`.

A.3 Phase Associator and Quality Control Parameter Dictionaries

Table [A.3](#) documents the parameters governing the Bayesian Gaussian Mixture Model Associator (`GammaAssociator`). Table [A.4](#) outlines the quality-control gates enforced by `OGSPickStatQC`.

Table A.3 Configuration parameter dictionary for Bayesian phase association (GammaAssociator).

Parameter	Data Type	Configured Value	Epistemic Tag	Operational Semantics / Description
crs.value	Proj4 String	+proj=sterea ...	[Analyst-Normalized]	Conformal stereographic projection centered at (12° E, 46° N)
dims	List[String]	[x(km), y(km), z(km)]	[System Architecture]	Spatial coordinate axes in kilometers for distance calculations
x(km), y(km)	Pair[Float]	[-300, 400], [-350, 250]	[Analyst-Normalized]	Lateral spatial bounding box enclosing regional network aperture
z(km)	Pair[Float]	[0, 40]	[Analyst-Normalized]	Seismogenic crustal depth search range (0–40 km)
vel.p, vel.s	Float [km/s]	5.85, 3.28	[Analyst-Normalized]	Apparent body-wave velocities for initial half-space travel times
method	String	BGMM	[Model Architecture]	Bayesian Gaussian Mixture Model with Dirichlet process priors
use_dbscan	Boolean	true	[System Architecture]	Enables spatiotemporal DBSCAN pre-clustering to discard noise
dbscan_eps	Float [s]	13.4	[Model Hypothesis]	Maximum spatiotemporal distance threshold for DBSCAN connectivity
dbscan_min_samples	Integer	3	[Model Hypothesis]	Minimum neighboring picks required to seed a DBSCAN cluster
oversample_factor	Integer	5	[System Architecture]	Time oversampling multiplier during EM initialization
min_picks_per_eq	Integer	5	[Analyst-Normalized]	Minimum total picks required to instantiate an earthquake event
min_p_picks_per_eq	Integer	3	[Analyst-Normalized]	Minimum P-wave arrivals required per associated event
min_s_picks_per_eq	Integer	2	[Analyst-Normalized]	Minimum S-wave arrivals required to constrain focal depth
covariance_prior_pre	Pair[Float]	[5.0, 5.0]	[Model Hypothesis]	Prior variance scale for horizontal and vertical travel-time misfits
max_sigma11	Float	1.0	[Model Hypothesis]	Maximum allowable primary eigenvalue for cluster spatial covariance

Target & Scope: Clustering and spatial domain constraints for Bayesian phase association.

What the reader should notice: The requirement for at least 3 P and 2 S picks ensures that candidate events possess sufficient degrees of freedom for initial hypocentral stability.

Evidence Anchor: The current source is OGS/config/group_modules/associator/ogsgamma.yaml; values in this table describe the historical example and require run-level confirmation.

Table A.4 Configuration parameter dictionary for pick quality control (OGSPickStatQC).

Parameter	Data Type	Configured Value	Operational Semantics / Description
<code>_target_</code>	Class Path	OGSPickStatQC	Post-association quality control module
<code>total_picks</code>	Integer	6	Minimum associated picks required for event survival into location
<code>p_and_s_picks</code>	Integer	3	Minimum stations recording both P and S arrivals

Target & Scope: Quality-control filter gating associated event hypotheses prior to NonLinLoc inversion.

What the reader should notice: Enforcing 3 co-located P and S stations eliminates unconstrained hypocenters with infinite trade-offs between depth and origin time.

Evidence Anchor: Implemented in `OGS/src/ogsqc.py`.

A.4 Hypocenter Location and Magnitude Parameter Dictionaries

Table [A.5](#) specifies the NonLinLoc inversion cards. Table [A.6](#) details local magnitude estimation and Dask parallel cluster parameters.

Table A.5 Configuration parameter dictionary for non-linear hypocenter location (NonLinLoc).

NLL Card	Data Type	Configured Value	Epistemic Tag	Operational Semantics / Description
TRANS	Statement	TRANS_MERC WGS-84 ...	[Analyst-Normalized]	Transverse Mercator projection; origin (9.5° E, 44.3° N)
VGGRID	Grid Spec	9000 × 550, 0.1 km	[System Architecture]	High-resolution 2D velocity lookup grid (100 m node spacing)
GTMODE	Mode	GRID2D ANGLES_YES	[System Architecture]	Computes take-off angles and ray paths via Podvin-Lecomte solver
LOCSEARCH	Search Spec	OCT 50 50 5 0.01 ...	[Model Architecture]	Oct-Tree importance sampling: initial cells, max nodes (50 000)
LOCGRID	Grid Bounds	600 × 450 × 50 km	[Analyst-Normalized]	3D search domain enclosing regional crustal volume down to 50 km
LOCMETH	Inversion	EDT_OT_WT 9999.0 4 ...	[Model Architecture]	Equal Differential Time formulation; minimum 4 station pairs
LOCGAU	Error Spec	0.15 s, 1.0 s	[Analyst-Normalized]	Correlation length and theoretical crustal travel-time error
LOCQUAL2ERR	List[Float]	[0.1, 0.5, 1.0, 2.0, ...]	[Analyst-Normalized]	Maps pick weight classes (0–4) to observational uncertainties in seconds

Target & Scope: NonLinLoc probabilistic relocation cards under the 1D crustal model.

What the reader should notice: The EDT formulation decouples origin time from spatial coordinates, rendering hypocentral convergence robust against outlier picks.

Evidence Anchor: The current source is `OGS/config/group_modules/nonlinloc/ogsnonlinloc1d.yaml`; values in this table describe the historical example and require run-level confirmation.

Table A.6 Configuration parameter dictionary for magnitude estimation and Dask execution.

Parameter	Data Type	Configured Value	Epistemic Tag	Operational Semantics / Description
magnitude._target_	Class Path	OGSLocalMagnitude	[System Architecture]	Synthetic Wood-Anderson Richter local magnitude engine
station_corrections	CSV Target	mag_correction.csv	[Analyst-Normalized]	Empirical station site correction terms ($c_{5,s}$)
ignore_stations	CSV Target	ignore_stations.csv	[Analyst-Normalized]	Blacklisted noisy or uncalibrated accelerometric stations
cluster._target_	Class Path	LocalCluster	[System Architecture]	Distributed memory Dask task scheduler
n_workers	Integer	1	[System Architecture]	Worker process count per MPI task allocation
processes	Boolean	false	[System Architecture]	Uses thread-based worker model to share in-memory waveform arrays
dashboard_address	Port String	:8989	[System Architecture]	Interactive web telemetry dashboard for tracking real-time execution

Target & Scope: Automated local magnitude calculation and asynchronous job scheduling.
 What the reader should notice: Thread-based worker execution eliminates multi-gigabyte memory copying of raw waveform streams during parallel station processing.
 Evidence Anchor: Implemented in OGS/src/ogsmagnitude.py and OGS/config/config.yaml.

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